

250 LIMITES AVEC CORRECTION

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Calculer les limites suivantes :

$$(1) \lim_{x \rightarrow +\infty} \frac{x}{\sqrt[3]{x} - 1}$$

$$(2) \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x-x^3}}{2-x}$$

$$(3) \lim_{x \rightarrow 0} \frac{\sqrt[3]{1-x}-1}{\sin x}$$

$$(4) \lim_{x \rightarrow 2} \frac{x - \sqrt[3]{x+6}}{3 - \sqrt{2x+5}}$$

$$(5) \lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt[4]{x}-1}$$

$$(6) \lim_{x \rightarrow 4} \frac{\sqrt[3]{5-x}-1}{2 - \sqrt[3]{x+4}}$$

$$(7) \lim_{x \rightarrow +\infty} x \left(\sqrt[3]{x} - \sqrt{x} \right)$$

$$(8) \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2}}{x-2}$$

$$(9) \lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - 2x$$

$$(10) \lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - x$$

$$(11) \lim_{x \rightarrow +\infty} \frac{\sqrt{x} - \sqrt[3]{x}}{\sqrt{x} + \sqrt[8]{x}}$$

$$(12) \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3-x} - \sqrt[3]{x^3+2x}}{x}$$

$$(13) \lim_{x \rightarrow 3} \frac{x^3 - 27}{\sqrt[3]{x} - \sqrt[3]{3}}$$

$$(14) \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3+5}}{\sqrt[3]{x^3+2}}$$

$$(15) \lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$$

$$(16) \lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 3} + 2x - 1)$$

$$(17) \lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 8} - x)$$

$$(18) \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - 2}{x^2 - 2x}$$

$$(19) \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 + x + 1} - \sqrt[4]{x^4 + 1}}{2x}$$

$$(20) \lim_{x \rightarrow +\infty} (x - \sqrt[3]{5x^2 + x + 1})$$

$$(21) \lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{5x^2 - 7x - 6}$$

$$(22) \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+2}} - x \right)$$

$$(23) \lim_{x \rightarrow +\infty} x^2 \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}} \right)$$

$$(24) \lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(8x^3 + x^2)^2}}$$

$$(25) \lim_{x \rightarrow +\infty} (\sqrt[3]{64x^3 + x^2} - 4x)$$

$$(26) \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 1} + 2x - 3)$$

$$(27) \lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}$$

$$(28) \lim_{x \rightarrow -1} \frac{\sqrt[3]{7x^2 - x} + \sqrt{x^2 - 3x - 4}}{x^2 - 1}$$

$$(29) \lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt{x+3}-2}$$

$$(30) \lim_{x \rightarrow -\infty} \sqrt[4]{\frac{x^3+5x^2-7}{16x^3-3x+1}}$$

$$(31) \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4+x^2+2}}{x+2}$$

$$(32) \lim_{x \rightarrow 0} \frac{1-\cos(2x)}{\sqrt[3]{x^2+1}-1}$$

$$(33) \lim_{x \rightarrow +\infty} \left(\sqrt[3]{x^3 + 5x^2 + 3} - 2x + 1 \right)$$

$$(34) \lim_{x \rightarrow 5} \frac{\sqrt[3]{4x+7}-3}{\sqrt{2x^2-1}-7}$$

$$(35) \lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt[3]{x^2+4}+5}$$

$$(36) \lim_{x \rightarrow -\infty} \frac{4x-1}{\sqrt{x^2+3x+5}+2x}$$

$$(37) \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1}-\sqrt{1-x}}{x}$$

$$(38) \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3}+\sqrt{2x+2}-4}{x-1}$$

$$(39) \lim_{x \rightarrow -\infty} \frac{x}{\sqrt[3]{2-x}}$$

$$(40) \lim_{x \rightarrow 0} \frac{1-\cos(4x)}{\tan x \sin 3x}$$

$$(41) \lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7)$$

$$(42) \lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3$$

$$(43) \lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5}$$

$$(44) \lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3}$$

$$(45) \lim_{x \rightarrow +\infty} \frac{x^2+5x}{(2x+3)^2}$$

$$(46) \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2}$$

$$(47) \lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2+3x-5}{x^2+x+2}$$

$$(48) \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{2}x^2}{x+1} - \sqrt{3}x \right)$$

$$(49) \lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8}$$

$$(50) \lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2-5x}{8x^2+7}}$$

$$(51) \lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x-5}}$$

$$(52) \lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}}$$

$$(53) \lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}}$$

$$(54) \lim_{x \rightarrow 3} \frac{x^2-5x+6}{9-x^2}$$

$$(55) \lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5}$$

$$(56) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}$$

$$(57) \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x - x^2}$$

$$(58) \lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x} - 2}$$

$$(59) \lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2}$$

$$(60) \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x+6} - 3}$$

$$(61) \lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}}$$

$$(62) \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2}$$

$$(63) \lim_{x \rightarrow 3} \frac{7 - 5x}{(x - 3)^2}$$

$$(64) \lim_{x \rightarrow -1} \frac{x+1}{\sqrt{x^2-1}}$$

$$(65) \lim_{x \rightarrow 1^-} \frac{x^2 - x - 3}{x^2 + 2x - 3}$$

$$(66) \lim_{x \rightarrow 1^-} \frac{\sqrt{x^2-3x+2}}{x-1}$$

$$(67) \lim_{x \rightarrow 1^-} \frac{8x-1}{\sqrt{1-x}}$$

$$(68) \lim_{x \rightarrow 4^+} \frac{x}{2 - \sqrt{x}}$$

$$(69) \lim_{x \rightarrow 4^+} \frac{\sqrt{x+12} - \sqrt{3x-4}}{x^2 - 8x + 16}$$

$$(70) \lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2 - 4}}$$

$$(71) \lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x - 2}$$

$$(72) \lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)}$$

$$(73) \lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}}$$

$$(74) \lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|}$$

$$(75) \lim_{x \rightarrow \left(\frac{1}{2}\right)^+} \frac{3x^2 - x - 1}{4x^2 - 1}$$

$$(76) \lim_{x \rightarrow +\infty} (2x - 5\sqrt{x})$$

$$(77) \lim_{x \rightarrow -\infty} (\sqrt{9x^2 + 3x - 5} + x + 3)$$

$$(78) \lim_{x \rightarrow +\infty} (\sqrt{x+3} - x)$$

$$(79) \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4)$$

$$(80) \lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x})$$

$$(81) \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3)$$

$$(82) \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x}$$

$$(83) \lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}}$$

$$(84) \lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2}$$

$$(85) \lim_{x \rightarrow +\infty} \frac{x+6\sqrt{x}}{2x+\sqrt{x}-3}$$

$$(86) \lim_{x \rightarrow +\infty} \frac{1}{x} (\sqrt{3x^2+1} - \sqrt{x^2+x})$$

$$(87) \lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2)$$

$$(88) \lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}}$$

$$(89) \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right)$$

$$(90) \lim_{x \rightarrow +\infty} \left(\frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right)$$

$$(91) \lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$$

$$(92) \lim_{x \rightarrow 0} \frac{2\sqrt{x+1} - \sqrt{x^2+4}}{x}$$

$$(93) \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x^3}}{\sqrt{1+x} - \sqrt{1+2x}}$$

$$(94) \lim_{x \rightarrow 0} \frac{\sqrt{4-x} - \sqrt{4+x}}{x}$$

$$(95) \lim_{x \rightarrow 1} \frac{x\sqrt{x}-1}{1-x^2};$$

$$(96) \lim_{z \rightarrow 0} \frac{\sqrt{x+1}-1}{x^2-x}$$

$$(97) \lim_{x \rightarrow 2} \frac{2-\sqrt{x+2}}{3-\sqrt{2x+5}};$$

$$(98) \lim_{z \rightarrow 2} \frac{x-\sqrt{x+2}}{x^2-4}$$

$$(99) \lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{x-4}$$

$$(100) \lim_{z \rightarrow 3} \frac{\sqrt{x+1}-\sqrt{7-x}}{\sqrt{2x+3}-\sqrt{15-2x}}$$

$$(101) \lim_{x \rightarrow 0^+} \frac{x+\sqrt{x}}{\sqrt{x^2+x}-x}$$

$$(102) \lim_{x \rightarrow +\infty} (\sqrt{x^2+x+3}-2x)$$

$$(103) \lim_{x \rightarrow -\infty} (\sqrt{3x^2-5x+7}+x-4)$$

$$(104) \lim_{x \rightarrow +\infty} \left(1 + \frac{x}{2} - \sqrt{1+x}\right)$$

$$(105) \lim_{x \rightarrow +\infty} \frac{x-1}{\sqrt{x^2+4}};$$

$$(106) \lim_{x \rightarrow -\infty} \frac{\sqrt{3x^2-x+8}}{5x-2};$$

$$(107) \lim_{x \rightarrow -\infty} (x + \sqrt{1-x})$$

$$(108) \lim_{x \rightarrow -\infty} (\sqrt{2x} + \sqrt{x^2 - x + 4})$$

$$(109) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 2x + 3}}{\sqrt{4-x}}$$

$$(110) \lim_{x \rightarrow -\infty} \sqrt{\frac{x^2 - 1}{x^4 + 2}}$$

$$(111) \lim_{x \rightarrow +\infty} (x - 2\sqrt{x} + 5)$$

$$(112) \lim_{x \rightarrow +\infty} (\sqrt{1+x} - \sqrt{1+4x})$$

$$(113) \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - x - 1)$$

$$(114) \lim_{x \rightarrow -\infty} (\sqrt{x^4 + x^2 + 1} - x^2 + x + 3)$$

$$(115) \lim_{x \rightarrow +\infty} (\sqrt{4x^2 - 3x + 7} - 2x + 5)$$

$$(116) \lim_{x \rightarrow +\infty} (2x - \sqrt{4x^2 + 1});$$

$$(117) \lim_{x \rightarrow -\infty} (\sqrt{9x^2 - 4x + 5} + 3x - 2)$$

$$(118) \lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 7} - 2)$$

$$(119) \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - \sqrt{x+1});$$

$$(120) \lim_{x \rightarrow +\infty} (\sqrt{2x+3} - \sqrt{x+4});$$

$$(121) \lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x}} - \sqrt{x})$$

$$(122) \lim_{x \rightarrow -\infty} (\sqrt{1-x} - \sqrt{3-x});$$

$$(123) \lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x})$$

$$(124) \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^2 + x + 5}}{x + 3} - \frac{x + 2}{\sqrt{x^2 + x + 5}} \right)$$

$$(125) \lim_{x \rightarrow 4} \left(\frac{\sqrt{x+5} - \sqrt{x-1}}{\sqrt{x+12} - \sqrt{x-2}} \right)$$

$$(126) \lim_{x \rightarrow 3^+} \left(\frac{\sqrt{x^2 - 9} + \sqrt{x} - \sqrt{3}}{\sqrt{x-3}} \right)$$

$$(127) \lim_{x \rightarrow \frac{1}{3}} \left| \frac{x^2 - 6x}{3x - 1} \right|$$

$$(128) \lim_{x \rightarrow -\infty} x(x + \sqrt{x^2 + 1})$$

$$(129) \lim_{x \rightarrow 9} \left(\frac{\sqrt{x} - 3}{\sqrt{x} + 7 - 4} \right)$$

$$(130) \lim_{x \rightarrow 2} \left(\frac{\sqrt{2x+1} - \sqrt{3+x}}{\sqrt{x^2-1} - \sqrt{2x^2-5}} \right)$$

$$(131) \lim_{x \rightarrow 0^+} \left(\frac{x + \sqrt{x^2 + x}}{\sqrt{x^2 + x + 1} - 1} \right)$$

$$(132) \lim_{x \rightarrow +\infty} (\sqrt{2x^2 + 1} - \sqrt{x^2 - x - 2})$$

$$(133) \lim_{x \rightarrow +\infty} \frac{x+1}{\sqrt{1+x^2}}$$

$$(134) \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}}{x^2}$$

$$(135) \lim_{x \rightarrow -\infty} (\sqrt{3x^2 - 6x - 1} + 2x - 5)^*$$

$$(136) \lim_{x \rightarrow +\infty} (\sqrt{x^3 + 3x^2 + 4} - x^2 + 2)$$

$$(137) \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - 3x)$$

$$(138) \lim_{x \rightarrow +\infty} \left(x \sqrt{\frac{x}{x-1}} - x - 1 \right)$$

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limites de fonctions trigonomitriques

1. $\lim_{x \rightarrow 0} \frac{\tan x}{x}$

2. $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

3. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

4. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$

5. $\lim_{x \rightarrow 0} \frac{\tan(5x)}{x}$

6. $\lim_{x \rightarrow 0} \frac{\sin x}{\tan x}$

7. $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$

8. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$

9. $\lim_{x \rightarrow 0} \frac{\sin(7x)}{\sin(2x)}$

10. $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$

11. $\lim_{x \rightarrow 0} \frac{\sin(3x) - 3\sin x}{x^3}$

12. $\lim_{x \rightarrow 0} \frac{\tan(x) - \sin(x)}{x^3}$

13. $\lim_{x \rightarrow 0} \frac{\sin(x)^x - x \cos(x)}{x^3}$

14. $\lim_{x \rightarrow 0} \frac{\cos(2x) - e^{-x^2}}{x^4}$

15. $\lim_{x \rightarrow 0} \frac{\arctan(\sin x) - \sin(\arctan x)}{x^3}$

16. $\lim_{x \rightarrow 0} \frac{\sin(\sqrt{x}) - \sqrt{x^3}}{x^{3/2}}$ | quad(Chgt de variable $t = \sqrt{x}$)

17. $\lim_{x \rightarrow 0} \frac{\tan(ax) - \sin(bx)}{x^3} \setminus \text{quad avec } a, b > 0$

18. $\lim_{x \rightarrow 0} \frac{\ln(\cos x) + \frac{x^2}{2}}{x^4}$

19. $\lim_{x \rightarrow 0} \frac{\sin(x + \sin x) - 2\sin x}{x^3}$
20. $\lim_{x \rightarrow 0} \frac{\sin(x)\sin(2x)\sin(3x)}{x^3}$
21. $\lim_{x \rightarrow 0} \frac{\arcsin(\tan x) - \tan(\arcsin x)}{x^3}$
22. $\lim_{\substack{x \rightarrow 0 \\ x \rightarrow \frac{\pi}{2}}} \left(\tan x - \frac{1}{\cos x} \right) \sin x$
23. $\lim_{x \rightarrow 0} \frac{\sin(x^2) - x\sin x}{x^3}$
24. $\lim_{x \rightarrow 0} \frac{\cos(x) - \cos(\sqrt{x})}{x} \setminus \text{quad}(\text{Chgt } t = \sqrt{x})$
25. $\lim_{x \rightarrow 0} \frac{\tan(2x)^x - 2\tan x}{x^3}$
26. $\lim_{x \rightarrow 0} \frac{\sin(ax)^{x^3} - a\sin x}{x^3} \setminus \text{quad}(a \in \mathbb{R} \text{ fixé})$
27. $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$
28. $\lim_{x \rightarrow 0} \frac{\cos(x\sin x) - 1 + \frac{x^2}{2}}{x^4}$
29. $\lim_{x \rightarrow 0} \frac{\tan(\sqrt{x}) - \sqrt{x}}{x^{3/2}} \setminus \text{quad}(\text{Chgt } t = \sqrt{x})$
30. $\lim_{x \rightarrow 0} \frac{\arctan(1 - \cos x) - \frac{x^2}{2}}{x^4}$
31. $\lim_{x \rightarrow 0} \cos\left(\frac{\sin(\pi x)}{x}\right)$
32. $\lim_{x \rightarrow 0} \cos\left(x^2 \cdot \cos\left(\frac{1}{x}\right) + x\right)$
33. $\lim_{x \rightarrow 0^+} \tan\left(\sqrt{x} \cdot \sin\left(\frac{1}{x}\right)\right)$
34. $\lim_{x \rightarrow 0} \sin\left(\frac{\tan^2 x - \pi \tan x}{2x}\right)$

limite fonction logarithme népérien

1. $\lim_{x \rightarrow 0^+} x \ln(x)$
2. $\lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x}$
3. $\lim_{x \rightarrow +\infty} \frac{\ln(x)}{\frac{x}{x}}$
4. $\lim_{x \rightarrow +\infty} \frac{x}{\ln(x)}$
5. $\lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x^2}$
6. $\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1}$
7. $\lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{\ln(x)}$
8. $\lim_{x \rightarrow +\infty} \frac{\ln(x^2+1)}{\ln(x+1)}$
9. $\lim_{x \rightarrow 0^+} \frac{\ln(1+x)-x}{x^2}$
10. $\lim_{x \rightarrow +\infty} \frac{\ln(x)}{\sqrt{x}}$
11. $\lim_{x \rightarrow 0^+} x^a \ln(x)$, avec $a > 0$
12. $\lim_{x \rightarrow +\infty} (\ln(x+1) - \ln(x))$
13. $\lim_{x \rightarrow +\infty} \frac{\ln(x^3+2x)}{\ln(x^4+5)}$
14. $\lim_{x \rightarrow 0^+} \frac{\ln(1+x^2)}{\ln(1+x)}$
15. $\lim_{x \rightarrow 0^+} \frac{\ln(1+\sin x)}{\ln(1+x)}$
16. $\lim_{x \rightarrow +\infty} \frac{x \ln(x)}{x^2+1}$
17. $\lim_{x \rightarrow 0^+} \frac{\ln(1+e^x)-x}{x^2}$
18. $\lim_{x \rightarrow +\infty} \frac{\ln(x!)}{x \ln(x)}$
19. $\lim_{x \rightarrow +\infty} (\ln(x+1))^{1/\ln(x)}$
20. $\lim_{x \rightarrow 0^+} \left(\frac{\ln(1+x)}{x} \right)^{1/x}$

21. $\lim_{x \rightarrow 1^+} \left(\frac{2x}{x-1} + \ln(x-1) \right)$
22. $\lim_{x \rightarrow +\infty} (2x^2 - x - \ln(x+1))$
23. $\lim_{x \rightarrow +\infty} (3x - \ln(x^2 + 1))$
24. $\lim_{x \rightarrow 0} \frac{1}{x^2} \cdot \ln(\cos x)$
25. $\lim_{x \rightarrow \infty} \frac{\ln(x+1+\sqrt{x^2+1})}{2x}$
26. $\lim_{x \rightarrow 2^+} (x-2) \ln(x^2 - 2x)$
27. $\lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln^2(x)$
28. $\lim_{x \rightarrow -\infty} \frac{\ln(x^2 - x + 1)}{2x+1}$
29. $\lim_{x \rightarrow +\infty} \frac{x^2}{\ln x} :$
30. $\lim_{x \rightarrow +\infty} \frac{(x+1)\ln x}{x^2} :$
31. $\lim_{x \rightarrow +\infty} x \ln \left(\frac{x+1}{x-1} \right) :$
32. $\lim_{x \rightarrow e} \frac{(\ln x)^2 + 2\ln x - 3}{x-e} :$
33. $\lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{1+x}{1-x} \right)$
34. $\lim_{x \rightarrow -\infty} \ln \left(\frac{2x-1}{x+2} \right) :$
35. $\lim_{x \rightarrow 2^+} \left(\frac{3x}{x-2} + \ln(x-2) \right)$
36. $\lim_{x \rightarrow +\infty} \ln(\sqrt{x^2 + 3} - x)$
37. $\lim_{x \rightarrow (-1)^+} (x+1) \ln(x^2 + x)$
38. $\lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln x$
39. $\lim_{x \rightarrow +\infty} x \cdot \ln \left(\frac{x-1}{x+1} \right)$

limites fonction exponentielle

1. $\lim_{x \rightarrow +\infty} 3xe^{-x}$

2. $\lim_{x \rightarrow -\infty} \frac{e^x - 1}{2e^x + 1}$

3. $\lim_{x \rightarrow -\infty} (x + 1)e^x$

4. $\lim_{x \rightarrow +\infty} \frac{2e^x + 1}{e^x - 1}$

5. $\lim_{x \rightarrow -\infty} x + 2 + xe^x$

6. $\lim_{x \rightarrow +\infty} 2x - e^x$

7. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3}$

8. $\lim_{x \rightarrow +\infty} x^3 - e^x$

9. $\lim_{x \rightarrow -\infty} \frac{e^x + e^{-x}}{3 + e^x}$

10. $\lim_{x \rightarrow +\infty} e^{2x} - x^2$

11. $\lim_{x \rightarrow 0} e^{1 - \frac{1}{x^2}}$

12. $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - x + 1}$

13. $\lim_{x \rightarrow +\infty} xe^{2x} - e^x$

14. $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3 + 1}$

15. $\lim_{x \rightarrow -\infty} (x^4 - x)e^x$

16. $\lim_{x \rightarrow -\infty} e^{x^2 - 3x - 5}$

17. $\lim_{x \rightarrow -\infty} \frac{e^x}{1 + x^2}$

18. $\lim_{x \rightarrow -\infty} x(e^{2x} - e^x)$

$$19. \quad \lim_{x \rightarrow -\infty} (2 - x)e^x - x$$

$$20. \quad \lim_{x \rightarrow -\infty} x^2(e^x - e^{2x})$$

$$21. \quad \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x}\sin 2x} - e^{\cos 2x}}{x^2}$$

$$22. \quad \lim_{x \rightarrow 0} \frac{e^{\tan^2 x + \cos 2x} - e^{1 + \sin^2 3x}}{x^2}$$

$$23. \quad \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^{2x} - 1) \dots (e^{nx} - 1)}{x^n}$$

$$24. \quad \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\tan 2x}}{x^3}$$

$$25. \quad \lim_{x \rightarrow 0} \frac{1 - e^x \cos 2x}{\tan^2 x}$$

$$26. \quad \lim_{x \rightarrow 0} \frac{e^{x^2 + 2x} - e^{x^2 - 2x}}{x}$$

$$27. \quad \lim_{x \rightarrow 0} \frac{1 - e^{\sin^2 x \sqrt{\cos x}}}{x^2}$$

$$28. \quad \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x^2}} - e^{\cos 2x}}{x^2}$$

$$29. \quad \lim_{x \rightarrow \infty} \left(2^{\frac{1}{x}} + 3^{\frac{1}{x}} - 2 \right) x$$

$$30. \quad \lim_{x \rightarrow 0} \frac{e^{1 - \sqrt{\cos 2x}} - e^{\tan^2 x}}{x^2}$$

Correction des limites rationnelles :

(1) Calculons $\lim_{x \rightarrow +\infty} \frac{x}{\sqrt[3]{x}-1}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x}{\sqrt[3]{x}-1} &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{x})^3}{\sqrt[3]{x} \left(1 - \frac{1}{\sqrt[3]{x}}\right)} \\ &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{x})^2}{\left(1 - \frac{1}{\sqrt[3]{x}}\right)} \end{aligned}$$

donc: $\lim_{x \rightarrow +\infty} \frac{x}{\sqrt[3]{x}-1} = +\infty$ car : $\lim_{x \rightarrow +\infty} \sqrt[3]{x} = +\infty$; $\lim_{x \rightarrow +\infty} X^2 = +\infty$ et $\lim_{t \rightarrow +\infty} \frac{1}{t} = 0$

(2) Calculons $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x-x^3}}{2-x}$:

On a : $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x-x^3}}{2-x} = \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{(-x^3)\left(\frac{-1}{x^2}+1\right)}}{x\left(\frac{2}{x}-1\right)}$ donc:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x-x^3}}{2-x} &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{(-x)^3 \left(\frac{-1}{x^2}+1\right)}}{x \left(\frac{2}{x}-1\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{(-x)^3} \times \sqrt[3]{\frac{-1}{x^2}+1}}{x \left(\frac{2}{x}-1\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{-x \times \sqrt[3]{\frac{-1}{x^2}+1}}{x \left(\frac{2}{x}-1\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt[3]{\frac{-1}{x^2}+1}}{\left(\frac{2}{x}-1\right)} \end{aligned}$$

donc: $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x-x^3}}{2-x} = 1$ car : $\lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0$ $\lim_{x \rightarrow 1} \sqrt[3]{x} = 1$ et $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$

(3) Calculons $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1-x}-1}{\sin x}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt[3]{1-x}-1}{\sin x} &= \lim_{x \rightarrow 0} \frac{(\sqrt[3]{1-x}-1)((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1)}{((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1)\sin x} \\ &= \lim_{x \rightarrow 0} \frac{(\sqrt[3]{1-x})^3 - 1^3}{((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1)\sin x} \\ &= \lim_{x \rightarrow 0} \frac{1-x-1}{((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1)\sin x} \\ &= \lim_{x \rightarrow 0} \frac{-x}{((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1)\sin x} \end{aligned}$$

donc:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1-x}-1}{\sin x} = \lim_{x \rightarrow 0} \frac{-1}{((\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1) \frac{\sin x}{x}}$$

donc:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1-x}-1}{\sin x} = \frac{-1}{3}$$

car :

$$\lim_{x \rightarrow 0} \sqrt[3]{1-x} = 1 \text{ et } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

(4) Calculons $\lim_{x \rightarrow 2} \frac{x - \sqrt[3]{x+6}}{3 - \sqrt{2x+5}}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 2} \frac{x - \sqrt[3]{x+6}}{3 - \sqrt{2x+5}} &= \lim_{x \rightarrow 2} \frac{(x - \sqrt[3]{x+6})(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)(3 + \sqrt{2x+5})}{(3 - \sqrt{2x+5})(3 + \sqrt{2x+5})(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \lim_{x \rightarrow 2} \frac{(x^3 - (\sqrt[3]{x+6})^3)(3 + \sqrt{2x+5})}{(3^2 - (\sqrt{2x+5})^2)(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \lim_{x \rightarrow 2} \frac{(x^3 - (x+6))(3 + \sqrt{2x+5})}{(9 - (2x+5))(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \lim_{x \rightarrow 2} \frac{(x^3 - x - 6)(3 + \sqrt{2x+5})}{(-2x+4)(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+3)(3 + \sqrt{2x+5})}{-2(x-2)(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \lim_{x \rightarrow 2} \frac{(x^2+2x+3)(3 + \sqrt{2x+5})}{-2(x^2 + x\sqrt[3]{x+6} + (\sqrt[3]{x+6})^2)} \\
 &= \frac{(4+4+3)(3 + \sqrt{9})}{-2(4 + 2\sqrt[3]{8} + (\sqrt[3]{8})^2)} \\
 &= \frac{11 \times 6}{-2 \times 12}
 \end{aligned}$$

donc :

$$\lim_{x \rightarrow 2} \frac{x - \sqrt[3]{x+6}}{3 - \sqrt{2x+5}} = \frac{-11}{4}$$

(5) Calculons $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt[5]{x}-1}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt[5]{x}-1} &= \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x+7}-2)(\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 2^2)(\sqrt[4]{x} + \sqrt[4]{x} + \sqrt[4]{x} + 1)}{(\sqrt[4]{x}-1)(\sqrt[4]{x} + \sqrt[4]{x}^2 + \sqrt[4]{x} + 1)((\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 2^2)} \\
 &= \lim_{x \rightarrow 1} \frac{((\sqrt[3]{x+7})^3 - 2^3)(\sqrt[4]{x} + \sqrt[4]{x} + \sqrt[4]{x} + 1)}{(\sqrt[4]{x}-1^4)((\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 4)} \\
 &= \lim_{x \rightarrow 1} \frac{(x+7-8)(\sqrt[4]{x} + \sqrt[4]{x} + \sqrt[4]{x} + 1)}{(x-1)((\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 4)} \\
 &= \lim_{x \rightarrow 1} \frac{(x-1)(\sqrt[4]{x} + \sqrt[4]{x} + \sqrt[4]{x} + 1)}{(x-1)((\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 4)} \\
 &= \lim_{x \rightarrow 1} \frac{\sqrt[4]{x}^3 + \sqrt[4]{x}\sqrt[4]{x}^2 + \sqrt[4]{x} + 1}{(\sqrt[3]{x+7})^2 + 2^3\sqrt[3]{x+7} + 4} \\
 &= \frac{1+1+1+1}{2^2 + 2 \times 2 + 4}
 \end{aligned}$$

donec :

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt[5]{x}-1} = \frac{1}{3}$$

(6) calculons $\lim_{x \rightarrow 4} \frac{\sqrt[3]{5-x}-1}{2-\sqrt[3]{x+4}}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 4} \frac{\sqrt[3]{5-x}-1}{2-\sqrt[3]{x+4}} &= \lim_{x \rightarrow 4} \frac{(\sqrt[3]{5-x}-1)(\sqrt[3]{5-x})^2 + \sqrt[3]{5-x} + 1)(2^2 + 2^3\sqrt[3]{x+4} + \sqrt[3]{x+4}^2)}{(2-\sqrt[3]{x+4})(2^2 + 2^3\sqrt[3]{x+4} + \sqrt[3]{x+4}^2)(\sqrt[3]{5-x}^2 + \sqrt[3]{5-x} + 1)} \\
 &= \lim_{x \rightarrow 4} \frac{((\sqrt[3]{5-x})^3 - 1^3)(4 + 2^3\sqrt[3]{x+4} + (\sqrt[3]{x+4})^2)}{(2^3 - (\sqrt[3]{x+4})^3)((\sqrt[3]{5-x})^2 + \sqrt[3]{5-x} + 1)} \\
 &= \lim_{x \rightarrow 4} \frac{(5-x-1)(4 + 2^3\sqrt[3]{x+4} + (\sqrt[3]{x+4})^2)}{(8 - (x+4))((\sqrt[3]{5-x})^2 + \sqrt[3]{5-x} + 1)} \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(4 + 2^3\sqrt[3]{x+4} + (\sqrt[3]{x+4})^2)}{(4-x)((\sqrt[3]{5-x})^2 + \sqrt[3]{5-x} + 1)}
 \end{aligned}$$

donc:

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{\sqrt[3]{5-x} - 1}{2 - \sqrt[3]{x+4}} &= \lim_{x \rightarrow 4} \frac{4 + 2\sqrt[3]{x+4} + (\sqrt[3]{x+4})^2}{(\sqrt[3]{5-x})^2 + \sqrt[3]{5-x} + 1} \\ &= \frac{4 + 2 \times 2 + 2^2}{1 + 1 + 1}\end{aligned}$$

donc:

$$\lim_{x \rightarrow 4} \frac{\sqrt[3]{5-x} - 1}{2 - \sqrt[3]{x+4}} = 4$$

(7) Calculons $\lim_{x \rightarrow +\infty} x - \sqrt[3]{x} - \sqrt{x}$:

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} x - \sqrt[3]{x} - \sqrt{x} &= \lim_{x \rightarrow +\infty} x \left(1 - \frac{\sqrt[3]{x}}{x} - \frac{\sqrt{x}}{x} \right) \\ &= \lim_{x \rightarrow +\infty} x \left(1 - \frac{\sqrt[3]{x}}{(\sqrt[3]{x})^3} - \frac{\sqrt{x}}{(\sqrt{x})^2} \right) \\ &= \lim_{x \rightarrow +\infty} x \left(1 - \frac{1}{(\sqrt[3]{x})^2} - \frac{1}{\sqrt{x}} \right)\end{aligned}$$

donc:

$$\lim_{x \rightarrow +\infty} x - \sqrt[3]{x} - \sqrt{x} = +\infty$$

car :

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x} = +\infty; \lim_{x \rightarrow +\infty} \sqrt{x} = +\infty \text{ et } \lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

(8) Calculons $\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2}}{x-2}$:

On a:

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2}}{x-2} &= \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2} - 2 + 2}{x-2} \\ &= \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - 2 - \sqrt{x+2} + 2}{x-2} \\ &= \lim_{x \rightarrow 2} \frac{(\sqrt[3]{x+6} - 2) - (\sqrt{x+2} - 2)}{x-2} \\ &= \lim_{x \rightarrow 2} \left(\frac{(\sqrt[3]{x+6} - 2)}{x-2} - \frac{(\sqrt{x+2} - 2)}{x-2} \right) \end{aligned}$$

donc:

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2}}{x-2} &= \lim_{x \rightarrow 2} \left(\frac{(\sqrt[3]{x+6} - 2)(\sqrt[3]{x+6}^2 + 2\sqrt[3]{x+6} + 2^2)}{(x-2)(\sqrt[3]{x+6}^2 + 2\sqrt[3]{x+6} + 2^2)} - \frac{(\sqrt{x+2} - 2)}{x-2} \right) \\ &= \lim_{x \rightarrow 2} \left(\frac{(\sqrt[3]{x+6}^3 - 2^3)}{(x-2)(\sqrt[3]{x+6}^2 + 2\sqrt[3]{x+6} + 4)} - \frac{(x+2-4)}{(x-2)(\sqrt{x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 2} \left(\frac{x-2}{(x-2)(\sqrt[3]{x+6}^2 + 2\sqrt[3]{x+6} + 4)} - \frac{(x-2)}{(x-2)(\sqrt{x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 2} \left(\frac{1}{(\sqrt[3]{x+6}^2 + 2\sqrt[3]{x+6} + 4)} - \frac{1}{(\sqrt{x+2} + 2)} \right) \\ &= \frac{1}{4+4+4} - \frac{1}{2+2} \end{aligned}$$

donc :

$$\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - \sqrt{x+2}}{x-2} = \frac{+1}{6}$$

(9) Calculons $\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - 2x$:

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - 2x &= \lim_{x \rightarrow +\infty} \sqrt[3]{x^3 \left(1 + \frac{1}{x} + \frac{1}{x^3}\right)} - 2x \\ &= \lim_{x \rightarrow +\infty} x \sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}} - 2x \\ &= \lim_{x \rightarrow +\infty} x \left(\sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}} - 2 \right) \end{aligned}$$

donc:

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - 2x = -\infty$$

car:

$$\lim_{x \rightarrow +\infty} x = +\infty ; \lim_{x \rightarrow +\infty} \frac{1}{x} = 0 \text{ et } \lim_{x \rightarrow +\infty} \frac{1}{x^3} = 0$$

(10) Calculons $\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - x :$

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - x &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{x^3 + x^2 + 1})^3 - x^3}{\sqrt[3]{x^3 + x^2 + 1}^2 + x\sqrt[3]{x^3 + x^2 + 1} + x^2} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^3 + x^2 + 1 - x^3}{\sqrt[3]{x^3 + x^2 + 1}^2 + x\sqrt[3]{x^3 + x^2 + 1} + x^2} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2 + 1}{\sqrt[3]{x^3 + x^2 + 1}^2 + x\sqrt[3]{x^3 + x^2 + 1} + x^2} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2 + 1}{\left(\sqrt[3]{x^3 \left(1 + \frac{1}{x} + \frac{1}{x^3}\right)}\right)^2 + x\sqrt[3]{x^3 \left(1 + \frac{1}{x} + \frac{1}{x^3}\right)} + x^2} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2 + 1}{x^2 \left(\sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}}\right)^2 + x^2 \sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}} + x^2} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2 \left(1 + \frac{1}{x^2}\right)}{x^2 \left(\left(\sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}}\right)^2 + \sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}} + 1\right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{\left(1 + \frac{1}{x^2}\right)}{\left(\left(\sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}}\right)^2 + \sqrt[3]{1 + \frac{1}{x} + \frac{1}{x^3}} + 1\right)}
 \end{aligned}$$

donc :

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + x^2 + 1} - x = \frac{1}{3}$$

(11) Calculons $\lim_{x \rightarrow +\infty} \frac{\sqrt{x} - \sqrt[3]{x}}{\sqrt{x} + \sqrt[5]{x}}$:

On a :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x} - \sqrt[3]{x}}{\sqrt{x} + \sqrt[5]{x}} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \left(1 - \frac{\sqrt[3]{x}}{\sqrt{x}}\right)}{\sqrt{x} \left(1 + \frac{\sqrt[5]{x}}{\sqrt{x}}\right)}$$

$$= \lim_{x \rightarrow +\infty} \frac{1 - \frac{x^{\frac{1}{3}}}{x^{\frac{1}{2}}}}{1 + \frac{x^{\frac{1}{5}}}{x^{\frac{1}{2}}}}$$

$$= \lim_{x \rightarrow +\infty} \frac{1 - \frac{1}{x^{\frac{1}{6}}}}{1 + \frac{1}{x^{\frac{1}{10}}}}$$

$$= \lim_{x \rightarrow +\infty} \frac{1 - \frac{1}{x^{\frac{1}{6}}}}{1 + \frac{1}{x^{\frac{1}{10}}}}$$

done:

$$\lim_{x \rightarrow (\infty)} \frac{\sqrt{x} - \sqrt[3]{x}}{\sqrt{x} + \sqrt[5]{x}} = 1$$

car :

$$\lim_{x \rightarrow +\infty} x^{\frac{1}{x}} = +\infty \left| \lim_{x \rightarrow +\infty} x^{\frac{1}{x}} = +\infty \text{ et } \lim_{x \rightarrow +\infty} \frac{1}{x} = 0 \right.$$

(12) Calculons $\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 - x} - \sqrt[3]{x^3 + 2x}}{x}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 - x} - \sqrt[3]{x^3 + 2x}}{x} &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 \left(2 - \frac{1}{x^2}\right)} - \sqrt[3]{x^3 \left(1 + \frac{2}{x^2}\right)}}{x} \\ &= \lim_{x \rightarrow +\infty} \frac{x \sqrt[3]{\left(2 - \frac{1}{x^2}\right)} - x \sqrt[3]{\left(1 + \frac{2}{x^2}\right)}}{x} \\ &= \lim_{x \rightarrow +\infty} \frac{x \left(\sqrt[3]{\left(2 - \frac{1}{x^2}\right)} - \sqrt[3]{\left(1 + \frac{2}{x^2}\right)} \right)}{x} \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 - x} - \sqrt[3]{x^3 + 2x}}{x} = \lim_{x \rightarrow +\infty} \left(\sqrt[3]{\left(2 - \frac{1}{x^2}\right)} - \sqrt[3]{\left(1 + \frac{2}{x^2}\right)} \right)$$

done:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 - x} - \sqrt[3]{x^3 + 2x}}{x} = \sqrt[3]{2} - 1$$

(13) Calculons $\lim_{x \rightarrow 3} \frac{x^3 - 27}{\sqrt[3]{x} - \sqrt[3]{3}}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{x^3 - 27}{\sqrt[3]{x} - \sqrt[3]{3}} &= \lim_{x \rightarrow 3} \frac{x^3 - 3^3}{\sqrt[3]{x} - \sqrt[3]{3}} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x^2 + 3x + 9)(\sqrt[3]{x^2} + \sqrt[3]{x}\sqrt[3]{3} + \sqrt[3]{3^2})}{(\sqrt[3]{x} - \sqrt[3]{3})(\sqrt[3]{x^2} + \sqrt[3]{x}\sqrt[3]{3} + \sqrt[3]{3^2})} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x^2 + 3x + 9)(\sqrt[3]{x^2} + \sqrt[3]{x}\sqrt[3]{3} + \sqrt[3]{3^2})}{\sqrt{x^3 - 3^3}} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x^2 + 3x + 9)(\sqrt[3]{x^2} + \sqrt[3]{x}\sqrt[3]{3} + \sqrt[3]{3^2})}{x - 3} \\
 &= \lim_{x \rightarrow 3} (x^2 + 3x + 9)(\sqrt[3]{x^2} + \sqrt[3]{x}\sqrt[3]{3} + \sqrt[3]{3^2}) \\
 &= (9 + 9 + 9)(\sqrt[3]{9} + \sqrt[3]{9} + \sqrt[3]{9})
 \end{aligned}$$

done:

$$\lim_{x \rightarrow 3} \frac{x^3 - 27}{\sqrt[3]{x} - \sqrt[3]{3}} = 81\sqrt[3]{9}$$

(14) Calculons $\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3 + 5}}{\sqrt[3]{x^3 + 2}}$:

On a :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3 + 5}}{\sqrt[3]{x^3 + 2}} = \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 \left(8 + \frac{5}{x^3}\right)}}{\sqrt[3]{x^3 \left(1 + \frac{2}{x^3}\right)}}$$

done:

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3 + 5}}{\sqrt[3]{x^3 + 2}} &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3} \sqrt[3]{8 + \frac{5}{x^3}}}{\sqrt[3]{x^3} \sqrt[3]{1 + \frac{2}{x^3}}} \\
 &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8 + \frac{5}{x^3}}}{\sqrt[3]{1 + \frac{2}{x^3}}} \\
 &= \frac{\sqrt[3]{8 + 0}}{\sqrt[3]{1 + 0}} \\
 &= \frac{\sqrt[3]{x^3}}{1}
 \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3 + 5}}{\sqrt[3]{x^3 + 2}} = 2$$

(15) Calculons $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$:

Posons $t = x - \pi$ d'où: $x = t + \pi$ et $x \rightarrow \pi \Rightarrow t \rightarrow 0$ done:

$$\begin{aligned}
 \lim_{t \rightarrow 0} \frac{\sin x}{x - \pi} &= \lim_{t \rightarrow 0} \frac{\sin(t + \pi)}{t} \\
 &= \lim_{t \rightarrow 0} \frac{-\sin(t)}{t}
 \end{aligned}$$

done:

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi} = -1$$

(16) Calculons $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 3} + 2x - 1)$:

Ona:

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 3} + 2x - 1) &= \lim_{x \rightarrow -\infty} \left(\sqrt{x^2 \left(1 + \frac{4}{x} + \frac{3}{x^2} \right)} + 2x - 1 \right) \\
 &= \lim_{x \rightarrow -\infty} \left(\sqrt{x^2} \times \sqrt{1 + \frac{4}{x} + \frac{3}{x^2}} + 2x - 1 \right)
 \end{aligned}$$

done:

$$\begin{aligned}\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 3} + 2x - 1) &= \lim_{x \rightarrow -\infty} \left(-x \times \sqrt{1 + \frac{4}{x} + \frac{3}{x^2}} + 2x - 1 \right) \\ &= \lim_{x \rightarrow -\infty} x \left(-\sqrt{1 + \frac{4}{x} + \frac{3}{x^2}} + 2 - \frac{1}{x} \right)\end{aligned}$$

done:

$$\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 3} + 2x - 1) = -\infty$$

(17) Calculons $\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 8} - x)$:

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 8} - x) &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{x^3 + 8} - x) \left((\sqrt[3]{x^3 + 8})^2 + \sqrt[3]{x^3 + 8} \times x + x^2 \right)}{\left((\sqrt[3]{x^3 + 8})^2 + \sqrt[3]{x^3 + 8} \times x + x^2 \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{x^3 + 8})^3 - x^3}{\left((\sqrt[3]{x^3 + 8})^2 + \sqrt[3]{x^3 + 8} \times x + x^2 \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{x^3 + 8 - x^3}{\left((\sqrt[3]{x^3 + 8})^2 + \sqrt[3]{x^3 + 8} \times x + x^2 \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{8}{\left((\sqrt[3]{x^3 + 8})^2 + \sqrt[3]{x^3 + 8} \times x + x^2 \right)}\end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 8} - x) = 0$$

(18) Calculons $\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6}-2}{x^2-2x}$:

Ons :

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6}-2}{x^2-2x} &= \lim_{x \rightarrow 2} \frac{(\sqrt[3]{x+6}-2)((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 2^2)}{x(x-2)((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 2^2)} \\ &= \lim_{x \rightarrow 2} \frac{(\sqrt[3]{x+6})^3 - 2^3}{x(x-2)((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 4)}\end{aligned}$$

done:

$$\begin{aligned}
\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - 2}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{x+6-8}{x(x-2)((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 4)} \\
&= \lim_{x \rightarrow 2} \frac{x-2}{x(x-2)((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 4)} \\
&= \lim_{x \rightarrow 2} \frac{1}{x((\sqrt[3]{x+6})^2 + 2 \times \sqrt[3]{x+6} + 4)} \\
&= \frac{1}{2(4+4+4)}
\end{aligned}$$

done:

$$\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - 2}{x^2 - 2x} = \frac{1}{24}$$

(19) Calculons $\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3+x+1} - \sqrt[4]{x^4+1}}{2x}$:

On a :

$$\begin{aligned}
\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3+x+1} - \sqrt[4]{x^4+1}}{2x} &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 \left(1 + \frac{1}{x^2} + \frac{1}{x^3}\right)} - \sqrt[4]{x^4 \left(1 + \frac{1}{x^4}\right)}}{2x} \\
&= \lim_{x \rightarrow +\infty} \frac{x \times \sqrt[3]{1 + \frac{1}{x^2} + \frac{1}{x^3}} - x \times \sqrt[4]{1 + \frac{1}{x^4}}}{2x} \\
&= \lim_{x \rightarrow +\infty} \frac{x \left(\sqrt[3]{1 + \frac{1}{x^2} + \frac{1}{x^3}} - \sqrt[4]{1 + \frac{1}{x^4}} \right)}{2x} \\
&= \lim_{x \rightarrow +\infty} \frac{\left(\sqrt[3]{1 + \frac{1}{x^2} + \frac{1}{x^3}} - \sqrt[4]{1 + \frac{1}{x^4}} \right)}{2} \\
&= \frac{(\sqrt[3]{1+0+0} - \sqrt[4]{1+0})}{2}
\end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3+x+1} - \sqrt[4]{x^4+1}}{2x} = 0$$

(20) Calculons $\lim_{x \rightarrow +\infty} (x - \sqrt[3]{5x^2 + x + 1})$:

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} (x - \sqrt[3]{5x^2 + x + 1}) &= \lim_{x \rightarrow +\infty} \left(x - \sqrt[3]{x^3 \left(\frac{5}{x} + \frac{1}{x^2} + \frac{1}{x^3} \right)} \right) \\ &= \lim_{x \rightarrow +\infty} \left(x - x \times \sqrt[3]{\frac{5}{x} + \frac{1}{x^2} + \frac{1}{x^3}} \right) \\ &= \lim_{x \rightarrow +\infty} x \left(1 - \sqrt[3]{\frac{5}{x} + \frac{1}{x^2} + \frac{1}{x^3}} \right)\end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} (x - \sqrt[3]{5x^2 + x + 1}) = +\infty$$

(21) Calculons $\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{5x^2 - 7x - 6}$

$x^3 - 3x^2 + 4$	$x - 2$
$x^3 - 2x^2$	
$-x^2 + 4$	$x^2 - x - 2$
$-x^2 + 2x$	
$-2x + 4$	
$-2x + 4$	

done:

$$x^3 - 3x^2 + 4 = (x - 2)(x^2 - x - 2)$$

on a:

$5x^2 - 7x - 6$	$x - 2$
$-5x^2 - 10x$	$5x + 3$
$3x - 6$	
$-3x - 6$	
$0 \quad 0$	

done:

$$5x^2 - 7x - 6 = (x - 2)(5x + 3)$$

done:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{5x^2 - 7x - 6} &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 - x - 2)}{(x-2)(5x+3)} \\ &= \lim_{x \rightarrow 2} \frac{(x^2 - x - 2)}{(5x+3)}\end{aligned}$$

done:

$$\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{5x^2 - 7x - 6} = \frac{4 - 2 - 2}{5 \times 2 + 3}$$

done:

$$\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{5x^2 - 7x - 6} = 0$$

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(22) Calculons $\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+2}} - x \right) :$

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+2}} - x \right) &= \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{\frac{x^3}{x+2}} - x \right) \left(\sqrt{\frac{x^3}{x+2}} + x \right)}{\left(\sqrt{\frac{x^3}{x+2}} + x \right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{\frac{x^3}{x+2} - x^2}{\sqrt{\frac{x^3}{x+2}} + x} \\
 &= \lim_{x \rightarrow +\infty} \frac{\frac{x^3 - x^2(x+2)}{x+2}}{\left(\sqrt{\frac{x^3}{x(1+\frac{2}{x})}} + x \right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^3 - x^3 - 2x^2}{(x+2) \left(\sqrt{\frac{x^2}{1+\frac{2}{x}}} + x \right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{-2x^2}{x \left(1 + \frac{2}{x} \right) \left(x \times \sqrt{\frac{1}{1+\frac{2}{x}}} + x \right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{-2x^2}{x(\sqrt{x})}
 \end{aligned}$$

done:

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+2}} - x \right) &= \lim_{x \rightarrow +\infty} \frac{-2x^2}{x^2 \left(1 + \frac{2}{x} \right) \left(\sqrt{\frac{1}{1+\frac{2}{x}}} + 1 \right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{-2}{\left(1 + \frac{2}{x} \right) \left(\sqrt{\frac{1}{1+\frac{2}{x}}} + 1 \right)} \\
 &= \frac{-2}{(1+0) \left(\sqrt{\frac{1}{1+0}} + 1 \right)}
 \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+2}} - x \right) = -1$$

23 Calculons $\lim_{x \rightarrow +\infty} x^2 \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}} \right)$:

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} x^2 \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}} \right) &= \lim_{x \rightarrow +\infty} x^2 \times \frac{\left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}} \right) \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)}{\left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)} \\ &= \lim_{x \rightarrow +\infty} x^2 \times \frac{\left(\frac{1}{x} - \frac{1}{x+1} \right)}{\left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)} \\ &= \lim_{x \rightarrow +\infty} x^2 \times \frac{(x+1-x)}{x(x+1) \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)} \\ &= \lim_{x \rightarrow +\infty} x^2 \times \frac{1}{x^2 \left(1 + \frac{1}{x} \right) \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{\left(1 + \frac{1}{x} \right) \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right)} \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} x^2 \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}} \right) = +\infty$$

car :

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0^+ \text{ et } \lim_{x \rightarrow +\infty} \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+1}} \right) = 0^+$$

(24) Calculons $\lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(8x^3 + x^2)^2}}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(8x^3 + x^2)^2}} &= \lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{\left(x^3 \left(8 + \frac{1}{x}\right)\right)^2}} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(x^3)^2 \left(8 + \frac{1}{x}\right)^2}} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(x^3)^2 \times \sqrt[3]{\left(8 + \frac{1}{x}\right)^2}}} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2}{x^2 \times \sqrt[3]{\left(8 + \frac{1}{x}\right)^2}} \\
 &= \frac{1}{\sqrt[3]{\left(8 + \frac{1}{x}\right)^2}} \\
 &= \frac{1}{\sqrt[3]{(2^3)^2}} \\
 &= \frac{1}{\sqrt[3]{2^6}} \\
 &= \frac{1}{2^2}
 \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} \frac{x^2}{\sqrt[3]{(8x^3 + x^2)^2}} = \frac{1}{4}$$

(25) Calculons $\lim_{x \rightarrow +\infty} (\sqrt[3]{64x^3 + x^2} - 4x)$

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} (\sqrt[3]{64x^3 + x^2} - 4x) &= \lim_{x \rightarrow +\infty} \frac{(\sqrt[3]{64x^3 + x^2})^3 - (4x)^3}{(\sqrt[3]{64x^3 + x^2} + 4x\sqrt[3]{64x^3 + x^2} + (4x)^2)} \\
 &= \lim_{x \rightarrow +\infty} \frac{64x^3 + x^2 - 64x^3}{(\sqrt[3]{64x^3 + x^2} + 4x\sqrt[3]{64x^3 + x^2} + (4x)^2)} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2}{\left(\sqrt[3]{x^3 \left(64 + \frac{1}{x}\right)^2} + 4x\sqrt[3]{x^3 \left(64 + \frac{1}{x}\right)} + 16x^2\right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{x^2}{x^2 \left(\sqrt[3]{\left(64 + \frac{1}{x}\right)^2} + 4\sqrt[3]{\left(64 + \frac{1}{x}\right)} + 16\right)} \\
 &= \lim_{x \rightarrow +\infty} \frac{1}{\left(\sqrt[3]{\left(64 + \frac{1}{x}\right)^2} + 4\sqrt[3]{\left(64 + \frac{1}{x}\right)} + 16\right)} \\
 &= \frac{1}{\sqrt[3]{64^2} + 4\sqrt[3]{64+16}} \\
 &= \frac{1}{\sqrt[3]{2^{12}} + 4\sqrt[3]{2^6} + 16} \\
 &= \frac{1}{2^4 + 4 \times 2^2 + 16}
 \end{aligned}$$

done:

$$\lim_{x \rightarrow +\infty} (\sqrt[3]{64x^3 + x^2} - 4x) = \frac{1}{48}$$

(26) Calculons $\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 1} + 2x - 3)$:

On a:

$$\begin{aligned} \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 1} + 2x - 3) &= \lim_{x \rightarrow -\infty} \left(\frac{(\sqrt{4x^2 + x + 1} + 2x)(\sqrt{4x^2 + x + 1} - 2x)}{(\sqrt{4x^2 + x + 1} - 2x)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{4x^2 + x + 1}^2 - (2x)^2}{(\sqrt{4x^2 + x + 1} - 2x)} - 3 \right) \end{aligned}$$

done:

$$\begin{aligned} \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 1} + 2x - 3) &= \lim_{x \rightarrow -\infty} \left(\frac{4x^2 + x + 1 - 4x^2}{(\sqrt{4x^2 + x + 1} - 2x)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{x + 1}{(\sqrt{4x^2 + x + 1} - 2x)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{x + 1}{\left(\sqrt{x^2 \left(4 + \frac{1}{x} + \frac{1}{x^2} \right)} - 2x \right)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{x \left(1 + \frac{1}{x} \right)}{\left(-x \sqrt{\left(4 + \frac{1}{x} + \frac{1}{x^2} \right)} - 2x \right)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{x \left(1 + \frac{1}{x} \right)}{x \left(-\sqrt{\left(4 + \frac{1}{x} + \frac{1}{x^2} \right)} - 2 \right)} - 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{\left(1 + \frac{1}{x} \right)}{\left(-\sqrt{\left(4 + \frac{1}{x} + \frac{1}{x^2} \right)} - 2 \right)} - 3 \right) \\ &= \frac{1 + 0}{-\sqrt{4 + 0 + 0} - 2} - 3 \\ & \quad x^1) \end{aligned}$$

done:

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 1} + 2x - 3) = \frac{-13}{4}$$

(27) Calculons $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}$:

On a :

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} &= \lim_{x \rightarrow 0} \frac{\sin x \left(1 - \frac{\tan x}{\sin x}\right)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{\sin x \left(1 - \frac{1}{\cos x}\right)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{\sin x (\cos x - 1)}{x^3 \cos x}\end{aligned}$$

done :

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} &= \lim_{x \rightarrow 0} -\frac{\sin x}{x} \times \frac{(1 - \cos x)}{x^2} \times \frac{1}{\cos x} \\ &= -1 \times \frac{1}{2} \times \frac{1}{\cos 0}\end{aligned}$$

done :

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} = \frac{-1}{2}$$

(28) Calculons $\lim_{x \rightarrow -1} \frac{\sqrt[3]{7x^2 - x} + \sqrt{x^2 - 3x - 4}}{x^2 - 1}$:

On a :

$$\begin{aligned}\lim_{x \rightarrow -1} \frac{\sqrt[3]{7x^2 - x} + \sqrt{x^2 - 3x - 4}}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{\sqrt[3]{7x^2 - x} - 2 + \sqrt{x^2 - 3x} - 2}{x^2 - 1} \\ &= \lim_{x \rightarrow -1} \frac{\frac{\sqrt[3]{7x^2 - x} - x^3}{\sqrt[3]{7x^2 - x^2} + 2\sqrt[3]{7x^2 - x} + 2^2} + \frac{\sqrt{x^2 - 3x} - 2^2}{\sqrt{x^2 - 3x} + 2}}{(x - 1)(x + 1)} \\ &= \lim_{x \rightarrow -1} \frac{\frac{7x^2 - x - 8}{\sqrt[3]{7x^2 - x^2} + 2\sqrt[3]{7x^2 - x} + 4} + \frac{x^2 - 3x - 4}{\sqrt{x^2 - 3x} + 2}}{(x - 1)(x + 1)} \\ &= \lim_{x \rightarrow -1} \frac{\frac{7(x + 1)\left(x - \frac{8}{7}\right)}{\sqrt[3]{7x^2 - x^2} + 2\sqrt[3]{7x^2 - x} + 4} + \frac{(x + 1)(x - 4)}{\sqrt{x^2 - 3x} + 2}}{(x - 1)(x + 1)} \\ &= \lim_{x \rightarrow -1} \frac{\frac{7}{\sqrt[3]{7x^2 - x}}^2 + 2\sqrt[3]{7x^2 - x} + 4}{(x - 1)} + \frac{(x - 4)}{\sqrt{x^2 - 3x} + 2} \\ &= \frac{-15}{4 + 4 + 4} + \frac{-5}{4}\end{aligned}$$

done:

$$\lim_{x \rightarrow -1} \frac{\sqrt[3]{7x^2 - x} + \sqrt{x^2 - 3x} - 4}{x^2 - 1} = \frac{5}{4}$$

(29) Calculons $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt{x+3}-2}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt{x+3}-2} &= \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x+7}-2)(\sqrt[3]{x+7}^2 + 2\sqrt[3]{x+7} + 2)(\sqrt{x+3}+2)}{(\sqrt{x+3}-2)(\sqrt{x+3}+2)(\sqrt[3]{x+7}^2 + 2\sqrt[3]{x+7} + 2)} \\ &= \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x+7}-2)(\sqrt{x+3}+2)}{(\sqrt{x+3}-2)(\sqrt[3]{x+7}^2 + 2\sqrt[3]{x+7} + 2)} \\ &= \lim_{x \rightarrow 1} \frac{(x+7-8)(\sqrt{x+3}+2)}{(x+3-4)(\sqrt{x+7})^2 + 2\sqrt{x+7}+4)} \\ &= \lim_{x \rightarrow 1} \frac{(x-1)(\sqrt{x+3}+2)}{(x-1)(\sqrt[3]{x+7}^2 + 2\sqrt[3]{x+7} + 2)} \\ &= \lim_{x \rightarrow 1} \frac{(\sqrt{x+3}+2)}{(\sqrt[3]{x+7})^2 + 2\sqrt[3]{x+7} + 2)} \\ &= \frac{2+2}{4+4+4} \end{aligned}$$

done:

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x+7}-2}{\sqrt{x+3}-2} = \frac{1}{3}$$

(30) Calculons $\lim_{x \rightarrow -\infty} \sqrt[4]{\frac{x^3 + 5x^2 - 7}{16x^3 - 3x + 1}} =$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \sqrt[4]{\frac{x^3 + 5x^2 - 7}{16x^3 - 3x + 1}} &= \lim_{x \rightarrow -\infty} \sqrt[4]{\frac{x^3 \left(1 + \frac{5}{x} - \frac{7}{x^3}\right)}{x^3 \left(16 - \frac{3}{x^2} + \frac{1}{x^3}\right)}} \\ &= \lim_{x \rightarrow -\infty} \sqrt[4]{\frac{\left(1 + \frac{5}{x} - \frac{7}{x^3}\right)}{\left(16 - \frac{3}{x^2} + \frac{1}{x^3}\right)}} \\ &= \sqrt[4]{\frac{(1 + 0 - 0)}{(16 - 0 + 0)}} \\ &= \sqrt[4]{\frac{1}{2^4}} \end{aligned}$$

done :

$$\lim_{x \rightarrow -\infty} \sqrt[4]{\frac{x^3 + 5x^2 - 7}{16x^3 - 3x + 1}} = \frac{1}{2}$$

(31) Calculons $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4 + x^2 + 2}}{x + 2} =$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4 + x^2 + 2}}{x + 2} &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4 \left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)}}{x \left(1 + \frac{2}{x}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4 \left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)}}{x \left(1 + \frac{2}{x}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4} \times \sqrt[3]{\left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)}}{x \left(1 + \frac{2}{x}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{(-x)^4} \times \sqrt[3]{\left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)}}{-\sqrt[3]{(-x)^3} \left(1 + \frac{2}{x}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt[3]{(-x)} \times \sqrt[3]{\left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)}}{\left(1 + \frac{2}{x}\right)} \end{aligned}$$

done:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^4 + x^2 + 2}}{x + 2} = -\infty$$

(32) Calculons $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{\sqrt[3]{x^2 + 1} - 1} :$

Ona:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{\sqrt[3]{x^2 + 1} - 1} &= \lim_{x \rightarrow 0} \frac{(1 - \cos(2x)) \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right)}{(\sqrt[3]{x^2 + 1} - 1) \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right)} \\ &= \lim_{x \rightarrow 0} \frac{(1 - \cos(2x)) \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right)}{(\sqrt[3]{x^2 + 1})^3 - 1^3} \end{aligned}$$

donc :

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{\sqrt[3]{x^2 + 1} - 1} &= \lim_{x \rightarrow 0} \frac{(1 - \cos(2x)) \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right)}{x^2 + 1 - 1} \\
 &= \lim_{x \rightarrow 0} \frac{(1 - \cos(2x)) \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right)}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(1 - \cos(2x))}{x^2} \times \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right) \\
 &= \lim_{x \rightarrow 0} 4 \times \frac{(1 - \cos(2x))}{(2x)^2} \times \left((\sqrt[3]{x^2 + 1})^2 + \sqrt[3]{x^2 + 1} + 1 \right) \\
 &= 4 \times \frac{1}{2} \times (1 + 1 + 1)
 \end{aligned}$$

donc :

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{\sqrt[3]{x^2 + 1} - 1} = 6$$

(33) Calculons $\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 5x^2 + 3} - 2x + 1)$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 5x^2 + 3} - 2x + 1) &= \lim_{x \rightarrow +\infty} \left(\sqrt[3]{x^3 \left(1 + \frac{5}{x} + \frac{3}{x^3} \right)} - 2x + 1 \right) \\
 &= \lim_{x \rightarrow +\infty} \left(\sqrt[3]{x^3} \times \sqrt[3]{1 + \frac{5}{x} + \frac{3}{x^3}} - 2x + 1 \right) \\
 &= \lim_{x \rightarrow +\infty} \left(x \times \sqrt[3]{1 + \frac{5}{x} + \frac{3}{x^3}} - 2x + 1 \right) \\
 &= \lim_{x \rightarrow +\infty} \left(x \times \left(\sqrt[3]{1 + \frac{5}{x} + \frac{3}{x^3}} - 2 \right) + 1 \right)
 \end{aligned}$$

donc:

$$\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 5x^2 + 3} - 2x + 1) = -\infty$$

(34) Calculons $\lim_{x \rightarrow 5} \frac{\sqrt[3]{4x+7}-3}{\sqrt{2x^2-1}-7}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 5} \frac{\sqrt[3]{4x+7}-3}{\sqrt{2x^2-1}-7} &= \lim_{x \rightarrow 5} \frac{(\sqrt[3]{4x+7}-3)(\sqrt[3]{4x+7}+3\sqrt[3]{4x+7}+3^2)(\sqrt{2x^2-1}+7)}{(\sqrt{2x^2-1}-7)(\sqrt{2x^2-1}+7)(\sqrt[3]{4x+7}+3\sqrt[3]{4x+7}+3^2)} \\
 &= \lim_{x \rightarrow 5} \frac{(\sqrt[3]{4x+7}-3^3)(\sqrt{2x^2-1}+7)}{(\sqrt{2x^2-1}^2-7^2)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \lim_{x \rightarrow 5} \frac{(4x+7-27)(\sqrt{2x^2-1}+7)}{(2x^2-1-49)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \lim_{x \rightarrow 5} \frac{(4x-20)(\sqrt{2x^2-1}+7)}{(2x^2-50)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \lim_{x \rightarrow 5} \frac{4(x-5)(\sqrt{2x^2-1}+7)}{2(x^2-25)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \lim_{x \rightarrow 5} \frac{4(x-5)(\sqrt{2x^2-1}+7)}{2(x-5)(x+5)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \lim_{x \rightarrow 5} \frac{4(\sqrt{2x^2-1}+7)}{2(x+5)(\sqrt[3]{4x+7^2}+3\sqrt[3]{4x+7}+9)} \\
 &= \frac{4 \times (7+7)}{2(5+5)(9+9+9)}
 \end{aligned}$$

donc:

$$\lim_{x \rightarrow 5} \frac{\sqrt[3]{4x+7}-3}{\sqrt{2x^2-1}-7} = \frac{14}{135}$$

(35) Calculons $\lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt[3]{x^2+4}+5}$:

On a :

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt[3]{x^2+4}+5} &= \lim_{x \rightarrow +\infty} \frac{x\left(2+\frac{3}{x}\right)}{\sqrt[3]{x^2\left(1+\frac{4}{x^2}\right)}+5} \\
 &= \lim_{x \rightarrow +\infty} \frac{x\left(2+\frac{3}{x}\right)}{\sqrt[3]{x^2} \times \sqrt[3]{\left(1+\frac{4}{x^2}\right)}+5}
 \end{aligned}$$

donc:

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt[3]{x^2+4}+5} &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3} \left(2 + \frac{3}{x}\right)}{\sqrt[3]{x^2} \times \left(\sqrt[3]{1 + \frac{4}{x^2}} + \frac{5}{\sqrt[3]{x^2}}\right)} \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x} \left(2 + \frac{3}{x}\right)}{\left(\sqrt[3]{1 + \frac{4}{x^2}} + \frac{5}{\sqrt[3]{x^2}}\right)}\end{aligned}$$

donc:

$$\lim_{x \rightarrow +\infty} \frac{2x+3}{\sqrt[3]{x^2+4}+5} = +\infty$$

(36) Calculons $\lim_{x \rightarrow -\infty} \frac{4x-1}{\sqrt{x^2+3x+5}+2x}$:

On a :

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{4x-1}{\sqrt{x^2+3x+5}+2x} &= \lim_{x \rightarrow -\infty} \frac{x \left(4 - \frac{1}{x}\right)}{\sqrt{x^2 \left(1 + \frac{3}{x} + \frac{5}{x^2}\right)} + 2x} \\ &= \lim_{x \rightarrow -\infty} \frac{x \left(4 - \frac{1}{x}\right)}{\sqrt{x^2} \times \sqrt{\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)} + 2x} \\ &= \lim_{x \rightarrow -\infty} \frac{x \left(4 - \frac{1}{x}\right)}{-x \times \sqrt{\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)} + 2x} \\ &= \lim_{x \rightarrow -\infty} \frac{x \left(4 - \frac{1}{x}\right)}{x \left(-\sqrt{\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)} + 2\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{\left(4 - \frac{1}{x}\right)}{\left(-\sqrt{\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)} + 2\right)} \\ &= \frac{4-0}{-1+2}\end{aligned}$$

donc:

$$\lim_{x \rightarrow -\infty} \frac{4x - 1}{\sqrt{x^2 + 3x + 5} + 2x} = 4$$

(37) Calculons $\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - \sqrt{1-x}}{x}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - \sqrt{1-x}}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - 1 - \sqrt{1-x} + 1}{x} \\ &= \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - 1 - (\sqrt{1-x} - 1)}{x} \\ &= \lim_{x \rightarrow 0} \left(\frac{\sqrt[3]{x+1} - 1}{x} - \frac{\sqrt{1-x} - 1}{x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(\sqrt[3]{x+1})^3 - 1^3}{x(\sqrt[3]{x+1}^2 + \sqrt[3]{x+1} + 1)} - \frac{(\sqrt{1-x})^2 - 1^2}{x(\sqrt{1-x} + 1)} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{x+1-1}{x(\sqrt[3]{x+1}^2 + \sqrt[3]{x+1} + 1)} - \frac{1-x-1}{x(\sqrt{1-x} + 1)} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{x}{x(\sqrt[3]{x+1}^2 + \sqrt[3]{x+1} + 1)} - \frac{-x}{x(\sqrt{1-x} + 1)} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1}{(\sqrt[3]{x+1} + \sqrt[3]{x+1} + 1)} + \frac{1}{(\sqrt{1-x} + 1)} \right) \\ &= \frac{1}{1+1+1} + \frac{1}{1+1} \end{aligned}$$

donc:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - \sqrt{1-x}}{x} = \frac{5}{6}$$

(38) Calculons $\lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} + \sqrt{2x+2} - 4}{x-1}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} + \sqrt{2x+2} - 4}{x-1} &= \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} - 2 + \sqrt{2x+2} - 2}{x-1} \\ &= \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+3} - 2}{x-1} + \frac{\sqrt{2x+2} - 2}{x-1} \right) \end{aligned}$$

donc :

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} + \sqrt{2x+2} - 4}{x-1} &= \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+3}^2 - 2^2}{(x-1)(\sqrt{x^2+3} + 2)} + \frac{(\sqrt{2x+2})^2 - 2^2}{(x-1)(\sqrt{2x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x^2 + 3 - 4}{(x-1)(\sqrt{x^2+3} + 2)} + \frac{(2x+2) - 4}{(x-1)(\sqrt{2x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{(x-1)(\sqrt{x^2+3} + 2)} + \frac{(2x-2)}{(x-1)(\sqrt{2x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{(x-1)(x+1)}{(x-1)(\sqrt{x^2+3} + 2)} + \frac{2(x-1)}{(x-1)(\sqrt{2x+2} + 2)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{(x+1)}{(\sqrt{x^2+3} + 2)} + \frac{2}{(\sqrt{2x+2} + 2)} \right) \\ &= \frac{2}{2+2} + \frac{2}{2+2} \end{aligned}$$

donc :

$$\lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} + \sqrt{2x+2} - 4}{x-1} = 1$$

(39) Calculons $\lim_{x \rightarrow -\infty} \frac{x}{\sqrt[3]{2-x}}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{x}{\sqrt[3]{2-x}} &= \lim_{x \rightarrow -\infty} \frac{x}{\sqrt[3]{-x \left(\frac{-2}{x} + 1 \right)}} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt[3]{(-x)^3}}{\sqrt[3]{-x} \times \sqrt[3]{\left(\frac{-2}{x} + 1 \right)}} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt[3]{(-x)^2}}{\sqrt[3]{\left(\frac{-2}{x} + 1 \right)}} \end{aligned}$$

donc:

$$\lim_{x \rightarrow -\infty} \frac{x}{\sqrt[3]{2-x}} = -\infty$$

(40) Calculons $\lim_{x \rightarrow 0} \frac{1 - \cos(4x)}{\tan x \sin 3x}$:

On a :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos(4x)}{\tan x \sin 3x} &= \lim_{x \rightarrow 0} \frac{(4x)^2 \times \frac{(1 - \cos(4x))}{(4x)^2}}{x \times \frac{\tan x}{x} \times 3x \times \frac{\sin 3x}{3x}} \\ &= \lim_{x \rightarrow 0} \frac{16x^2 \times \frac{(1 - \cos(4x))}{(4x)^2}}{3x^2 \times \frac{\tan x}{x} \times \frac{\sin 3x}{3x}} \\ &= \lim_{x \rightarrow 0} \frac{16 \times \frac{(1 - \cos(4x))}{(4x)^2}}{3 \times \frac{\tan x}{x} \times \frac{\sin 3x}{3x}} \\ &= \frac{16 \times \frac{1}{2}}{3 \times 1 \times 1} \end{aligned}$$

donc:

$$\lim_{x \rightarrow 0} \frac{1 - \cos(4x)}{\tan x \sin 3x} = \frac{8}{3}$$

(41) Calculons $\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7)$

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7)$$

On a:

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7) = \lim_{x \rightarrow -\infty} 4x^5$$

Donc:

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7) = -\infty$$

Car:

$$\lim_{x \rightarrow -\infty} x^5 = -\infty \text{ et } 4 > 0$$

(42) Calculons $\lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3 &= \lim_{x \rightarrow +\infty} (-2x^4)^3 \\ &= \lim_{x \rightarrow +\infty} -8x^{12} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3 = -\infty$$

Car: $\lim_{x \rightarrow -\infty} x^{12} = +\infty$ et $-8 < 0$

(43) Calculons: $\lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5}$

On a:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5} &= \lim_{x \rightarrow -\infty} \frac{x}{x^2} \\ &= \lim_{x \rightarrow -\infty} \frac{1}{x} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5} = 0$$

(44) Calculons: $\lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3}$

On a :

Donc:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3} &= \lim_{x \rightarrow +\infty} \frac{-x^3}{x^3} \\ &= \lim_{x \rightarrow +\infty} -1 \end{aligned}$$

$$\lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3} = -1$$

(45) Calculons: $\lim_{x \rightarrow +\infty} \frac{x^2+5x}{(2x+3)^2}$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x^2+5x}{(2x+3)^2} &= \lim_{x \rightarrow +\infty} \frac{x^2}{4x^2} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{4} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{x^2+5x}{(2x+3)^2} = \frac{1}{4}$$

(46) Calculons: $\lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2}$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2} &= \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2} \\ &= \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{-2x^2-4x-1} \end{aligned}$$

Donc:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2} &= \lim_{x \rightarrow +\infty} \frac{2x^2}{-2x^2} \\ &= \lim_{x \rightarrow +\infty} -1 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2} = -1$$

(47) Calculons: $\lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2 + 3x - 5}{x^2 + x + 2}$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2 + 3x - 5}{x^2 + x + 2} &= \lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2}{x^2} \\ &= \lim_{x \rightarrow +\infty} (1-\sqrt{2}) \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2 + 3x - 5}{x^2 + x + 2} = (1-\sqrt{2})$$

(48) Calculons: $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{2}x^2}{x+1} - \sqrt{3}x \right)$

On a:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{2}x^2}{x+1} - \sqrt{3}x \right) &= \lim_{x \rightarrow -\infty} \frac{\sqrt{2}x^2 - \sqrt{3}x(x+1)}{x+1} \\ &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{2} - \sqrt{3})x^2 - \sqrt{3}x}{x+1} \\ &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{2} - \sqrt{3})x^2}{x} \\ &= \lim_{x \rightarrow -\infty} (\sqrt{2} - \sqrt{3})x \end{aligned}$$

$$\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{2}x^2}{x+1} - \sqrt{3}x \right) = +\infty$$

Car:

$$\lim_{x \rightarrow -\infty} x = -\infty \text{ et } (\sqrt{2} - \sqrt{3}) < 0$$

(49) Calculons: $\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8}$

On a:

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8} = \lim_{x \rightarrow +\infty} \sqrt{x^3 \left(1 - \frac{2}{x} + \frac{3}{x^2} - \frac{8}{x^3} \right)}$$

Donc:

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8} = +\infty$$

Car:

$$\left\{ \begin{array}{l} \lim_{x \rightarrow +\infty} \frac{2}{x} = \lim_{x \rightarrow +\infty} \frac{3}{x^2} = \lim_{x \rightarrow +\infty} \frac{8}{x^3} = 0 \\ \lim_{x \rightarrow +\infty} x^3 = +\infty \\ \lim_{x \rightarrow +\infty} \sqrt{x} = +\infty \end{array} \right.$$

(50) Calculons: $\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}}$

On a:

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{x^2 \left(2 - \frac{5}{x}\right)}{x^2 \left(8 + \frac{7}{x^2}\right)}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{\left(2 - \frac{5}{x}\right)}{\left(8 + \frac{7}{x^2}\right)}} \text{ Donc:}$$

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}} = \frac{1}{2}$$

Car:

$$\lim_{x \rightarrow -\infty} \frac{5}{x} = \lim_{x \rightarrow -\infty} \frac{7}{x^2} = 0$$

(51) Calculons : $\lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x-5}}$

On a:

$$\lim_{x \rightarrow +\infty} \frac{1}{2x-5} = 0^+$$

et

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

Donc:

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x-5}} = 0$$

(52) Calculons: $\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}}$

On a:

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{x^3 \left(5 - \frac{1}{x^3}\right)}{x^3 \left(2 + \frac{7}{x^3}\right)}}$$

Donc:

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{\left(5 - \frac{1}{x^3}\right)}{\left(2 + \frac{7}{x^3}\right)}}$$

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}} = \frac{\sqrt{10}}{2}$$

Car:

$$\lim_{x \rightarrow -\infty} \frac{1}{x^3} = \lim_{x \rightarrow -\infty} \frac{7}{x^3} = 0$$

(53) Calculons: $\lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}}$

On a:

$$\begin{aligned} \lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}} &= \lim_{x \rightarrow \sqrt{2}} \frac{x^2 - (\sqrt{2})^2}{x - \sqrt{2}} \\ &= \lim_{x \rightarrow \sqrt{2}} \frac{(x - \sqrt{2})(x + \sqrt{2})}{x - \sqrt{2}} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}} = \lim_{x \rightarrow \sqrt{2}} (x + \sqrt{2})$$

Donc:

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}} = 2\sqrt{2}$$

(54) Calculons: $\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2} &= \lim_{x \rightarrow 3} \frac{(x - 2)(x - 3)}{(3 - x)(3 + x)} \\ &= \lim_{x \rightarrow 3} \frac{-(x - 2)}{(3 + x)} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2} = -\frac{1}{6}$$

(55) Calculons: $\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5}$

On a:

$$\begin{aligned} \lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} &= \lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} \\ &= \lim_{x \rightarrow -5} \frac{(x + 1)(x + 5)}{(x - 1)(x + 5)} \\ &= \lim_{x \rightarrow -5} \frac{(x + 1)}{(x - 1)} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} = \frac{2}{3}$$

(56) Calculons: $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}$

On a:

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x(x - 2)}$$

Donc:

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{x(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x} \\ \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} &= 6 \end{aligned}$$

(57) Calculons: $\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x-x^2}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x-x^2} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+4}-2)(\sqrt{x+4}+2)}{x(1-x)(\sqrt{x+4}+2)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(1-x)(\sqrt{x+4}+2)} \\ &= \lim_{x \rightarrow 0} \frac{1}{(1-x)(\sqrt{x+4}+2)} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x-x^2} = \frac{1}{4}$$

(58) Calculons: $\lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{\sqrt{x}-2}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{\sqrt{x}-2} &= \lim_{x \rightarrow 4} \frac{(\sqrt{2x+1}-3)(\sqrt{2x+1}+3)(\sqrt{x}+2)}{(\sqrt{x}-2)(\sqrt{x}+2)(\sqrt{2x+1}+3)} \\ &= \lim_{x \rightarrow 4} \frac{(2x+1-9)(\sqrt{x}+2)}{(x-4)(\sqrt{2x+1}+3)} \\ &= \lim_{x \rightarrow 4} \frac{2(x-4)(\sqrt{x}+2)}{(x-4)(\sqrt{2x+1}+3)} \\ &= \lim_{x \rightarrow 4} \frac{2(\sqrt{x}+2)}{\sqrt{2x+1}+3} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{\sqrt{x}-2} = \frac{4}{3}$$

(59) Calculons: $\lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2}$

On a:

$$\begin{array}{r|l}
 2x^3 + 3x^2 + 4x + 3 & x + 1 \\
 -2x^3 + 2x^2 & \\
 \hline
 -x^2 + 4x & \\
 x^2 + x & 2x^2 + x + 3 \\
 \hline
 3x + 3 & \\
 -3x + 3 & \\
 \hline
 0 &
 \end{array}$$

Donc:

$$\begin{aligned}
 \lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2} &= \lim_{x \rightarrow -1} \frac{(2x^2 + x + 3)(x + 1)}{(x + 1)(x + 2)} \\
 &= \lim_{x \rightarrow -1} \frac{2x^2 + x + 3}{x + 2}
 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2} = 4$$

(60) Calculons: $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3}$

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3} &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)(\sqrt{x + 6} + 3)}{(\sqrt{x + 6} - 3)(\sqrt{x + 6} + 3)} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)(\sqrt{x + 6} + 3)}{x - 3} \\
 &= \lim_{x \rightarrow 3} (x + 2)(\sqrt{x + 6} + 3)
 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3} = 30$$

(61) Calculons: $\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}} &= \lim_{x \rightarrow 3} \frac{(\sqrt{2x+3} - \sqrt{x+6})(\sqrt{2x+3} + \sqrt{x+6})(\sqrt{x-1} + \sqrt{8-2x})}{(\sqrt{x-1} - \sqrt{8-2x})(\sqrt{x-1} + \sqrt{8-2x})(\sqrt{2x+3} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(\sqrt{x-1} + \sqrt{8-2x})}{3(x-3)(\sqrt{2x+3} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{\sqrt{x-1} + \sqrt{8-2x}}{3(\sqrt{2x+3} + \sqrt{x+6})} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}} = \frac{\sqrt{2}}{9}$$

(62) Calculons: $\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2} &= \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - (\sqrt{x} + 1)}{\sqrt{x+12} - (\sqrt{x} + 2)} \\ &= \lim_{x \rightarrow 4} \frac{[\sqrt{x+5} - (\sqrt{x} + 1)][\sqrt{x+5} + (\sqrt{x} + 1)][\sqrt{x+12} + (\sqrt{x} + 2)]}{[\sqrt{x+12} - (\sqrt{x} + 2)][\sqrt{x+12} + (\sqrt{x} + 2)][\sqrt{x+5} + (\sqrt{x} + 1)]} \\ &= \lim_{x \rightarrow 4} \frac{-2[\sqrt{x} - 2][\sqrt{x+12} + (\sqrt{x} + 2)]}{-4[\sqrt{x} - 2][\sqrt{x+5} + (\sqrt{x} + 1)]} \\ &= \lim_{x \rightarrow 4} \frac{\sqrt{x+12} + (\sqrt{x} + 2)}{2[\sqrt{x+5} + (\sqrt{x} + 1)]} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2} = \frac{2}{3}$$

(63) Calculons: $\lim_{x \rightarrow 3} \frac{7-5x}{(x-3)^2}$

On a:

$$\lim_{x \rightarrow 3} 7 - 5x = -8$$

et

$$\lim_{x \rightarrow 3} (x-3)^2 = 0^+$$

Donc:

$$\lim_{x \rightarrow 3} \frac{7 - 5x}{(x - 3)^2} = -\infty$$

Car:

$$\lim_{x \rightarrow 0^+} \frac{1}{X} = +\infty \text{ et } -8 < 0$$

(64) Calculons: $\lim_{x \rightarrow -1^-} \frac{x+1}{\sqrt{x^2-1}}$

On a:

$$\begin{aligned} \lim_{x \rightarrow -1^-} \frac{x+1}{\sqrt{x^2-1}} &= \lim_{x \rightarrow -1^-} \frac{-\sqrt{(x+1)^2}}{\sqrt{(x-1)(x+1)}} \\ &= \lim_{x \rightarrow -1^-} -\sqrt{\frac{x+1}{x-1}} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -1^-} \frac{x+1}{\sqrt{x^2-1}} = 0$$

Car:

$$\begin{cases} \lim_{x \rightarrow -1^-} x+1 = 0^- \\ \lim_{x \rightarrow -1^-} x-1 = -2 \\ \lim_{x \rightarrow 0^+} \sqrt{X} = 0 \end{cases}$$

(65) Calculons: $\lim_{x \rightarrow 1^-} \frac{x^2-x-3}{x^2+2x-3}$

Le discriminant du trinôme $x^2 + 2x - 3$ est : $\Delta = 2^2 - 4 \times 1 \times (-3) = 16$. Donc le trinôme $x^2 + 2x - 3$ admet deux racines distinctes qui sont : $x_1 = -3$ et $x_2 = 1$.
Le tableau de signe du trinôme $x^2 + 2x - 3$ est:

x	$-\infty$	-3		1	$+\infty$
$x^2 + 2x - 3$	+	0	-	0	+

D'où :

$$\lim_{x \rightarrow 1^-} x^2 + 2x - 3 = 0^-$$

On a:

$$\lim_{x \rightarrow 1^-} x^2 - x - 3 = -3$$

Donc:

$$\lim_{x \rightarrow 1^-} \frac{x^2 - x - 3}{x^2 + 2x - 3} = +\infty \quad \text{Car} \quad \lim_{x \rightarrow 0^-} \frac{1}{X} = -\infty \quad \text{et} \quad -3 < 0$$

(66) Calculons: $\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1}$

On a:

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1} = \lim_{x \rightarrow 1^-} \frac{\sqrt{(x-1)(x-2)}}{-\sqrt{(x-1)^2}}$$

Donc:

$$= \lim_{x \rightarrow 1^-} -\sqrt{\frac{x-2}{x-1}}$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1} = -\infty$$

Car:

$$\begin{cases} \lim_{x \rightarrow 1^-} x - 2 = -1 \\ \lim_{x \rightarrow 1^-} x - 1 = 0^- \\ \lim_{x \rightarrow 0^-} \frac{-1}{x} = +\infty \end{cases}$$

(67) Calculons: $\lim_{x \rightarrow 1^-} \frac{8x-1}{\sqrt{1-x}}$

On a:

$$\lim_{x \rightarrow 1^-} 8x - 1 = 7$$

et

$$\lim_{x \rightarrow 1^-} \sqrt{1-x} = 0^+$$

Donc:

$$\lim_{x \rightarrow 1^-} \frac{8x-1}{\sqrt{1-x}} = +\infty$$

Car:

$$\lim_{x \rightarrow 0^+} \frac{7}{X} = +\infty$$

(68) Calculons: $\lim_{x \rightarrow 4^+} \frac{x}{2 - \sqrt{x}}$

On a:

$$\lim_{x \rightarrow 4^+} x = 4$$

et

$$\lim_{x \rightarrow 4^+} 2 - \sqrt{x} = 0^-$$

Donc:

$$\lim_{x \rightarrow 4^+} \frac{x}{2 - \sqrt{x}} = -\infty$$

Car:

$$\lim_{x \rightarrow 0^-} \frac{4}{X} = -\infty$$

(69) Calculons: $\lim_{x \rightarrow 4^+} \frac{\sqrt{x+12} - \sqrt{3x-4}}{x^2 - 8x + 16}$

On a:

$$\lim_{x \rightarrow 4^+} \sqrt{x+12} - \sqrt{3x-4} = 4 - 2\sqrt{2}$$

et

$$\lim_{x \rightarrow 4^+} x^2 - 8x + 16 = \lim_{x \rightarrow 4^+} (x-4)^2 = 0^+$$

Donc:

$$\lim_{x \rightarrow 4^+} \frac{\sqrt{x+12} - \sqrt{3x-4}}{x^2 - 8x + 16} = +\infty$$

Car

$$\lim_{x \rightarrow 0^+} \frac{4 - 2\sqrt{2}}{X} = +\infty$$

(70) Calculons: $\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2-4}}$

On a:

$$\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2-4}}$$

$$\begin{aligned}
 \lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2-4}} &= \lim_{x \rightarrow 2^+} \frac{\sqrt{x} - \sqrt{2}}{\sqrt{x^2-4}} + \frac{\sqrt{x-2}}{\sqrt{x^2-4}} \\
 &= \lim_{x \rightarrow 2^+} \frac{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})}{\sqrt{x^2-4}(\sqrt{x} + \sqrt{2})} + \frac{\sqrt{x-2}}{\sqrt{(x-2)(x+2)}} \\
 &= \lim_{x \rightarrow 2^+} \frac{\sqrt{x-2}}{\sqrt{(x+2)}(\sqrt{x} + \sqrt{2})} + \frac{1}{\sqrt{(x+2)}}
 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2-4}} = \frac{1}{2}$$

(71) Calculons: $\lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2}$

On a:

$$\begin{aligned}
 \lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2} &= \lim_{x \rightarrow 2^-} \left(\frac{\sqrt{4-x^2}}{x-2} + 1 \right) \\
 &= \lim_{x \rightarrow 2^-} \left(\frac{\sqrt{(2-x)(2+x)}}{-\sqrt{(x-2)^2}} + 1 \right) \\
 &= \lim_{x \rightarrow 2^-} \left(-\sqrt{\frac{2+x}{2-x}} + 1 \right)
 \end{aligned}$$

$$\lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2} = -\infty$$

Car:

$$\begin{cases} \lim_{x \rightarrow 2^-} 2-x = 0^+ \\ \lim_{x \rightarrow 0^+} \frac{4}{X} = +\infty \\ \lim_{Y \rightarrow +\infty} \sqrt{Y} = +\infty \end{cases}$$

(72) Calculons: $\lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)}$

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)} &= \lim_{x \rightarrow -1} \frac{x^3 - (-1)^3}{(x+1)^3(x-2)} \\ &= \lim_{x \rightarrow -1} \frac{(x+1)(x^2 - x + 1)}{(x+1)^3(x-2)} \\ &= \lim_{x \rightarrow -1} \frac{x^2 - x + 1}{(x+1)^2(x-2)} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)} = -\infty$$

Car:

$$\begin{cases} \lim_{x \rightarrow -1} x^2 - x + 1 = 3 \\ \lim_{x \rightarrow -1} (x+1)^2(x-2) = 0^- \\ \lim_{x \rightarrow 0^-} \frac{1}{X} = -\infty \end{cases}$$

(73) Calculons: $\lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}}$

On a:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}} &= \lim_{x \rightarrow 0^+} \frac{(2 - \sqrt{x^2 + 4})(2 + \sqrt{x^2 + 4})}{\sqrt{x}(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \\ &= \lim_{x \rightarrow 0^+} \frac{-x^2 \times \sqrt{x}}{\sqrt{x} \times \sqrt{x}(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \\ &= \lim_{x \rightarrow 0^+} \frac{-x\sqrt{x}}{(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}} = 0$$

Car:

$$\begin{cases} \lim_{x \rightarrow 0^+} -x\sqrt{x} = 0 \\ \lim_{x \rightarrow 0^+} (1 - \sqrt{2x})(2 + \sqrt{x^2 + 4}) = 4 \end{cases}$$

(74) Calculons: $\lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|}$

On a:

$$\lim_{x \rightarrow -3^-} x^3 = -27$$

et

$$\lim_{x \rightarrow -3^-} |x^2 - 9| = 0^+$$

Donc:

$$\lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|} = -\infty$$

Car:

$$\lim_{x \rightarrow 0^+} \frac{-27}{X} = -\infty$$

(75) Calculons: $\lim_{x \rightarrow (\frac{1}{2})^+} \frac{3x^2 - x - 1}{4x^2 - 1}$

On a:

$$\lim_{x \rightarrow (\frac{1}{2})^+} 3x^2 - x - 1 = \frac{1}{4}$$

et

$$\lim_{x \rightarrow (\frac{1}{2})^+} 4x^2 - 1 = 0^+$$

Donc:

$$\lim_{x \rightarrow (\frac{1}{2})^+} \frac{3x^2 - x - 1}{4x^2 - 1} = +\infty$$

Car : $\lim_{x \rightarrow 0^+} \frac{1}{X} = +\infty$

(76) Calculons: $\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x})$

On a:

$$\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x}) = \lim_{x \rightarrow +\infty} \sqrt{x}(2\sqrt{x} - 5)$$

Donc:

$$\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x}) = +\infty$$

Car:

$$\lim_{x \rightarrow +\infty} \sqrt{x} = +\infty$$

(77) Calculons: $\lim_{x \rightarrow -\infty} (\sqrt{9x^2 + 3x - 5} + x + 3)$

On a:

$$\begin{aligned} \lim_{x \rightarrow -\infty} (\sqrt{9x^2 + 3x - 5} + x + 3) &= \lim_{x \rightarrow -\infty} \left(\sqrt{x^2 \left(9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(|x| \sqrt{\left(9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left(-x \sqrt{\left(9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left[x \left(-\sqrt{\left(9 + \frac{3}{x} - \frac{5}{x^2} \right)} + 1 \right) + 3 \right] \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -\infty} (\sqrt{9x^2 + 3x - 5} + x + 3) = +\infty$$

Car:

$$\lim_{x \rightarrow -\infty} \frac{3}{x} = \lim_{x \rightarrow -\infty} \frac{5}{x} = 0 \text{ et } \lim_{x \rightarrow -\infty} x = -\infty$$

(78) Calculons: $\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x)$

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x+3} - x) &= \lim_{x \rightarrow +\infty} \sqrt{x \left(1 + \frac{3}{x} \right)} - x \\ &= \lim_{x \rightarrow +\infty} \sqrt{x} \sqrt{\left(1 + \frac{3}{x} \right)} - \sqrt{x^2} \\ &= \lim_{x \rightarrow +\infty} \sqrt{x} \left(\sqrt{\left(1 + \frac{3}{x} \right)} - \sqrt{x} \right) \end{aligned}$$

$$\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x) = -\infty$$

Car:

$$\lim_{x \rightarrow +\infty} \sqrt{x} = +\infty \text{ et } \lim_{x \rightarrow +\infty} \frac{3}{x} = 0$$

(79) Calculons: $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4)$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4) &= \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - (x - 4)) \\ &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 + 5x + 2} - (x - 4))(\sqrt{x^2 + 5x + 2} + (x - 4))}{(\sqrt{x^2 + 5x + 2} + (x - 4))} \\ &= \lim_{x \rightarrow +\infty} \frac{13 - \frac{14}{x}}{\sqrt{1 + \frac{5}{x} + \frac{2}{x^2}} + 1 - \frac{4}{x}} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4) = \frac{13}{2}$$

(80) Calculons: $\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x})$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x}) &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{(\sqrt{x+1} + \sqrt{x})} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{(\sqrt{x+1} + \sqrt{x})} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x}) = 0$$

Car:

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} + \sqrt{x}) = +\infty$$

et

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

(81) Calculons: $\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3)$

On a:

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3) &= \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + (2x - 3)) \\
 &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{4x^2 + x + 3} + (2x - 3))(\sqrt{4x^2 + x + 3} - (2x - 3))}{(\sqrt{4x^2 + x + 3} - (2x - 3))} \\
 &= \lim_{x \rightarrow -\infty} \frac{x \left(13 - \frac{6}{x}\right)}{x \left(-\sqrt{4 + \frac{1}{x} + \frac{3}{x^2}} - 2 + \frac{3}{x}\right)} \\
 &= \lim_{x \rightarrow -\infty} \frac{13 - \frac{6}{x}}{-\sqrt{4 + \frac{1}{x} + \frac{3}{x^2}} - 2 + \frac{3}{x}} \\
 \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3) &= -\frac{13}{4}
 \end{aligned}$$

(82) Calculons: $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x}$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 \left(1 + \frac{5}{x} - \frac{3}{x^2}\right)}}{x} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x \times \sqrt{\left(1 + \frac{5}{x} - \frac{3}{x^2}\right)}}{x} \\
 &= \lim_{x \rightarrow -\infty} -\sqrt{\left(1 + \frac{5}{x} - \frac{3}{x^2}\right)}
 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x} = -1$$

(83) Calculons: $\lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}}$

On a:

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{4x^2 + 3}{x^2 + 8x} &= \lim_{x \rightarrow +\infty} \frac{4x^2}{x^2} \\
 &= \lim_{x \rightarrow +\infty} 4 \\
 &= 4
 \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}} = 2$$

(84) Calculons: $\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2}$

On a:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \times \sqrt{1 + \frac{1}{x}}}{x \left(1 + \frac{2}{x}\right)}$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = \lim_{x \rightarrow +\infty} \frac{\sqrt{1 + \frac{1}{x}}}{\sqrt{x} \left(1 + \frac{2}{x}\right)}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = 0$$

(85) Calculons: $\lim_{x \rightarrow +\infty} \frac{x+6\sqrt{x}}{2x+\sqrt{x}-3}$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x+6\sqrt{x}}{2x+\sqrt{x}-3} &= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{6}{\sqrt{x}}\right)}{x \left(2 + \frac{1}{\sqrt{x}} - \frac{3}{x}\right)} \\ &= \lim_{x \rightarrow +\infty} \frac{1 + \frac{6}{\sqrt{x}}}{2 + \frac{1}{\sqrt{x}} - \frac{3}{x}} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{x+6\sqrt{x}}{2x+\sqrt{x}-3} = \frac{1}{2}$$

(86) Calculons: $\lim_{x \rightarrow +\infty} \frac{1}{x} (\sqrt{3x^2+1} - \sqrt{x^2+x})$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{1}{x} (\sqrt{3x^2+1} - \sqrt{x^2+x}) &= \lim_{x \rightarrow +\infty} \frac{1}{x} \left(\sqrt{x^2 \left(3 + \frac{1}{x^2}\right)} - \sqrt{x^2 \left(1 + \frac{1}{x}\right)} \right) \\ &= \lim_{x \rightarrow +\infty} \left(\sqrt{3 + \frac{1}{x^2}} - \sqrt{1 + \frac{1}{x}} \right) \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{1}{x} (\sqrt{3x^2 + 1} - \sqrt{x^2 + x}) = \sqrt{3} - 1$$

(87) Calculons: $\lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2)$

$$\lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2)$$

On a:

$$\lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2) = \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^4 - x^3} - x^2)(\sqrt{x^4 - x^3} + x^2)}{(\sqrt{x^4 - x^3} + x^2)}$$

Donc:

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2) &= \lim_{x \rightarrow +\infty} \frac{-x^3}{x^2 \left(\sqrt{1 - \frac{1}{x}} + 1 \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{-x}{\sqrt{1 - \frac{1}{x}} + 1} \\ \lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2) &= -\infty \end{aligned}$$

(88) Calculons: $\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}}$

On a:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}} &= \lim_{x \rightarrow +\infty} \left(\frac{3x\sqrt{x^2+x} - 3x^2}{5\sqrt{x^2+3}} + \frac{x}{5\sqrt{x^2+3}} \right) \\ &= \lim_{x \rightarrow +\infty} \left(\frac{(3x\sqrt{x^2+x} - 3x^2)(3x\sqrt{x^2+x} + 3x^2)}{5\sqrt{x^2+3}(3x\sqrt{x^2+x} + 3x^2)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right) \end{aligned}$$

(88) Calculons: $\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}}$

$$\begin{aligned}
\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}} &= \lim_{x \rightarrow +\infty} \left(\frac{(3x\sqrt{x^2+x} - 3x^2)(3x\sqrt{x^2+x} + 3x^2)}{5\sqrt{x^2+3}(3x\sqrt{x^2+x} + 3x^2)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right) \\
&= \lim_{x \rightarrow +\infty} \left(\frac{9x^3}{5x\sqrt{1 + \frac{3}{x^2}} \times x^2 \left(3\sqrt{1 + \frac{1}{x}} + 3 \right)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right) \\
&= \lim_{x \rightarrow +\infty} \left(\frac{9}{5 \times \sqrt{1 + \frac{3}{x^2}} \left(3\sqrt{1 + \frac{1}{x}} + 3 \right)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right)
\end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2+x} - 3x^2 + x}{5\sqrt{x^2+3}} = \frac{1}{2}$$

(89) Calculons: $\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right)$

On a:

$$\begin{aligned}
&\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right) \\
&\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right) = \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{\frac{x^3}{x+1}} - x \right) \left(\sqrt{\frac{x^3}{x+1}} + x \right)}{\left(\sqrt{\frac{x^3}{x+1}} + x \right)} \\
&= \lim_{x \rightarrow +\infty} \frac{\frac{x^3}{x+1} - x^2}{\left(\sqrt{\frac{x^3}{x+1}} + x \right)} \\
&= \lim_{x \rightarrow +\infty} \frac{\frac{-x^2}{x+1}}{\left(\sqrt{\frac{x^2}{1 + \frac{1}{x}}} + x \right)}
\end{aligned}$$

Donc:

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right) &= \lim_{x \rightarrow +\infty} \frac{-x^2}{(x+1) \left(x \times \sqrt{\frac{1}{1+\frac{1}{x}}} + x \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{-1}{\left(1 + \frac{1}{x} \right) \left(\sqrt{\frac{1}{1+\frac{1}{x}}} + 1 \right)}\end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x+1}} - x \right) = -\frac{1}{2}$$

(90) Calculons: $\lim_{x \rightarrow +\infty} \left(\frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right)$

On a:

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left(\frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right) &= \lim_{x \rightarrow +\infty} \frac{x\sqrt{x+2} - x\sqrt{x-2}}{\sqrt{x-2}\sqrt{x+2}} \\ &= \lim_{x \rightarrow +\infty} \frac{(x\sqrt{x+2} - x\sqrt{x-2})(x\sqrt{x+2} + x\sqrt{x-2})}{\sqrt{x-2}\sqrt{x+2}(x\sqrt{x+2} + x\sqrt{x-2})} \\ &= \lim_{x \rightarrow +\infty} \frac{4x^2}{x^2 \times \sqrt{1-\frac{2}{x}} \sqrt{1+\frac{2}{x}} (\sqrt{x+2} + \sqrt{x-2})} \\ &= \lim_{x \rightarrow +\infty} \frac{4}{\sqrt{1-\frac{2}{x}} \sqrt{1+\frac{2}{x}} (\sqrt{x+2} + \sqrt{x-2})}\end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} \left(\frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right) = 0$$

(91) Calculons $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$

On pose $u = \sqrt{x}$, donc $x = u^2$.
 Quand $x \rightarrow 1$, alors $u \rightarrow 1$.
 L'expression devient :

$$\frac{u^2 - 1}{u - 1} = \frac{(u - 1)(u + 1)}{u - 1} = u + 1 \text{ (pour } u \neq 1\text{)}$$

Donc:

$$\lim_{u \rightarrow 1} (u + 1) = 1 + 1 = 2$$

Réponse: 2

(92) Calculons. $\lim_{x \rightarrow 0} \frac{2\sqrt{x+1} - \sqrt{x^2+4}}{x}$

On ne peut pas appliquer L'Hôpital. Essayons de séparer ou de manipuler.
 On remarque que :

- $\sqrt{x+1} \approx 1 + \frac{x}{2}$ (développement limité, mais on va éviter cela).
- Utilisons la rationalisation en séparant:

On écrit :

$$\frac{2\sqrt{x+1} - 2 + 2 - \sqrt{x^2+4}}{x} = 2 \cdot \frac{\sqrt{x+1} - 1}{x} - \frac{\sqrt{x^2+4} - 2}{x}$$

Calculons chaque limite.

1. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$

Rationalisons:

$$\frac{\sqrt{x+1} - 1}{x} \cdot \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} = \frac{(x+1) - 1}{x(\sqrt{x+1} + 1)} = \frac{x}{x(\sqrt{x+1} + 1)} = \frac{1}{\sqrt{x+1} + 1} \rightarrow \frac{1}{2}$$

2. $\lim_{x \rightarrow 0} \frac{\sqrt{x^2+4} - 2}{x}$

Rationalisons:

$$\frac{\sqrt{x^2+4} - 2}{x} \cdot \frac{\sqrt{x^2+4} + 2}{\sqrt{x^2+4} + 2} = \frac{(x^2+4) - 4}{x(\sqrt{x^2+4} + 2)} = \frac{x^2}{x(\sqrt{x^2+4} + 2)} = \frac{x}{\sqrt{x^2+4} + 2} \rightarrow 0$$

Donc:

$$2 \cdot \frac{1}{2} - 0 - 1$$

Réponse:

1

(93) Calculons $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x} - \sqrt{1+2x}}$

On a une forme $\frac{0}{0}$. Rationalisons numérateur et dénominateur.

Numérateur : $\sqrt{1+x^2} - \sqrt{1-x^2}$

Multiplions par le conjugué :

$$\frac{(\sqrt{1+x^2} - \sqrt{1-x^2})(\sqrt{1+x^2} + \sqrt{1-x^2})}{\sqrt{1+x^2} + \sqrt{1-x^2}} = \frac{(1+x^2) - (1-x^2)}{\sqrt{1+x^2} + \sqrt{1-x^2}} = \frac{2x^2}{\sqrt{1+x^2} + \sqrt{1-x^2}}$$

Dénominateur : $\sqrt{1+x} - \sqrt{1+2x}$

Conjugué :

$$\frac{(\sqrt{1+x} - \sqrt{1+2x})(\sqrt{1+x} + \sqrt{1+2x})}{\sqrt{1+x} + \sqrt{1+2x}} = \frac{(1+x) - (1+2x)}{\sqrt{1+x} + \sqrt{1+2x}} = \frac{-x}{\sqrt{1+x} + \sqrt{1+2x}}$$

Donc l'expression devient:

$$\frac{\frac{2x^2}{\sqrt{1+x^2} + \sqrt{1-x^2}}}{\frac{-x}{\sqrt{1+x} + \sqrt{1+2x}}} = \frac{2x^2}{\sqrt{1+x^2} + \sqrt{1-x^2}} \cdot \frac{\sqrt{1+x} + \sqrt{1+2x}}{-x} = -2x \cdot \frac{\sqrt{1+x} + \sqrt{1+2x}}{\sqrt{1+x^2} + \sqrt{1-x^2}}$$

Quand $x \rightarrow 0$:

- $-2x \rightarrow 0$
- Le reste tend vers $\frac{1+1}{1+1} = 1$

Donc la limite est $0 \times 1 = 0$

Réponse: 0

(94) lim $\lim_{x \rightarrow 0} \frac{\sqrt{4-x} - \sqrt{4+x}}{x}$

Rationalisons le numérateur:

$$\frac{\sqrt{4-x} - \sqrt{4+x}}{x} \cdot \frac{\sqrt{4-x} + \sqrt{4+x}}{\sqrt{4-x} + \sqrt{4+x}} = \frac{(4-x) - (4+x)}{x(\sqrt{4-x} + \sqrt{4+x})} = \frac{-2x}{x(\sqrt{4-x} + \sqrt{4+x})}$$

Simplifions $x \neq 0$:

$$= \frac{-2}{\sqrt{4-x} + \sqrt{4+x}}$$

Quand $x \rightarrow 0$:

$$\frac{-2}{\sqrt{4} + \sqrt{4}} = \frac{-2}{2+2} = -\frac{1}{2}$$

Réponse: $-\frac{1}{2}$

(95) calculons $\lim_{x \rightarrow 1} \frac{x\sqrt{x}-1}{1-x^2}$

Posons $u = \sqrt{x}$, donc $x = u^2, x\sqrt{x} = u^3, x^2 = u^4$

Quand $x \rightarrow 1, u \rightarrow 1$

Expression :

$$\frac{u^3 - 1}{1 - u^4} = \frac{(u - 1)(u^2 + u + 1)}{-(u^4 - 1)} = \frac{(u - 1)(u^2 + u + 1)}{-(u^2 - 1)(u^2 + 1)} = \frac{(u - 1)(u^2 + u + 1)}{-(u - 1)(u + 1)(u^2 + 1)}$$

Simplifions $u - 1 \neq 0$:

$$= \frac{u^2 + u + 1}{-(u + 1)(u^2 + 1)}$$

En $u = 1$:

$$\frac{1 + 1 + 1}{-(1 + 1)(1 + 1)} = \frac{3}{-2 \cdot 2} = -\frac{3}{4}$$

Réponse: $-\frac{3}{4}$

(96) calculons $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x^2-x}$

On a $x^2 - x = x(x - 1)$

Rationalisons le numérateur:

$$\begin{aligned} \frac{\sqrt{x+1}-1}{x(x-1)} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} &= \frac{(x+1)-1}{x(x-1)(\sqrt{x+1}+1)} = \frac{x}{x(x-1)(\sqrt{x+1}+1)} \\ &= \frac{1}{(x-1)(\sqrt{x+1}+1)} \end{aligned}$$

Quand $x \rightarrow 0$:

$$\frac{1}{(-1)(1+1)} = \frac{1}{-2} = -\frac{1}{2}$$

Réponse: $-\frac{1}{2}$

(97) calculons $\lim_{x \rightarrow 2} \frac{2-\sqrt{x+2}}{3-\sqrt{2x+5}}$

Forme $\frac{0}{0}$. Rationalisons numérateur et dénominateur.

Multiplions numérateur et dénominateur par les conjugués :

- Numérateur : $2 + \sqrt{x+2}$
- Dénominateur : $3 + \sqrt{2x+5}$

$$\frac{(2 - \sqrt{x+2})(2 + \sqrt{x+2})}{(3 - \sqrt{2x+5})(3 + \sqrt{2x+5})} \cdot \frac{3 + \sqrt{2x+5}}{2 + \sqrt{x+2}} = \frac{4 - (x+2)}{9 - (2x+5)} \cdot \frac{3 + \sqrt{2x+5}}{2 + \sqrt{x+2}}$$

$$= \frac{2-x}{4-2x} \cdot \frac{3 + \sqrt{2x+5}}{2 + \sqrt{x+2}}$$

Mais $4 - 2x = 2(2 - x)$, donc:

$$\frac{2-x}{2(2-x)} \cdot \frac{3 + \sqrt{2x+5}}{2 + \sqrt{x+2}} = \frac{1}{2} \cdot \frac{3 + \sqrt{2x+5}}{2 + \sqrt{x+2}} \quad (x \neq 2)$$

En $x = 2$:

$$\frac{1}{2} \cdot \frac{3+3}{2+2} = \frac{1}{2} \cdot \frac{6}{4} = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

Réponse: $\frac{3}{4}$

(98) calculons $\lim_{x \rightarrow 2} \frac{x - \sqrt{x+2}}{x^2 - 4}$

$$x^2 - 4 = (x - 2)(x + 2)$$

Rationalisons le numérateur :

$$\frac{x - \sqrt{x+2}}{(x - 2)(x + 2)} \cdot \frac{x + \sqrt{x+2}}{x + \sqrt{x+2}} = \frac{x^2 - (x + 2)}{(x - 2)(x + 2)(x + \sqrt{x+2})} = \frac{x^2 - x - 2}{(x - 2)(x + 2)(x + \sqrt{x+2})}$$

$$\text{Factorisons : } x^2 - x - 2 = (x - 2)(x + 1)$$

Donc:

$$\frac{(x - 2)(x + 1)}{(x - 2)(x + 2)(x + \sqrt{x+2})} = \frac{x + 1}{(x + 2)(x + \sqrt{x+2})} \quad (x \neq 2)$$

En $x = 2$:

$$\frac{3}{(4)(2+2)} = \frac{3}{4 \cdot 4} = \frac{3}{16}$$

Réponse: $\frac{3}{16}$

(99) calculons $\lim_{x \rightarrow 4} \frac{2x+1-3}{x-4}$

Simplifions le numérateur :

$$2x + 1 - 3 = 2x - 2 = 2(x - 1)$$

Donc:

$$\frac{2(x-1)}{x-4}$$

Quand $x \rightarrow 4$, numérateur $\rightarrow 6$, dénominateur $\rightarrow 0$

- Si $x \rightarrow 4^-$, dénominateur $< 0 \rightarrow$ limite $\rightarrow -\infty$
- Si $x \rightarrow 4^+$, dénominateur $> 0 \rightarrow$ limite $\rightarrow +\infty$

Les deux infinis sont de signes opposés $\rightarrow$ la limite n'existe pas

Réponse: N'existe pas

(100) Calculons $\lim_{x \rightarrow 3} \frac{\sqrt{x+1}-\sqrt{7-x}}{\sqrt{2x+3}-\sqrt{15-2x}}$

Forme $\frac{0}{0}$. Rationalisons numérateur et dénominateur.

Multiplions par les conjugués:

- Numérateur : $\sqrt{x+1} + \sqrt{7-x}$
- Dénominateur : $\sqrt{2x+3} + \sqrt{15-2x}$

Alors:

$$\frac{[(x+1) - (7-x)](\sqrt{2x+3} + \sqrt{15-2x})}{[(2x+3) - (15-2x)](\sqrt{x+1} + \sqrt{7-x})} = \frac{(2x-6)(\sqrt{2x+3} + \sqrt{15-2x})}{(4x-12)(\sqrt{x+1} + \sqrt{7-x})}$$

Or $2x - 6 = 2(x - 3)$, $4x - 12 = 4(x - 3)$

Donc:

$$\frac{2(x-3)(\sqrt{2x+3} + \sqrt{15-2x})}{4(x-3)(\sqrt{x+1} + \sqrt{7-x})} = \frac{2}{4} \cdot \frac{\sqrt{2x+3} + \sqrt{15-2x}}{\sqrt{x+1} + \sqrt{7-x}} = \frac{1}{2} \cdot \frac{\sqrt{2x+3} + \sqrt{15-2x}}{\sqrt{x+1} + \sqrt{7-x}}$$

En $x = 3$:

- Numérateur : $\sqrt{9} + \sqrt{9} = 3 + 3 = 6$
- Dénominateur : $\sqrt{4} + \sqrt{4} = 2 + 2 = 4$
- Donc : $\frac{1}{2} \cdot \frac{6}{4} = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$

Réponse: $\frac{3}{4}$

(101) Calculons $\lim_{x \rightarrow 0^+} \frac{x+\sqrt{x}}{\sqrt{x^2+x-x}}$

Posons $t = \sqrt{x}$, donc $x = t^2, t \rightarrow 0^+$
Alors:

- Numérateur: $t^2 + t$
- Dénominateur: $\sqrt{t^4 + t^2} - t^2 = \sqrt{t^2(t^2 + 1)} - t^2 = t\sqrt{t^2 + 1} - t^2$

Expression :

$$\frac{t^2 + t}{t\sqrt{t^2 + 1} - t^2} = \frac{t(t + 1)}{t(\sqrt{t^2 + 1} - t)} = \frac{t + 1}{\sqrt{t^2 + 1} - t}$$

Quand $t \rightarrow 0^+$:

- Numérateur $\rightarrow 1$
- Dénominateur $\rightarrow \sqrt{1} - 0 = 1$

Donc limite $= \frac{1}{1} = 1$

Réponse: 1

(102) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + x + 3} - 2x)$

On a une forme indéterminée $\infty - \infty$. On utilise la rationalisation.
Multiplions par le conjugué :

$$\begin{aligned} \sqrt{x^2 + x + 3} - 2x &= \frac{(\sqrt{x^2 + x + 3} - 2x)(\sqrt{x^2 + x + 3} + 2x)}{\sqrt{x^2 + x + 3} + 2x} = \frac{(x^2 + x + 3) - 4x^2}{\sqrt{x^2 + x + 3} + 2x} \\ &= \frac{-3x^2 + x + 3}{\sqrt{x^2 + x + 3} + 2x} \end{aligned}$$

Divisons numérateur et dénominateur par x (car $x > 0$):

Numérateur:

$$\frac{-3x^2 + x + 3}{x} = -3x + 1 + \frac{3}{x}$$

Dénominateur:

$$\frac{\sqrt{x^2 + x + 3}}{x} + 2 = \sqrt{1 + \frac{1}{x} + \frac{3}{x^2}} + 2$$

Donc l'expression devient:

$$\frac{-3x + 1 + \frac{3}{x}}{\sqrt{1 + \frac{1}{x} + \frac{3}{x^2}} + 2}$$

Quand $x \rightarrow +\infty$:

- Numérateur $\rightarrow -\infty$
- Dénominateur $\rightarrow \sqrt{1} + 2 - 3$

Donc limite: $\frac{-\infty}{3} = -\infty$

Réponse: $-\infty$

(103) $\lim_{x \rightarrow -\infty} (\sqrt{3x^2 - 5x + 7} + x - 4)$

Forme $\infty + (-\infty)$. On traite la racine carrée.

Comme $x \rightarrow -\infty$, on pose $x = -t$, avec $t \rightarrow +\infty$

Alors:

$$\sqrt{3(-t)^2 - 5(-t) + 7} + (-t) - 4 = \sqrt{3t^2 + 5t + 7} - t - 4$$

On étudie $\lim_{t \rightarrow +\infty} (\sqrt{3t^2 + 5t + 7} - t) - 4$

Rationalisons $\sqrt{3t^2 + 5t + 7} - t$:

$$\frac{(\sqrt{3t^2 + 5t + 7} - t)(\sqrt{3t^2 + 5t + 7} + t)}{\sqrt{3t^2 + 5t + 7} + t} = \frac{(3t^2 + 5t + 7) - t^2}{\sqrt{3t^2 + 5t + 7} + t} = \frac{2t^2 + 5t + 7}{\sqrt{3t^2 + 5t + 7} + t}$$

Divisons numérateur et dénominateur par t :

Numérateur: $2t + 5 + \frac{7}{t}$

Dénominateur: $\sqrt{3 + \frac{5}{t} + \frac{7}{t^2}} + 1$

Quand $t \rightarrow +\infty$:

- Numérateur $\rightarrow +\infty$
- Dénominateur $\rightarrow \sqrt{3} + 1 \approx 2.732$

Donc $\sqrt{3t^2 + 5t + 7} - t \rightarrow +\infty$, donc expression totale $\rightarrow +\infty - 4 = +\infty$

Réponse: $+\infty$

(104) $\lim_{x \rightarrow +\infty} \left(1 + \frac{x}{2} - \sqrt{1+x}\right)$

Forme $\infty - \infty$. On regroupe:

$$\left(1 + \frac{x}{2}\right) - \sqrt{1+x}$$

On met en facteur x pour analyser le comportement.

Posons $u = \sqrt{1+x}$, donc $u \rightarrow +\infty$, $x = u^2 - 1$

Alors:

$$1 + \frac{u^2 - 1}{2} - u - 1 + \frac{u^2}{2} - \frac{1}{2} - u - \frac{u^2}{2} - u + \frac{1}{2}$$

Quand $u \rightarrow +\infty$, $\frac{u^2}{2}$ domine $\rightarrow +\infty$

Réponse: $\mathbb{R} + \infty$

(105) $\lim_{x \rightarrow +\infty} \frac{x-1}{\sqrt{x^2+4}}$

On divise numérateur et dénominateur par x (puisque $x > 0$):

$$\frac{1 - \frac{1}{x}}{\sqrt{1 + \frac{4}{x^2}}} \rightarrow \frac{1 - 0}{\sqrt{1 + 0}} = 1$$

Réponse: $\mathbb{R} 1$

(106) $\lim_{x \rightarrow -\infty} \frac{\sqrt{3(x)^2 - (x) + 8}}{5x - 2}$

On factorise sous la racine et dans le dénominateur.

$x \rightarrow -\infty$, donc $x < 0$. On écrit $x = -t$, $t \rightarrow +\infty$

Alors:

- $\sqrt{3(-t)^2 - (-t) + 8} = \sqrt{3t^2 + t + 8}$
- $5(-t) - 2 = -5t - 2$

Expression :

$$\frac{\sqrt{3t^2 + t + 8}}{-5t - 2}$$

Divisons numérateur et dénominateur par t :

Numérateur : $\frac{\sqrt{3t^2 + t + 8}}{t} = \sqrt{3 + \frac{1}{t} + \frac{8}{t^2}} \rightarrow \sqrt{3}$

Dénominateur : $-5 - \frac{2}{t} \rightarrow -5$

Donc limite : $\frac{\sqrt{3}}{-5} = -\frac{\sqrt{3}}{5}$

Réponse: $-\frac{\sqrt{3}}{5}$

(107) $\lim_{x \rightarrow -\infty} (x + \sqrt{1-x})$

$x \rightarrow -\infty$, donc $-x \rightarrow +\infty$

On pose $x = -t$, $t \rightarrow +\infty$

Alors:

$$-t + \sqrt{1 - (-t)} = -t + \sqrt{1 + t}$$

On étudie $-t + \sqrt{1 + t}$

On rationalise ou compare croissance :

- $\sqrt{1+t} \ll t$ quand $t \rightarrow +\infty$
- Donc $-t + \sqrt{1+t} \rightarrow -\infty$

Réponse: $\mathbb{R} - \infty$

$$(108) \lim_{x \rightarrow -\infty} (\sqrt{2x} + \sqrt{x^2 - x + 4})$$

Attention: $\sqrt{2x}$ n'est pas défini si $x < 0$, sauf si on considère $x \rightarrow -\infty$ avec $x < 0$

Mais $2x < 0$, donc $\sqrt{2x}$ n'existe pas dans $\mathbb{R}$

Donc la fonction n'est pas définie pour $x < 0$, donc la limite n'existe pas (ou n'est pas définie)

Réponse: $\mathbb{R}$ N'existe pas (fonction non définie)

$$(109) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 2x + 3}}{\sqrt{4 - x}}$$

Analysons le domaine:

- $\sqrt{4 - x}$ est défini seulement si $4 - x \geq 0 \Rightarrow x \leq 4$
- Mais $x \rightarrow +\infty$, donc impossible

Donc la fonction n'est pas définie pour $x > 4$

La limite n'existe pas

Réponse: N'existe pas

$$(110) \lim_{x \rightarrow -\infty} \frac{x^2 - 1}{x^4 + 2}$$

On divise numérateur et dénominateur par x^4 :

$$\frac{\frac{1}{x^2} - \frac{1}{x^4}}{1 + \frac{2}{x^4}} \rightarrow \frac{0 - 0}{1 + 0} = 0$$

Réponse: $\mathbb{R}$ 0

$$(111) \lim_{x \rightarrow +\infty} (x - 2\sqrt{x} + 5)$$

On pose $t = \sqrt{x}$, donc $x = t^2$, $t \rightarrow +\infty$

Expression : $t^2 - 2t + 5 \rightarrow +\infty$

Réponse: $\mathbb{R}$ $+\infty$

$$(112) \lim_{x \rightarrow +\infty} (\sqrt{1+x} - \sqrt{1+4x})$$

Forme $\infty - \infty$. Rationalisons:

$$\begin{aligned} \sqrt{1+x} - \sqrt{1+4x} &= \frac{(\sqrt{1+x} - \sqrt{1+4x})(\sqrt{1+x} + \sqrt{1+4x})}{\sqrt{1+x} + \sqrt{1+4x}} = \frac{(1+x) - (1+4x)}{\sqrt{1+x} + \sqrt{1+4x}} \\ &= \frac{-3x}{\sqrt{1+x} + \sqrt{1+4x}} \end{aligned}$$

Divisons numérateur et dénominateur par x :

Numérateur : -3

Dénominateur: $\frac{\sqrt{1+x}}{x} + \frac{\sqrt{1+4x}}{x} - \frac{1}{\sqrt{x}} \cdot \sqrt{1+\frac{1}{x}} + \frac{1}{\sqrt{x}} \cdot \sqrt{1+\frac{1}{4x}} \rightarrow 0$

Donc:

$$\frac{-3}{0^+} \rightarrow -\infty$$

Réponse:

$-\infty$

(113) Calculons $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - x - 1)$

Forme indéterminée $\infty - \infty$. On utilise la rationalisation.

On regroupe d'abord:

$$\sqrt{x^2 + 1} - x - 1 - (\sqrt{x^2 + 1} - x) - 1$$

On étudie d'abord $\sqrt{x^2 + 1} - x$

Rationalisons:

$$\sqrt{x^2 + 1} - x - \frac{(\sqrt{x^2 + 1} - x)(\sqrt{x^2 + 1} + x)}{\sqrt{x^2 + 1} + x} - \frac{(x^2 + 1) - x^2}{\sqrt{x^2 + 1} + x} - \frac{1}{\sqrt{x^2 + 1} + x}$$

Donc:

$$\sqrt{x^2 + 1} - x - 1 - \frac{1}{\sqrt{x^2 + 1} + x} - 1$$

Quand $x \rightarrow +\infty$:

- $\sqrt{x^2 + 1} \sim x$, donc $\sqrt{x^2 + 1} + x \sim 2x \rightarrow +\infty$
- Donc $\frac{1}{\sqrt{x^2 + 1} + x} \rightarrow 0$
- Alors $\frac{1}{\sqrt{x^2 + 1} + x} - 1 \rightarrow -1$

Réponse: -1

$$(114) \lim_{x \rightarrow -\infty} (\sqrt{x^4 + x^2 + 1} - x^2 + x + 3)$$

On a une forme $\infty - \infty$. On traite la racine carrée.

Comme $x \rightarrow -\infty$, on pose $x = -t$, avec $t \rightarrow +\infty$

Alors :

- $x^4 - t^4$
- $x^2 - t^2$
- $x - -t$

Expression :

$$\sqrt{t^4 + t^2 + 1} - t^2 - t + 3$$

On étudie $\sqrt{t^4 + t^2 + 1} - t^2$

Rationalisons :

$$\sqrt{t^4 + t^2 + 1} - t^2 = \frac{(t^4 + t^2 + 1) - t^4}{\sqrt{t^4 + t^2 + 1} + t^2} = \frac{t^2 + 1}{\sqrt{t^4 + t^2 + 1} + t^2}$$

Divisons numérateur et dénominateur par t^2 :

Numérateur : $1 + \frac{1}{t^2} \rightarrow 1$

Dénominateur : $\frac{\sqrt{t^5 + t^2 + 1}}{t^2} + 1 - \sqrt{1 + \frac{1}{t^2} + \frac{1}{t^2}} + 1 \rightarrow \sqrt{1} + 1 - 2$

Donc :

$$\sqrt{t^4 + t^2 + 1} - t^2 \rightarrow \frac{1}{2}$$

Et l'expression totale:

$$(\sqrt{t^4 + t^2 + 1} - t^2) - t + 3 \rightarrow \frac{1}{2} - t + 3 \rightarrow -\infty \text{ (car } -t \rightarrow -\infty)$$

Réponse: $-\infty$

$$(115) \text{ Calculons } \lim_{x \rightarrow +\infty} (\sqrt{4x^2 - 3x + 7} - 2x + 5)$$

Forme $\infty - \infty$. On regroupe:

$$\sqrt{4x^2 - 3x + 7} - 2x + 5 - (\sqrt{4x^2 - 3x + 7} - 2x) + 5$$

On étudie $\sqrt{4x^2 - 3x + 7} - 2x$

Rationalisons:

$$\frac{(\sqrt{4x^2 - 3x + 7} - 2x)(\sqrt{4x^2 - 3x + 7} + 2x)}{\sqrt{4x^2 - 3x + 7} + 2x} - \frac{(4x^2 - 3x + 7) - 4x^2}{\sqrt{4x^2 - 3x + 7} + 2x} - \frac{-3x + 7}{\sqrt{4x^2 - 3x + 7} + 2x}$$

Divisons numérateur et dénominateur par x (puisque $x > 0$):

Numérateur : $-3 + \frac{7}{x} \rightarrow -3$

Dénominateur: $\frac{\sqrt{4x^2-3x+7}}{x} + 2 - \sqrt{4 - \frac{3}{x} + \frac{7}{x^2}} + 2 \rightarrow \sqrt{4} + 2 - 2 + 2 - 4$

Donc:

$$\sqrt{4x^2 - 3x + 7} - 2x \rightarrow \frac{-3}{4}$$

Alors:

$$\sqrt{4x^2 - 3x + 7} - 2x + 5 \rightarrow -\frac{3}{4} + 5 - \frac{17}{4}$$

Réponse: $\frac{17}{4}$

(116) Calculons $\lim_{x \rightarrow +\infty} (2x - \sqrt{4x^2 + 1})$

Forme $\infty - \infty$. On rationalise:

$$2x - \sqrt{4x^2 + 1} = \frac{(2x - \sqrt{4x^2 + 1})(2x + \sqrt{4x^2 + 1})}{2x + \sqrt{4x^2 + 1}} = \frac{4x^2 - (4x^2 + 1)}{2x + \sqrt{4x^2 + 1}} = \frac{-1}{2x + \sqrt{4x^2 + 1}}$$

Quand $x \rightarrow +\infty$, le dénominateur $\rightarrow +\infty$, donc:

$$\frac{-1}{+\infty} \rightarrow 0$$

Réponse: 0

(117) $\lim_{x \rightarrow -\infty} (\sqrt{9x^2 - 4x + 5} + 3x - 2)$

On pose $x = -t$, avec $t \rightarrow +\infty$

Alors:

- $\sqrt{9(-t)^2 - 4(-t) + 5} - \sqrt{9t^2 + 4t + 5}$
- $3(-t) - 2 = -3t - 2$

Expression:

$$\sqrt{9t^2 + 4t + 5} - 3t - 2$$

Étudions $\sqrt{9t^2 + 4t + 5} - 3t$

Rationalisons:

$$\frac{(\sqrt{9t^2 + 4t + 5} - 3t)(\sqrt{9t^2 + 4t + 5} + 3t)}{\sqrt{9t^2 + 4t + 5} + 3t} \downarrow - \frac{(9t^2 + 4t + 5) - 9t^2}{\sqrt{9t^2 + 4t + 5} + 3t} = \frac{4t + 5}{\sqrt{9t^2 + 4t + 5} + 3t}$$

Divisons numérateur et dénominateur par t :

Numérateur : $4 + \frac{5}{t} \rightarrow 4$

Dénominateur : $\sqrt{9 + \frac{4}{t} + \frac{5}{t^2}} + 3 \rightarrow 3 + 3 - 6$

Donc:

$$\sqrt{9t^2 + 4t + 5} - 3t \rightarrow \frac{4}{6} = \frac{2}{3}$$

Et l'expression totale:

$$(\sqrt{9t^2 + 4t + 5} - 3t) - 2 \rightarrow \frac{2}{3} - 2 = -\frac{4}{3}$$

Réponse: $-\frac{4}{3}$

(118) Calculons $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 7} - 2)$

On pose $x = -t, t \rightarrow +\infty$

Alors:

- $x = -t$
- $\sqrt{x^2 + 7} - \sqrt{t^2 + 7}$

Expression:

$$-t + \sqrt{t^2 + 7} - 2 - (\sqrt{t^2 + 7} - t) - 2$$

Rationalisons $\sqrt{t^2 + 7} - t$:

$$\frac{(\sqrt{t^2 + 7} - t)(\sqrt{t^2 + 7} + t)}{\sqrt{t^2 + 7} + t} = \frac{t^2 + 7 - t^2}{\sqrt{t^2 + 7} + t} = \frac{7}{\sqrt{t^2 + 7} + t} \rightarrow 0 \text{ (car dénominateur } \rightarrow +\infty)$$

Donc:

$$(\sqrt{t^2 + 7} - t) - 2 \rightarrow 0 - 2 = -2$$

Réponse: $\boxed{-2}$

(119) Calculons $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - \sqrt{x} + 1)$

On a $\sqrt{x^2 + 1} \sim x, \sqrt{x} \rightarrow +\infty$

Donc:

$$\sqrt{x^2 + 1} - \sqrt{x} + 1 \sim x - \sqrt{x} + 1$$

Comme x croît plus vite que $\sqrt{x}$, on a :

- $x - \sqrt{x} \rightarrow +\infty$
- Donc expression $\rightarrow +\infty$

Réponse: $+\infty$

(120) Calculons $\lim_{x \rightarrow +\infty} (\sqrt{2x+3} - \sqrt{x+4})$

Forme $\infty - \infty$. Rationalisons:

$$\frac{(\sqrt{2x+3} - \sqrt{x+4})(\sqrt{2x+3} + \sqrt{x+4})}{\sqrt{2x+3} + \sqrt{x+4}} - \frac{(2x+3) - (x+4)}{\sqrt{2x+3} + \sqrt{x+4}} - \frac{x-1}{\sqrt{2x+3} + \sqrt{x+4}}$$

Divisons numérateur et dénominateur par $\sqrt{x}$:

Numérateur : $\sqrt{x} \cdot \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) \sim \sqrt{x} \cdot \sqrt{x} - x$

Mais mieux : divisons tout par $\sqrt{x}$:

Numérateur : $\frac{x-1}{\sqrt{x}} - \sqrt{x} - \frac{1}{\sqrt{x}} \rightarrow +\infty$

Dénominateur : $\frac{\sqrt{2x+3}}{\sqrt{x}} + \frac{\sqrt{x+4}}{\sqrt{x}} - \sqrt{2 + \frac{3}{x}} + \sqrt{1 + \frac{4}{x}} \rightarrow \sqrt{2} + 1 \approx 2.414$

Donc:

$$\frac{\sqrt{x} - \frac{1}{\sqrt{x}}}{\sqrt{2 + \frac{3}{x}} + \sqrt{1 + \frac{4}{x}}} \rightarrow \frac{+\infty}{\text{constante}} = +\infty$$

Réponse:

$\boxed{+\infty}$

(121) Calculons $\lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x}} - \sqrt{x})$

On met $\sqrt{x}$ en facteur.

Posons $t = \sqrt{x}$, donc $x = t^2, t \rightarrow +\infty$

Alors:

$$\sqrt{t^2 + t} - t$$

Rationalisons:

$$\frac{(\sqrt{t^2 + t} - t)(\sqrt{t^2 + t} + t)}{\sqrt{t^2 + t} + t} - \frac{(t^2 + t) - t^2}{\sqrt{t^2 + t} + t} - \frac{t}{\sqrt{t^2 + t} + t}$$

Divisons numérateur et dénominateur par t :

$$-\frac{1}{\sqrt{1 + \frac{1}{t}} + 1} \rightarrow \frac{1}{\sqrt{1} + 1} = \frac{1}{2}$$

Réponse: $\frac{1}{2}$

(122) Calculons $\lim_{x \rightarrow -\infty} (\sqrt{1-x} - \sqrt{3-x})$

On pose $x = -t, t \rightarrow +\infty$

Alors:

- $\sqrt{1 - (-t)} - \sqrt{1 + t}$
- $\sqrt{3 - (-t)} - \sqrt{3 + t}$

Expression:

$$\sqrt{1+t} - \sqrt{3+t}$$

Rationalisons:

$$\frac{(\sqrt{1+t} - \sqrt{3+t})(\sqrt{1+t} + \sqrt{3+t})}{\sqrt{1+t} + \sqrt{3+t}} = \frac{(1+t) - (3+t)}{\sqrt{1+t} + \sqrt{3+t}} = \frac{-2}{\sqrt{1+t} + \sqrt{3+t}} \rightarrow 0$$

Réponse: $\boxed{0}$

(123) Calculons $\lim_{x \rightarrow +\infty} \sqrt{x + \sqrt{x - \sqrt{x}}}$

On étudie le comportement de l'expression à l'intérieur.

Soit $f(x) = \sqrt{x + \sqrt{x - \sqrt{x}}}$

Analysons $\sqrt{x - \sqrt{x}}$:

On a $\sqrt{x} \ll x$, donc $x - \sqrt{x} \sim x$, donc $\sqrt{x - \sqrt{x}} \sim \sqrt{x}$

Donc:

$$x + \sqrt{x - \sqrt{x}} \sim x + \sqrt{x}$$

Donc:

$$f(x) \sim \sqrt{x + \sqrt{x}} \sim \sqrt{x} \cdot \sqrt{1 + \frac{1}{\sqrt{x}}} \sim \sqrt{x} \cdot \left(1 + \frac{1}{2\sqrt{x}}\right) \sim \sqrt{x} + \frac{1}{2}$$

Donc $f(x) \rightarrow +\infty$

Réponse: $+\infty$

(124) Calculons $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^2 + x + 5}}{x + 3} - \frac{x + 2}{\sqrt{x^2 + x + 5}} \right)$

On traite chaque terme.

Premier terme : $\frac{\sqrt{x^2 + x + 5}}{x + 3}$

$x \rightarrow -\infty$, donc $x < 0$. On écrit $x = -t$, $t \rightarrow +\infty$

Alors:

- $\sqrt{(-t)^2 + (-t) + 5} - \sqrt{t^2 - t + 5}$
- $x + 3 = -t + 3$

Donc:

Réponse: 0

$$\begin{aligned}
 (125) \text{ Calculons } \lim_{x \rightarrow 4} & \left(\frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2} \right) \\
 &= \lim_{x \rightarrow 4} \frac{(\sqrt{x+5})^2 - (\sqrt{x} + 1)^2}{(\sqrt{x+12})^2 - (\sqrt{x} + 2)^2} \cdot \left(\frac{\sqrt{x+12} + \sqrt{x} + 2}{\sqrt{x+5} + \sqrt{x} + 1} \right) \\
 &= \lim_{x \rightarrow 4} \left(\frac{4 - 2\sqrt{x}}{8 - 4\sqrt{x}} \right) \cdot \left(\frac{\sqrt{x+12} + \sqrt{x} - 2}{\sqrt{x+5} + \sqrt{x} - 1} \right) \\
 &= \lim_{x \rightarrow 4} \frac{1}{2} \left(\frac{4 - 2\sqrt{x}}{4 - 2\sqrt{x}} \right) \cdot \left(\frac{\sqrt{x+12} + \sqrt{x} - 2}{\sqrt{x+5} + \sqrt{x} - 1} \right) \\
 &= \frac{1}{2} \lim_{x \rightarrow 4} \left(\frac{\sqrt{x+12} + \sqrt{x} - 2}{\sqrt{x+5} + \sqrt{x} - 1} \right) \\
 &= \frac{1}{2} \left(\frac{\sqrt{4+12} + \sqrt{4} - 2}{\sqrt{4+5} + \sqrt{4} - 1} \right) = \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 (126) \text{ Calculons } \lim_{x \rightarrow 3^+} & \left(\frac{\sqrt{x^2 - 9} + \sqrt{x} - \sqrt{3}}{\sqrt{x} - 3} \right) \\
 &= \lim_{x \rightarrow 3^+} \frac{\sqrt{x^2 - 9}}{\sqrt{x} - 3} + \lim_{x \rightarrow 3^+} \frac{\sqrt{x} - \sqrt{3}}{\sqrt{x} - 3} \\
 &= \lim_{x \rightarrow 3^+} \frac{\sqrt{x-3} \cdot \sqrt{x+3}}{\sqrt{x} - 3} \\
 &+ \lim_{x \rightarrow 3^+} \frac{(\sqrt{x} - \sqrt{3})(\sqrt{x} + \sqrt{3})}{\sqrt{x} - 3(\sqrt{x} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 3^+} \sqrt{x+3} + \lim_{x \rightarrow 3^+} \frac{(x-3)}{\sqrt{x-3}(\sqrt{x} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 3^+} \sqrt{x+3} + \lim_{x \rightarrow 3^+} \frac{\sqrt{x-3} \cdot \sqrt{x-3}}{\sqrt{x-3}(\sqrt{x} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 3^+} \sqrt{x+3} + \lim_{x \rightarrow 3^+} \frac{\sqrt{x-3}}{\sqrt{x} + \sqrt{3}} \\
 &= \sqrt{6} + \frac{\sqrt{3-3}}{\sqrt{3} + \sqrt{3}} = \sqrt{6}
 \end{aligned}$$

$$\begin{aligned}
 (127) \text{ Calculons } \lim_{x \rightarrow \frac{1}{3}} \left| \frac{x^2 - 6x}{3x - 1} \right| \\
 &= \lim_{x \rightarrow \frac{1}{3}} \left| \frac{x(x-6)}{x\left(x - \frac{1}{3}\right)} \right| = \lim_{x \rightarrow \frac{1}{3}} \left| \frac{x-6}{x - \frac{1}{3}} \right| \\
 &= \lim_{x \rightarrow \frac{1}{3}} |x-6| \times \lim_{x \rightarrow \frac{1}{3}} \frac{1}{\left|x - \frac{1}{3}\right|} \\
 &= \frac{17}{3} \lim_{\substack{t \rightarrow 0 \pm \frac{1}{3} \\ t = x - \frac{1}{3}}} \frac{1}{|t|} = \frac{17}{3} \left(\frac{1}{|0 \pm|} \right) = +\infty
 \end{aligned}$$

$$(128) \text{ Calculons } \lim_{x \rightarrow -\infty} x(x + \sqrt{x^2 + 1})$$

1.

$$\begin{aligned}
 &= -\lim_{x \rightarrow -\infty} x \left(x + \sqrt{x^2 \left(1 + \frac{1}{x^2} \right)} \right) \\
 &= \lim_{x \rightarrow -\infty} x \left(x + \sqrt{x^2} \cdot \sqrt{1 + \frac{1}{x^2}} \right) \\
 &= \lim_{x \rightarrow -\infty} x \left(x + |x| \cdot \sqrt{1 + \frac{1}{x^2}} \right) \\
 &= \lim_{x \rightarrow -\infty} x \left(x - x \sqrt{1 + \frac{1}{x^2}} \right) \\
 &= \lim_{x \rightarrow -\infty} x x^2 \left(1 - \sqrt{1 + \frac{1}{x^2}} \right) \\
 &= \lim_{x \rightarrow -\infty} \frac{x^2 \left(1 - \sqrt{1 + \frac{1}{x^2}} \right) \left(1 + \sqrt{1 + \frac{1}{x^2}} \right)}{\left(1 + \sqrt{1 + \frac{1}{x^2}} \right)} \\
 &= \lim_{x \rightarrow -\infty} \frac{x^2 \left(1x^2 - \left(\sqrt{1 + \frac{1}{x^2}} \right)^2 \right)}{\left(1 + \sqrt{1 + \frac{1}{x^2}} \right)} \\
 &= \lim_{x \rightarrow -\infty} \left(\frac{-1}{1 + \sqrt{1 + \frac{1}{x^2}}} \right) = \frac{-1}{1 + \sqrt{1+0}} = \frac{-1}{2}
 \end{aligned}$$

(129) Calculons $\lim_{x \rightarrow 9} \left(\frac{\sqrt{x} - 3}{\sqrt{x+7} - 4} \right)$

$$\begin{aligned}
 &= \lim_{x \rightarrow 9} \frac{(\sqrt{x} - 3)(\sqrt{x} + 3)(\sqrt{x+7} + 4)}{(\sqrt{x+7} - 4)(\sqrt{x+7} + 4)(\sqrt{x} + 3)} \\
 &= \lim_{x \rightarrow 9} \frac{((\sqrt{x})^2 - 3^2)(\sqrt{x+7} + 4)}{((\sqrt{x+7})^2 - 4^2)(\sqrt{x} + 3)} \\
 &= \lim_{x \rightarrow 9} \frac{(x - 9)(\sqrt{x+7} + 4)}{(x - 9)(\sqrt{x} + 3)} \\
 &= \lim_{x \rightarrow 9} \left(\frac{\sqrt{x+7} + 4}{\sqrt{x} + 3} \right) = \frac{\sqrt{9+7} + 4}{\sqrt{9} + 3} = \frac{4}{3}
 \end{aligned}$$

(130) Calculons $\lim_{x \rightarrow 2} \left(\frac{\sqrt{2x+1} - \sqrt{3+x}}{\sqrt{x^2-1} - \sqrt{2x^2-5}} \right)$

$$\begin{aligned}
 &= \lim_{x \rightarrow 2} \frac{(\sqrt{2x+1} - \sqrt{3+x})(\sqrt{2x+1} + \sqrt{3+x})}{(\sqrt{x^2-1} - \sqrt{2x^2-5})(\sqrt{x^2-1} + \sqrt{2x^2-5})} \\
 &\quad \times \frac{(\sqrt{x^2-1} + \sqrt{2x^2-5})}{(\sqrt{2x+1} + \sqrt{3+x})} \\
 &= \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x^2-1} + \sqrt{2x^2-5})}{-(x-2)(x+2)(\sqrt{2x+1} + \sqrt{3+x})} \\
 &= \lim_{x \rightarrow 2} \frac{-(\sqrt{x^2-1} + \sqrt{2x^2-5})}{(x+2)(\sqrt{2x+1} + \sqrt{3+x})} \\
 &= \frac{-(\sqrt{4-1} + \sqrt{8-5})}{(2+2)(\sqrt{5} + \sqrt{5})} = \frac{-1}{4} \sqrt{\frac{3}{5}}
 \end{aligned}$$

$$\begin{aligned}
(131) \quad & \lim_{x \rightarrow 0^+} \left(\frac{x + \sqrt{x^2 + x}}{\sqrt{x^2 + x + 1} - 1} \right) \\
&= \lim_{x \rightarrow 0^+} \frac{(x + \sqrt{x^2 + x})(\sqrt{x^2 + x + 1} + 1)}{(\sqrt{x^2 + x + 1} - 1)(\sqrt{x^2 + x + 1} + 1)} \\
&= \lim_{x \rightarrow 0^+} \frac{(x + \sqrt{x^2 + x})(\sqrt{x^2 + x + 1} + 1)}{(\sqrt{x^2 + x + 1})^2 - 1^2} \\
&= \lim_{x \rightarrow 0^+} \frac{(x + \sqrt{x^2 + x})(\sqrt{x^2 + x + 1} + 1)}{x^2 + x} \\
&= \lim_{x \rightarrow 0^+} \left(\frac{x + \sqrt{x^2 + x}}{x^2 + x} \right) (\sqrt{x^2 + x + 1} + 1) \\
&= \lim_{x \rightarrow 0^+} \left(\frac{1}{x + 1} + \frac{1}{\sqrt{x^2 + x}} \right) (\sqrt{x^2 + x + 1} + 1) \\
&= \left(\frac{1}{0 + 1} + \frac{1}{0^+} \right) (\sqrt{0^2 + 0 + 1} + 1) \\
&= (+\infty) \times 2 = +\infty
\end{aligned}$$

$$\begin{aligned}
(132) \text{ Calculons } & \lim_{x \rightarrow +\infty} (\sqrt{2x^2 + 1} - \sqrt{x^2 - x - 2}) \\
&= \lim_{x \rightarrow +\infty} (\sqrt{2x^2 + 1} - \sqrt{x^2 - x - 2}) \\
&\quad \times \frac{(\sqrt{2x^2 + 1} + \sqrt{x^2 - x - 2})}{(\sqrt{2x^2 + 1} + \sqrt{x^2 - x - 2})} \\
&= \lim_{x \rightarrow +\infty} \frac{(\sqrt{2x^2 + 1})^2 - (\sqrt{x^2 - x - 2})^2}{\sqrt{x^2 \left(2 + \frac{1}{x^2}\right)} + \sqrt{x^2 \left(1 - \frac{1}{x} - \frac{2}{x^2}\right)}} \\
&= \lim_{x \rightarrow +\infty} \frac{(2x^2 + 1) - (x^2 - x - 2)}{\sqrt{x^2} \times \left(\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x} - \frac{2}{x^2}}\right)} \\
&= \lim_{x \rightarrow +\infty} \frac{x^2 + x + 3}{|x| \times \left(\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x} - \frac{2}{x^2}}\right)} \\
&= \lim_{x \rightarrow +\infty} \frac{x \cdot \left(x + 1 + \frac{3}{x}\right)}{x \cdot \left(\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x} - \frac{2}{x^2}}\right)} \\
&= \lim_{x \rightarrow +\infty} \left(\frac{x + 1 + \frac{3}{x}}{\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x} - \frac{2}{x^2}}} \right) \\
&\quad \frac{+\infty + 1 + 0}{\sqrt{2 + 0} + \sqrt{1 - 0 - 0}} = +\infty
\end{aligned}$$

(133) Calculons $\lim_{x \rightarrow +\infty} \frac{x+1}{\sqrt{1+x^2}}$

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{x+1}{\sqrt{1+x^2}} &= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{1}{x}\right)}{\sqrt{x^2} \cdot \sqrt{\frac{1}{x^2} + 1}} \\
 &= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{1}{x}\right)}{|x| - \sqrt{1 + \frac{1}{x^2}}} \\
 &= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{1}{x}\right)}{x \cdot \sqrt{1 + \frac{1}{x^2}}} \\
 &= \lim_{x \rightarrow +\infty} \frac{\left(1 + \frac{1}{x}\right)}{\sqrt{1 + \frac{1}{x^2}}} \\
 &= \frac{(1 + 0^+)}{\sqrt{1 + 0^+}} = 1 + = 1
 \end{aligned}$$

(134) Calculons $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}}{x^2}$

$$\frac{\sqrt{1+x^2}}{x^2} = \lim_{x \rightarrow 0} \frac{\sqrt{x^2 \left(1 + \frac{1}{x^2}\right)}}{x^2} = \lim_{x \rightarrow 0} \frac{|x| \sqrt{1 + \frac{1}{x^2}}}{x^2} \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{\sqrt{1+x^2}}{x^2} = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{1}{x} \sqrt{1 + \frac{1}{x^2}} = +\infty \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{\sqrt{1+x^2}}{x^2} =$$

$$\lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{-1}{x} \sqrt{1 + \frac{1}{x^2}} = +\infty$$

$$2) \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{x^2+1} \right) = \frac{1}{0^-} - 1 = -\infty$$

$$3) \lim_{x \rightarrow +\infty} \left(\frac{1}{x} + 2 + \sin \frac{1}{x} \right) = 0 + 2 + \sin 0 = 2$$

(135) Calculons $\lim_{x \rightarrow -\infty} (\sqrt{3x^2 - 6x - 1} + 2x - 5)$

$$\begin{aligned}
 & \times \frac{(\sqrt{3x^2 - 6x - 1} - (2x - 5))}{(\sqrt{3x^2 - 6x - 1} - (2x - 5))} \\
 &= \lim_{x \rightarrow -\infty} \frac{(3x^2 - 6x - 1) - (2x - 5)^2}{\sqrt{3x^2 - 6x - 1} - (2x - 5)} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x^2 + 14x - 26}{\sqrt{x^2 \left(3 - \frac{6}{x} - \frac{1}{x^2}\right)} - 2x + 5} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x^2 + 14x - 26}{\sqrt{x^2} \sqrt{3 - \frac{6}{x} - \frac{1}{x^2}} - 2x + 5} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x^2 + 14x - 26}{|x| \sqrt{3 - \frac{6}{x} - \frac{1}{x^2}} - 2x + 5} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x^2 + 14x - 26}{-x \sqrt{3 - \frac{6}{x} - \frac{1}{x^2}} - 2x + 5} \\
 &= \lim_{x \rightarrow -\infty} \frac{-x \left(x - 14 + \frac{26}{x}\right)}{-x \left(\sqrt{3 - \frac{6}{x} - \frac{1}{x^2}} + 2 - \frac{5}{x}\right)} \\
 &= \lim_{x \rightarrow -\infty} \left(\frac{x - 14 + \frac{26}{x}}{\sqrt{3 - \frac{6}{x} - \frac{1}{x^2}} + 2 - \frac{5}{x}} \right) \\
 &= \left(\frac{-\infty - 14 + 0}{\sqrt{3 - 0 - 0} + 2 - 0} \right) = -\infty
 \end{aligned}$$

$$\begin{aligned}
(136) \text{ Calculons } & \lim_{x \rightarrow +\infty} \left(\sqrt{x^3 + 3x^2 + 4} - x^2 + 2 \right) \\
& \times \frac{\left(\sqrt{x^3 + 3x^2 + 4} + (x^2 - 2) \right)}{\left(\sqrt{x^3 + 3x^2 + 4} + (x^2 - 2) \right)} \\
& = \lim_{x \rightarrow +\infty} \frac{\left(\left(\sqrt{x^3 + 3x^2 + 4} \right)^2 - (x^2 - 2)^2 \right)}{\sqrt{x^3 + 3x^2 + 4} + (x^2 - 2)} \\
& = \lim_{x \rightarrow +\infty} \frac{x^3 + 3x^2 + 4 - (x^2 - 2)^2}{\sqrt{x^3 + 3x^2 + 4} + (x^2 - 2)} \\
& = \lim_{x \rightarrow +\infty} \frac{-x^4 + x^3 + 7x^2}{\sqrt{x^4 \left(\frac{1}{x} + \frac{3}{x^2} + \frac{4}{x^4} \right)} + (x^2 - 2)} \\
& = \lim_{x \rightarrow +\infty} \frac{-x^4 + x^3 + 7x^2}{x^2 \sqrt{\frac{1}{x} + \frac{3}{x^2} + \frac{4}{x^4}} + x^2 - 2} \\
& = \lim_{x \rightarrow +\infty} \left(\frac{-x^2 + x + 7}{\sqrt{\frac{1}{x} + \frac{3}{x^2} + \frac{4}{x^4}} + 1 - \frac{2}{x^2}} \right) \\
& = \left(\frac{-\infty}{\sqrt{0 + 0 + 0} + 1 - 0} \right) = -\infty
\end{aligned}$$

(137) Calculons $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - 3x)$

$$\begin{aligned}
 &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 + 1} - 3x)(\sqrt{x^2 + 1} + 3x)}{(\sqrt{x^2 + 1} + 3x)} \\
 &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 + 1} - 3x)(\sqrt{x^2 + 1} + 3x)}{(\sqrt{x^2 + 1} + 3x)} \\
 &= \lim_{x \rightarrow +\infty} \frac{(x^2 + 1) - (3x)^2}{\sqrt{x^2 + 1} + 3x} \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{-8x^2 + 1}{\sqrt{x^2 + 1} + 3x} \right) \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{-8x^2 + 1}{\sqrt{x^2 \left(1 + \frac{1}{x^2}\right)} + 3x} \right) \\
 1. &= \lim_{x \rightarrow +\infty} \left(\frac{-8x^2 + 1}{|x| - \sqrt{\left(1 + \frac{1}{x^2}\right)} + 3x} \right) \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{-8x^2 + 1}{x \cdot \sqrt{\left(1 + \frac{1}{x^2}\right)} + 3x} \right) \\
 &= \lim_{x \rightarrow +\infty} \frac{x \cdot \left(-8x + \frac{1}{x}\right)}{x \cdot \left(\sqrt{\left(1 + \frac{1}{x^2}\right)} + 3\right)} \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{-8x + \frac{1}{x}}{\sqrt{1 + \frac{1}{x^2}} + 3} \right) = -\infty
 \end{aligned}$$

$$\begin{aligned}
(138) \text{ Calculons } \lim_{x \rightarrow +\infty} \left(x \sqrt{\frac{x}{x-1}} - x - 1 \right) \\
&= -1 + \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{x^3}{x-1}} - x \right) \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{\frac{x^3}{x-1}} - x \right) \left(\sqrt{\frac{x^3}{x-1}} + x \right)}{\left(\sqrt{\frac{x^3}{x-1}} + x \right)} \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{\frac{x^3}{x-1}} \right)^2 - x^2}{\left(\sqrt{\frac{x^3}{x-1}} + x \right)} \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{\frac{x^3 - x^3 + x^2}{x-1}}{\left(\sqrt{\frac{x^3}{x-1}} + x \right)} \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{\left(\frac{x^2}{x-1} \right)}{\left(\sqrt{\frac{x^3}{x-1}} + x \right)} \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{x \left(\frac{x}{x-1} \right)}{x \left(\frac{1}{x} \sqrt{\frac{x^3}{x-1}} + 1 \right)} \\
&= -1 + \lim_{x \rightarrow +\infty} \frac{\left(\frac{x}{x-1} \right)}{\left(\sqrt{\frac{x^3}{x^3 - x^2}} + 1 \right)} \\
&= -1 + \frac{1}{\sqrt{1} + 1} = \frac{-1}{2}
\end{aligned}$$

Correction limites de fonctions trigonométriques

pourquoi :

$$\lim_{x \rightarrow +\infty} \frac{\sin x}{x^2} = \lim_{x \rightarrow +\infty} \frac{\cos x}{x^2} = 0 ?$$

$$\text{Car } -1 \leq \sin x \leq 1 \text{ et } -1 \leq \cos x \leq 1$$

$$\Rightarrow \frac{-1}{x^2} \leq \frac{\sin x}{x^2} \leq \frac{1}{x^2} \text{ et } \frac{-1}{x^2} \leq \frac{\cos x}{x^2} \leq \frac{1}{x^2}$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \left(\frac{\sin x}{x^2} \right) = \lim_{x \rightarrow +\infty} \left(\frac{\cos x}{x^2} \right) = 0$$

$$3) \lim_{x \rightarrow +\infty} \left(\frac{2x + \cos x}{3x + \sin x} \right) = \lim_{x \rightarrow +\infty} \frac{x \left(2 + \frac{\cos x}{x} \right)}{x \left(3 + \frac{\sin x}{x} \right)}$$

$$= \lim_{x \rightarrow +\infty} \left(\frac{2 + \frac{\cos x}{x}}{3 + \frac{\sin x}{x}} \right) = \frac{2 + 0}{3 + 0} = \frac{2}{3}$$

$$\text{Pourquoi } \lim_{x \rightarrow +\infty} \left(\frac{\cos x}{x} \right) = \lim_{x \rightarrow +\infty} \left(\frac{\sin x}{x} \right) = 0$$

$$1) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

Rappel : $\tan x \sim x$ quand $x \rightarrow 0$. Donc $\frac{\tan x}{x} \rightarrow 1$.

$$2) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Limite Classique : $\sin x \sim x$, donc $\frac{\sin x}{x} \rightarrow 1$.

$$3) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

DL de $\cos x$ à l'ordre 2: $\cos x = 1 - \frac{x^2}{2} + o(x^2)$.

Donc $1 - \cos x = \frac{x^2}{2} + o(x^2)$.

En divisant par x^2 on obtient $\frac{1}{2}$.

$$4) \lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = 3$$

$$\sin(3x) \sim 3x, \text{ donc } \frac{\sin(3x)}{x} \rightarrow 3.$$

$$5) \lim_{x \rightarrow 0} \frac{\tan(5x)}{x} = 5$$

$$\tan(5x) \sim 5x, \text{ donc } \frac{\tan(5x)}{x} \rightarrow 5.$$

$$6) \lim_{x \rightarrow 0} \frac{\sin x}{\tan x} = 1$$

$$\frac{\sin x}{\tan x} = \cos x. \text{ Or } \cos x \rightarrow 1 \text{ quand } x \rightarrow 0. \text{ Donc limite} = 1$$

$$7) \lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2} = 2$$

On utilise $1 - \cos(2x) = 2\sin^2 x$. Et $\sin x \sim x$ donc $2\sin^2 x \sim 2x^2$.
Donc le quotient tend vers 2.

$$8) \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} = 1$$

$\sin x \sim x \Rightarrow \sin^2 x \sim x^2$. Le quotient vaut 1.

$$9) \lim_{x \rightarrow 0} \frac{\sin(7x)}{\sin(2x)} = \frac{7}{2}$$

$\sin(7x) \sim 7x$ et $\sin(2x) \sim 2x$. Donc $\frac{7x}{2x} = \frac{7}{2}$.

$$10) \lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)} = \frac{3}{2}$$

$\tan(3x) \sim 3x$, $\sin(2x) \sim 2x$. Donc $\frac{3x}{2x} = \frac{3}{2}$.

$$11) \lim_{x \rightarrow 0} \frac{\sin(3x) - 3\sin x}{x^3} = -4$$

On prend les DL de $\sin$ à l'ordre 3 :

$$\sin t = t - \frac{t^3}{6} + o(t^3)$$

- $\sin(3x) = 3x - \frac{(3x)^3}{6} + o(x^3) = 3x - \frac{27x^3}{6} + o(x^3) = 3x - \frac{9}{2}x^3 + o(x^3).$
- $\sin x = x - \frac{x^3}{6} + o(x^3).$

Donc

$$\sin(3x) - 3\sin x = \left(3x - \frac{9}{2}x^3\right) - 3\left(x - \frac{x^3}{6}\right) + o(x^3) = -4x^3 + o(x^3).$$

Divisant par x^3 donne -4 .

$$12) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \frac{1}{2}$$

On utilise les DL à l'ordre 3 :

- $\tan x - x + \frac{x^3}{3} + o(x^3)$.
- $\sin x - x + \frac{x^3}{6} + o(x^3)$.

$$\text{Donc } \tan x - \sin x - \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + o(x^3) - \left(\frac{1}{3} + \frac{1}{6}\right)x^3 + o(x^3) - \frac{1}{2}x^3 + o(x^3).$$

Divisé par $x^3 \rightarrow \frac{1}{2}$.

$$13) \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3} = \frac{1}{3}$$

DL à l'ordre 3:

- $\sin x - x + \frac{x^3}{6} + o(x^3)$.
- $\cos x - 1 - \frac{x^2}{2} + o(x^2) \rightarrow x \cos x - x - \frac{x^3}{2} + o(x^3)$.

Donc

$$\sin x - x \cos x - \left(x - \frac{x^3}{6}\right) - \left(x - \frac{x^3}{2}\right) + o(x^3) - \left(-\frac{1}{6} + \frac{1}{2}\right)x^3 + o(x^3) - \frac{1}{3}x^3 + o(x^3).$$

Divisé par x^3 donne $\frac{1}{3}$.

$$14) \lim_{x \rightarrow 0} \frac{\cos(2x) - e^{-x^2}}{x^4} = -\infty$$

Calcul des DL jusqu'à l'ordre 4:

- $\cos(2x) - 1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} + o(x^4) - 1 - 2x^2 + \frac{2}{3}x^4 + o(x^4)$.
- $e^{-x^2} - 1 - x^2 + \frac{x^4}{2} + o(x^4)$.

La différence:

$$\begin{aligned} \cos(2x) - e^{-x^2} - \left(1 - 2x^2 + \frac{2}{3}x^4\right) - \left(1 - x^2 + \frac{1}{2}x^4\right) + o(x^4) - -x^2 + \left(\frac{2}{3} - \frac{1}{2}\right)x^4 + o(x^4) \\ - -x^2 + \frac{1}{6}x^4 + o(x^4) \end{aligned}$$

Donc le numérateur vaut $-x^2 + \frac{1}{6}x^4 + o(x^4)$. En divisant par x^4 on obtient

$$\frac{-x^2 + \frac{1}{6}x^4 + o(x^4)}{x^4} = -\frac{1}{x^2} + \frac{1}{6} + o(1) \rightarrow -\infty.$$

Remarque : si le dénominateur avait été x^2 au lieu de x^4 , la limite aurait été -1.

$$15) \lim_{x \rightarrow 0} \frac{\arctan(\sin x) - \sin(\arctan x)}{x^3} = 0$$

Il faut comparer les DL jusqu'à l'ordre x^3 .
Rappels de DL (petites notations) :

- $\sin x - x - \frac{x^3}{6} + o(x^3)$.
- $\arctan y - y - \frac{y^3}{3} + o(y^3)$.

Calculons chaque terme:

$$1. \arctan(\sin x): \text{ posons } s = \sin x - x - \frac{x^3}{6} + o(x^3).$$

$$\text{Alors } \arctan(s) - s - \frac{s^3}{3} + o(x^3).$$

$$\text{Or } s^3 = (x + o(x))^3 = x^3 + o(x^3). \text{ Donc}$$

$$\arctan(\sin x) - \left(x - \frac{x^3}{6}\right) - \frac{x^3}{3} + o(x^3) = x - \left(\frac{1}{6} + \frac{1}{3}\right)x^3 + o(x^3) = x - \frac{1}{2}x^3 + o(x^3).$$

$$2. \sin(\arctan x): \text{ posons } a = \arctan x - x - \frac{x^3}{3} + o(x^3).$$

$$\text{Alors } \sin(a) - a - \frac{a^3}{6} + o(x^3).$$

$$\text{Comme } a^3 = x^3 + o(x^3),$$

$$\sin(\arctan x) - \left(x - \frac{x^3}{3}\right) - \frac{x^3}{6} + o(x^3) = x - \left(\frac{1}{3} + \frac{1}{6}\right)x^3 + o(x^3) = x - \frac{1}{2}x^3 + o(x^3).$$

Les deux expressions coïncident jusqu'à l'ordre x^3 . Donc

$$\arctan(\sin x) - \sin(\arctan x) = o(x^3),$$

et en divisant par x^3 on obtient 0.

$$16) \lim_{x \rightarrow 0} \frac{\sin(\sqrt{x}) - (\sqrt{x})^3}{x^{3/2}} \text{ (chgmt de variable } t = \sqrt{x} \text{)}$$

Posons $t = \sqrt{x}$. Alors $x = t^2$ et $x^{3/2} = t^3$.
L'expression devient

$$\frac{\sin t - t^3}{t^3}$$

DL: $\sin t - t - \frac{t^3}{6} + o(t^3)$.

Donc $\sin t - t^3 - t - \frac{t^3}{6} - t^3 + o(t^3) - t - \frac{7}{6}t^3 + o(t^3)$.

Divisant par t^3 :

$$\frac{\sin t - t^3}{t^3} - \frac{t}{t^3} - \frac{7}{6} + o(1) - \frac{1}{t^2} - \frac{7}{6} + o(1).$$

Comme $t \rightarrow 0^+$, $\frac{1}{t^2} \rightarrow +\infty$.

La limite diverge: $+\infty$.

17) $\lim_{x \rightarrow 0} \frac{\tan(ax) - \sin(bx)}{x^3}, a, b > 0$

DL à l'ordre 3 :

$$\tan(ax) - ax + \frac{(ax)^3}{3} + o(x^3), \sin(bx) - bx - \frac{(bx)^3}{6} + o(x^3)$$

Donc

$$\tan(ax) - \sin(bx) - (a-b)x + \left(\frac{a^3}{3} + \frac{b^3}{6}\right)x^3 + o(x^3)$$

Divisant par x^3 :

$$\frac{\tan(ax) - \sin(bx)}{x^3} - \frac{a-b}{x^2} + \frac{a^3}{3} + \frac{b^3}{6} + o(1)$$

- Si $a \neq b$, le terme $\frac{a-b}{x^2}$ diverge : la limite est $+\infty$ si $a > b$ et $-\infty$ si $a < b$.
- Si $a = b$, le terme en $1/x^2$ s'annule et la limite vaut

$$\frac{a^3}{3} + \frac{a^3}{6} - \frac{a^3}{2}$$

Donc :
$$\begin{cases} +\infty & \text{si } a > b, \\ -\infty & \text{si } a < b, \\ \frac{a^3}{2} & \text{si } a = b. \end{cases}$$

18) $\lim_{x \rightarrow 0} \frac{\ln(\cos x) + \frac{x^2}{2}}{x^4}$

DLs:

$$\cos x - 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

Posons $u = \cos x - 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$.

Alors $\ln(\cos x) - \ln(1+u) - u - \frac{u^2}{2} + o(x^4)$. Calcul :

$$u^2 - \left(-\frac{x^2}{2}\right)^2 + o(x^4) - \frac{x^4}{4} + o(x^4)$$

Donc

$$\ln(\cos x) - \left(-\frac{x^2}{2} + \frac{x^4}{24}\right) - \frac{1}{2} \cdot \frac{x^4}{4} + o(x^4) - \frac{x^2}{2} - \frac{x^4}{12} + o(x^4).$$

Alors $\ln(\cos x) + \frac{x^2}{2} = -\frac{x^4}{12} + o(x^4)$.

Divisant par x^4 donne $-\frac{1}{12}$.

$$-\frac{1}{12}$$

$$19) \lim_{x \rightarrow 0} \frac{\sin(x + \sin x) - 2\sin x}{x^3}$$

DL: $\sin x - x - \frac{x^3}{6} + o(x^3)$. Donc

$$x + \sin x - 2x - \frac{x^3}{6} + o(x^3)$$

Alors

$$\sin(x + \sin x) - \sin\left(2x - \frac{x^3}{6} + o(x^3)\right) - \sin(2x) - \frac{x^3}{6} \cos(2x) + o(x^3).$$

Développons $\sin(2x) - 2x - \frac{(2x)^3}{6} + o(x^3) - 2x - \frac{8}{6}x^3 + o(x^3) - 2x - \frac{4}{3}x^3 + o(x^3)$, et $\cos(2x) - 1 + o(x^2)$.
Donc

$$\sin(x + \sin x) - 2x - \frac{4}{3}x^3 - \frac{x^3}{6} + o(x^3) - 2x - \frac{9}{6}x^3 + o(x^3) - 2x - \frac{3}{2}x^3 + o(x^3).$$

Maintenant $2\sin x - 2x - \frac{2x^3}{6} + o(x^3) - 2x - \frac{x^3}{3} + o(x^3)$.

Donc la différence :

$$\sin(x + \sin x) - 2\sin x - \left(2x - \frac{3}{2}x^3\right) - \left(2x - \frac{1}{3}x^3\right) + o(x^3) - \frac{7}{6}x^3 + o(x^3)$$

Divisé par x^3 donne $-\frac{7}{6}$.

$$-\frac{7}{6}$$

$$20) \lim_{x \rightarrow 0} \frac{\sin x \sin(2x) \sin(3x)}{x^3}$$

Pour $x \rightarrow 0$, $\sin(kx) \sim kx$. Donc produit $\sim x \cdot 2x \cdot 3x = 6x^3$.
Divisant par x^3 donne 6.
6.

$$21) \lim_{x \rightarrow 0} \frac{\arcsin(\tan x) - \tan(\arcsin x)}{x^3}$$

On calcule les DL jusqu'à l'ordre x^3 .
DLs de base:

$$\tan x = x + \frac{x^3}{3} + o(x^3), \arcsin y = y + \frac{y^3}{6} + o(y^3)$$

Donc

$$\arcsin(\tan x) = \tan x + \frac{(\tan x)^3}{6} + o(x^3) = \left(x + \frac{x^3}{3}\right) + \frac{x^3}{6} + o(x^3) = x + \frac{x^3}{2} + o(x^3).$$

De même,

$$\tan(\arcsin x) = \arcsin x + \frac{(\arcsin x)^3}{3} + o(x^3) = \left(x + \frac{x^3}{6}\right) + \frac{x^3}{3} + o(x^3) = x + \frac{x^3}{2} + o(x^3).$$

Les deux développements coïncident à l'ordre x^3 , donc la différence est $o(x^3)$.

Après division par x^3 on obtient 0.

0.

$$22) \lim_{x \rightarrow \frac{\pi}{2}} \left(\tan x - \frac{1}{\cos x} \right) \sin x$$

Remarquons d'abord que $\tan x = \frac{\sin x}{\cos x} = \frac{\sin x}{\cos x} - \frac{1}{\cos x} + \frac{\sin x - 1}{\cos x}$.

Donc l'expression vaut $\frac{\sin x(\sin x - 1)}{\cos x}$.

Posons $h = \frac{\pi}{2} - x$, alors $h \rightarrow 0^+$. On a

$$\sin x = \sin\left(\frac{\pi}{2} - h\right) = \cos h = 1 - \frac{h^2}{2} + o(h^2), \cos x = \sin h = h + o(h)$$

Donc

$$\frac{\sin x(\sin x - 1)}{\cos x} \sim \frac{(1) \left(-\frac{h^2}{2}\right)}{h} = -\frac{h}{2} \rightarrow 0.$$

0.

$$23) \lim_{x \rightarrow 0} \frac{\sin(x^2) - x \sin x}{x^3}$$

DL:

$$\sin(x^2) - x^2 + o(x^3) \left(\text{en fait } x^2 - \frac{x^5}{6} + \dots \right), \sin x - x - \frac{x^3}{6} + o(x^3)$$

Donc $x \sin x - x^2 - \frac{x^4}{6} + o(x^4)$.

Ainsi

$$\sin(x^2) - x \sin x - (x^2) - \left(x^2 - \frac{x^4}{6} \right) + o(x^4) - \frac{x^4}{6} + o(x^4).$$

Divisant par x^3 donne $\frac{x}{6} + o(x) \rightarrow 0$.

o.

$$24) \lim_{x \rightarrow 0} \frac{\cos x - \cos(\sqrt{x})}{x} \text{ (changert = } \sqrt{x} \text{)}$$

Posons $t = \sqrt{x}$ (donc $x = t^2, t \rightarrow 0^+$). L'expression devient

$$\frac{\cos(t^2) - \cos t}{t^2}$$

DLs:

$$\cos(t^2) - 1 - \frac{t^4}{2} + o(t^4), \cos t - 1 - \frac{t^2}{2} + \frac{t^4}{24} + o(t^4)$$

La différence:

$$\cos(t^2) - \cos t - \left(1 - \frac{t^4}{2} \right) - \left(1 - \frac{t^2}{2} + \frac{t^4}{24} \right) + o(t^4) - \frac{t^2}{2} - \left(\frac{1}{2} + \frac{1}{24} \right) t^4 + o(t^4)$$

$$\text{Donc } \frac{\cos(t^2) - \cos t}{t^2} - \frac{1}{2} - \left(\frac{1}{2} + \frac{1}{24} \right) t^2 + o(t^2) \rightarrow \frac{1}{2}$$

$$\frac{1}{2}$$

$$25) \lim_{x \rightarrow 0} \frac{\tan(2x) - 2\tan x}{x^3}$$

(hypothese: l'expression est $\tan(2x) - 2\tan x$ au numérateur)

DLs à l'ordre 3 :

$$\tan(kx) = kx + \frac{(kx)^3}{3} + o(x^3).$$

Donc

$$\tan(2x) = 2x + \frac{8x^3}{3} + o(x^3), 2\tan x = 2\left(x + \frac{x^3}{3}\right) + o(x^3) = 2x + \frac{2x^3}{3} + o(x^3).$$

Difference:

$$\tan(2x) - 2\tan x = \left(\frac{8}{3} - \frac{2}{3}\right)x^3 + o(x^3) = 2x^3 + o(x^3).$$

Divisant par x^3 donne 2.

$$26) \lim_{x \rightarrow 0} \frac{(\sin(ax))^3 - a \sin x}{x^3}$$

(hypothèse : $(\sin(ax))^3 - a \sin x$ avec $a \in \mathbb{R}$ fixé

DLs:

$$\sin(ax) = ax - \frac{a^3 x^3}{6} + o(x^3) \Rightarrow (\sin(ax))^3 = (ax)^3 + o(x^3) = a^3 x^3 + o(x^8)$$

Et

$$a \sin x = a \left(x - \frac{x^3}{6} \right) + o(x^3) = ax - \frac{ax^3}{6} + o(x^3).$$

Donc le numérateur vaut

$$a^3 x^3 - ax + \frac{ax^3}{6} + o(x^3) = -ax + \left(a^3 + \frac{a}{6}\right)x^3 + o(x^3)$$

Le terme dominant est $-ax$. Donc, après division par x^3 .

$$\frac{(\sin(ax))^3 - a \sin x}{x^3} = \frac{-a}{x^2} + \left(a^3 + \frac{a}{6}\right) + o(1).$$

Ainsi :

- si $a \neq 0$ la limite diverge (sens dépend du signe de $-a$):
 $\rightarrow -\infty$ si $a > 0$, $\rightarrow +\infty$ si $a < 0$.

- si $a = 0$, le numérateur est 0 et la limite est 0.
 ($\rightarrow$ Si tu voulais une autre écriture pour l'énoncé, dis-moi.)

- $\frac{\sin x - x + \frac{x^3}{6}}{x^5}$

DL de $\sin x$ jusqu'à x^5 :

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5).$$

Donc le numérateur :

$$\sin x - x + \frac{x^3}{6} = \frac{x^5}{120} + o(x^5).$$

Divisé par x^5 donne $\frac{1}{120}$.

$$27) \lim_{x \rightarrow 10} \frac{\cos(x \sin x) - 1 + \frac{x^2}{2}}{x^4}$$

Calculons $u = x \sin x$. Avec $\sin x = x - \frac{x^3}{6} + o(x^3)$.

$$u = x \sin x = x^2 - \frac{x^4}{6} + o(x^4).$$

Pour $\cos u$ on a $\cos u = 1 - \frac{u^2}{2} + o(u^2)$. Or

$$u^2 = \left(x^2 - \frac{x^4}{6}\right)^2 = x^4 + o(x^4)$$

Donc

$$\cos(x \sin x) = 1 - \frac{x^4}{2} + o(x^4)$$

Le numérateur :

$$\cos(x \sin x) - 1 + \frac{x^2}{2} = \frac{x^2}{2} - \frac{x^4}{2} + o(x^4)$$

Divisant par x^4 :

$$\frac{1}{2} \cdot \frac{1}{x^2} - \frac{1}{2} + o(1) \rightarrow +\infty$$

Donc $+\infty$.

$$28) \lim_{x \rightarrow 0} \frac{\tan(\sqrt{x}) - \sqrt{x}}{x^{3/2}} \text{ (chgt de variable } t = \sqrt{x} \text{)}$$

Posons $t = \sqrt{x}$ donne $x = t^2$ et $x^{3/2} = t^3$. Alors

$$\frac{\tan(\sqrt{x}) - \sqrt{x}}{x^{3/2}} = \frac{\tan t - t}{t^3}$$

DL $\tan t = t + \frac{t^3}{3} + o(t^3)$ donne $\tan t - t = \frac{t^3}{3} + o(t^3)$.
Donc limite $= \frac{1}{3}$.

(29) $\lim_{x \rightarrow 0} \frac{\tan(\sqrt{x}) - \sqrt{x}}{x^{3/2}}$ (chgt de variable $t = \sqrt{x}$)

Posons $t = \sqrt{x}$ donne $x = t^2$ et $x^{3/2} = t^3$. Alors

$$\frac{\tan(\sqrt{x}) - \sqrt{x}}{x^{3/2}} = \frac{\tan t - t}{t^3}.$$

DL $\tan t = t + \frac{t^3}{3} + o(t^3)$ donne $\tan t - t = \frac{t^3}{3} + o(t^3)$.
Donc limite $= \frac{1}{3}$.

30) $\lim_{x \rightarrow 0} \frac{\arctan(1 - \cos x) - \frac{x^2}{2}}{x^4}$

On utilise $1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + o(x^4)$. Pour y petit,

$\arctan y = y - \frac{y^3}{3} + o(y^3)$; ici $y = O(x^2) \Rightarrow y^3 = O(x^6)$ négligeable.
Donc

$$\arctan(1 - \cos x) = \frac{x^2}{2} - \frac{x^4}{24} + o(x^4).$$

Après soustraction:

$$\arctan(1 - \cos x) - \frac{x^2}{2} = -\frac{x^4}{24} + o(x^4)$$

Divisé par x^4 donne $-\frac{1}{24}$.

31) $\lim_{x \rightarrow 0} \cos\left(\frac{\sin(\pi x)}{x}\right)$

Pour $x \rightarrow 0$, $\sin(\pi x) \sim \pi x$. Donc $\frac{\sin(\pi x)}{x} \rightarrow \pi$. D'aú

$$\cos\left(\frac{\sin(\pi x)}{x}\right) \rightarrow \cos(\pi) = -1$$

32) $\lim_{x \rightarrow 0} \cos(x^2 \cos(1/x) + x)$

Quand $x \rightarrow 0$ on a $x^2 \cos(1/x) \rightarrow 0$ et $x \rightarrow 0$. Donc l'argument tend vers 0 et la cosinus tend vers $\cos 0 = 1$.

1.

33) $\lim_{x \rightarrow 0^+} \tan(\sqrt{x} \sin(1/x))$

Ici $\sqrt{x} \rightarrow 0$ et $\sin(1/x)$ est bornée ($\in [-1, 1]$), donc l'argument $\sqrt{x} \sin(1/x) \rightarrow 0$. Pour $y \rightarrow 0$, $\tan y \sim y$. Donc la limite est

$$\lim_{x \rightarrow 0^+} \sqrt{x} \sin(1/x) = 0.$$

Donc 0.

34) $\lim_{x \rightarrow 0} \sin\left(\frac{\tan^2 x - \tan x}{2x}\right)$

On étudie l'intérieur :

$$\tan x = x + \frac{x^3}{3} + o(x^3), \tan^2 x = (\tan x)^2 = x^2 + \frac{2}{3}x^4 + o(x^4).$$

Donc le numérateur

$$\tan^2 x - \tan x = -x + x^2 - \frac{x^3}{3} + o(x^3)$$

(Dominante : $-x$.) D'où

$$\frac{\tan^2 x - \tan x}{2x} = -\frac{1}{2} + \frac{x}{2} - \frac{x^2}{6} + o(x^2) \quad x \rightarrow 0 \quad -\frac{1}{2}$$

Enfin $\sin$ est continue, donc la limite vaut $\sin\left(-\frac{1}{2}\right) = -\sin\left(\frac{1}{2}\right)$.

$$-\sin\left(\frac{1}{2}\right).$$

$$35) \lim_{x \rightarrow 0^-} \left(\frac{\cos x - \sqrt{1 + \sin x}}{x} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{\cos x - \sqrt{1 + \sin x}}{x} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\frac{\cos x - \sqrt{1 + \sin x}}{x} \right) \left(\frac{\cos x + \sqrt{1 + \sin x}}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\frac{\cos^2(x) - (1 + \sin x)}{x} \right) \left(\frac{1}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\frac{1 - \sin^2 x - (1 + \sin x)}{x} \right) \left(\frac{1}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\frac{-\sin^2 x - \sin x}{x} \right) \left(\frac{1}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\frac{\sin^2 x}{x} + \frac{\sin x}{x} \right) \left(\frac{-1}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= \lim_{x \rightarrow 0^-} \left(\sin x \cdot \left(\frac{\sin x}{x} \right) + \frac{\sin x}{x} \right) \left(\frac{-1}{\cos x + \sqrt{1 + \sin x}} \right) \\
 &= (\sin 0 \times (1) + 1) \left(\frac{-1}{\cos 0 + \sqrt{1 + \sin 0}} \right) = \frac{-1}{2}
 \end{aligned}$$

$$36) \lim_{x \rightarrow 1^\pm} \left(\frac{1 - \cos(2\pi x)}{x(x-1)} \right)$$

$$\begin{aligned}
 \lim_{x \rightarrow 1^\pm} k(x) &= \lim_{x \rightarrow 1^\pm} \left(\frac{1 - \cos(2\pi x)}{x(x-1)} \right) \\
 &= \lim_{\substack{t \rightarrow 0^\pm \\ t=x-1}} \left(\frac{1 - \cos(2\pi t + 2\pi)}{t(t+1)} \right) \\
 &= \lim_{t \rightarrow 0^\pm} \left(\frac{1 - \cos(2\pi t)}{t(t+1)} \right) \\
 &= \lim_{t \rightarrow 0^\pm} \left(\frac{1 - \cos(2\pi t)}{(2\pi t)^2} \right) \left(\frac{(2\pi t)^2}{t(t+1)} \right) \\
 &= \lim_{t \rightarrow 0^\pm} \left(\frac{1 - \cos(2\pi t)}{(2\pi t)^2} \right) \left(\frac{4\pi^2}{1 + \frac{1}{t}} \right) \\
 &= \left(\frac{1}{2} \right) \left(\frac{4\pi^2}{1 + (\pm\infty)} \right) = 0^\pm = 0
 \end{aligned}$$

$$37) \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x+2)}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x+2)} \\
 &= \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)} \cdot \frac{1}{(x+2)} \\
 &= \left[\lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)} \right] \left[\lim_{x \rightarrow 1} \frac{1}{(x+2)} \right] \\
 &= \left[\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right] \left[\lim_{x \rightarrow 1} \frac{1}{(x+2)} \right] \quad \text{change of variable } \theta = x - 1 \\
 &= (1) \left(\frac{1}{3} \right) \\
 &= \frac{1}{3}
 \end{aligned}$$

$$36) \lim_{x \rightarrow 5} \frac{2 \tan(x-5)}{(x-5)(x-1)}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 5} \frac{2 \tan(x-5)}{(x-5)(x-1)} \\
 &= \lim_{x \rightarrow 5} \frac{2 \frac{\sin(x-5)}{\cos(x-5)}}{(x-5)(x-1)} = \lim_{x \rightarrow 5} \frac{2 \sin(x-5)}{(x-5)(x-1) \cos(x-5)} \\
 &= \lim_{x \rightarrow 5} 2 \frac{\sin(x-5)}{(x-5)} \cdot \frac{1}{(x-1) \cos(x-5)} \\
 &= 2 \left[\lim_{x \rightarrow 5} \frac{\sin(x-5)}{(x-5)} \right] \left[\lim_{x \rightarrow 5} \frac{1}{(x-1) \cos(x-5)} \right] \\
 &= 2 \left[\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right] \left[\lim_{x \rightarrow 5} \frac{1}{(x-1) \cos(x-5)} \right] \quad \text{change of variable } \theta = x - 5 \\
 &= 2(1) \frac{1}{(4) \cos(0)} \\
 &= 2(1) \frac{1}{(4)(1)} = \frac{1}{2}
 \end{aligned}$$

$$37) \lim_{x \rightarrow 0} \left\{ \frac{\sin 3x}{3x} \frac{3x}{5x^3 - 4x} \right\}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \left\{ \frac{\sin 3x}{3x} \frac{3x}{5x^3 - 4x} \right\} \\ &= \left\{ \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \right\} \left\{ \lim_{x \rightarrow 0} \frac{3x}{5x^3 - 4x} \right\} \\ &= \left\{ \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right\} \left\{ \lim_{x \rightarrow 0} \frac{3x}{5x^3 - 4x} \right\} \quad \text{change of variable } \theta = 3x \\ &= (1) \left\{ \lim_{x \rightarrow 0} \frac{3x}{5x^3 - 4x} \right\} \\ &= \left\{ \lim_{x \rightarrow 0} \frac{3}{5x^2 - 4} \right\} \quad \text{canceling out } x \\ &= -\frac{3}{4} \end{aligned}$$

$$38) \lim_{x \rightarrow +\infty} \left(\frac{\cos x}{x^2 + 1} \right)$$

On a; $-1 \leq \cos x \leq 1$

$$\Rightarrow \frac{-1}{x^2 + 1} \leq \frac{\cos x}{x^2 + 1} \leq \frac{1}{x^2 + 1}$$

$\underbrace{\quad}_{x^2+1}$
terel Vers 0

$$\Rightarrow \lim_{x \rightarrow +\infty} \left(\frac{\cos x}{x^2 + 1} \right) = 0$$

$$39) \lim_{x \rightarrow +\infty} \left(\frac{x + \sin x}{x^2 + \cos x} \right)$$

$$\lim_{x \rightarrow +\infty} \left(\frac{x + \sin x}{x^2 + \cos x} \right) = \lim_{x \rightarrow +\infty} \frac{x^2 \left(\frac{1}{x} + \frac{\sin x}{x^2} \right)}{x^2 \left(1 + \frac{\cos x}{x^2} \right)}$$

$$= \lim_{x \rightarrow +\infty} \left(\frac{\frac{1}{x} + \frac{\sin x}{x^2}}{1 + \frac{\cos x}{x^2}} \right) = \frac{0^+ + 0}{1 + 0} = 0^+ = 0$$

$$\lim_{x \rightarrow +\infty} \frac{|\sin x|}{\sqrt{1 + x^2}} = 0$$

car on a: $|\sin x| \leq 1$

$$\Rightarrow \frac{|\sin x|}{\sqrt{1 + x^2}} \leq \frac{1}{\sqrt{1 + x^2}}$$

$$\Rightarrow \int_{\text{quand}}^0 \leq \frac{|\sin x|}{\sqrt{1+x^2}} \leq \frac{1}{\underbrace{\sqrt{1+x^2}}_{\text{tent ber2 quand } x \rightarrow +\infty}}$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \frac{|\sin x|}{\sqrt{1+x^2}} = 0$$

$$40) \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x} \right)$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(\sqrt{1+\sin x})^2 - (\sqrt{1-\sin x})^2}{\tan x \cdot (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(1+\sin x) - (1-\sin x)}{\tan x \cdot (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2\sin x}{\frac{\sin x}{\cos x} \cdot (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2\cos x}{\sqrt{1+\sin x} + \sqrt{1-\sin x}} \right) \\ &= \left(\frac{2\cos 0}{\sqrt{1+\sin 0} + \sqrt{1-\sin 0}} \right) = 1 \end{aligned}$$

$$41) \lim_{x \rightarrow 0} \left(\frac{\tan x - \sin x}{x^3} \right)$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \left(\frac{\tan x - \sin x}{x^3} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin x}{\cos x} - \sin x}{x \cdot x^2} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \left(\frac{\frac{1}{\cos x} - 1}{x^2} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \left(\frac{1 - \cos x}{x^2} \right) \left(\frac{1}{\cos x} \right) \\ &= 1 \times \frac{1}{2} \times \frac{1}{1} = \frac{1}{2} \end{aligned}$$

$$42) \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\cos x - \sin x}{1 - \sqrt{2} \cos x} \right)$$

$$\begin{aligned} & \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\cos x - \sin x}{1 - \sqrt{2} \cos x} \right) \\ &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\cos x - \sin x)(\cos x + \sin x)}{(1 - \sqrt{2} \cos x)(\cos x + \sin x)} \\ &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos^2 x - \sin^2 x}{(1 - \sqrt{2} \cos x)(\cos x + \sin x)} \\ &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \cos x^2 - 1}{(1 - \sqrt{2} \cos x)(\cos x + \sin x)} \\ &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{-(1 - \sqrt{2} \cos x)(1 + \sqrt{2} \cos x)}{(1 - \sqrt{2} \cos x)(\cos x + \sin x)} \\ &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{-(1 + \sqrt{2} \cos x)}{(\cos x + \sin x)} = \frac{-(1 + \sqrt{2} \cos \frac{\pi}{4})}{(\cos \frac{\pi}{4} + \sin \frac{\pi}{4})} \\ &= \frac{-\left(1 + \sqrt{2} \times \frac{\sqrt{2}}{2}\right)}{\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right)} = -\sqrt{2} \end{aligned}$$

$$43) \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\cos x - \sin x}{1 - \sqrt{2} \cos x} \right)$$

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \left(\frac{\sqrt{\cos x} - \cos x}{\sin(2x) \cdot \tan(3x)} \right) \\
 &= \lim_{x \rightarrow 0} \frac{(\sqrt{\cos x} - \cos x)(\sqrt{\cos x} + \cos x)}{\sin(2x) \cdot \tan(3x) \cdot (\sqrt{\cos x} + \cos x)} \\
 &= \lim_{x \rightarrow 0} \frac{\cos x - \cos^2 x}{\sin(2x) \cdot \tan(3x) \cdot (\sqrt{\cos x} + \cos x)} \\
 &= \lim_{x \rightarrow 0} \frac{(1 - \cos x) \cos x}{\sin(2x) \cdot \tan(3x) \cdot (\sqrt{\cos x} + \cos x)} \\
 &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{\sin(2x) \cdot \tan(3x)} \right) \left(\frac{\cos x}{\sqrt{\cos x} + \cos x} \right) \\
 &= \frac{\lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{\sin(2x) \cdot \tan(3x)} \right) \left(\frac{1}{2} \right)}{2} = \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \right) \left(\frac{2x}{\sin 2x} \right) \left(\frac{3x}{\tan 3x} \right) \left(\frac{x^2}{6x^2} \right) \\
 &= \frac{1}{2} \times \left(\frac{1}{1} \right) \times \left(\frac{1}{1} \right) \times \left(\frac{1}{1} \right) \times \left(\frac{1}{6} \right) = \frac{1}{24}
 \end{aligned}$$

$$44) \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos \sqrt{x}}{\sin x} \right)$$

$$\begin{aligned}
 & \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos \sqrt{x}}{\sin x} \right) \\
 &= \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos \sqrt{x}}{x} \right) \left(\frac{x}{\sin x} \right) \\
 &= \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos \sqrt{x}}{(\sqrt{x})^2} \right) \cdot \lim_{x \rightarrow 0^+} \left(\frac{1}{\frac{\sin x}{x}} \right) \\
 &= \lim_{t \rightarrow 0^+} \left(\frac{1 - \cos t}{t^2} \right) \cdot \lim_{x \rightarrow 0^+} \left(\frac{1}{\frac{\sin x}{x}} \right) \\
 &= \left(\frac{1}{2} \right) \times \left(\frac{1}{1} \right) = \frac{1}{2}
 \end{aligned}$$

Solution limite fonction logarithme népérien

$$1) \lim_{x \rightarrow 0^+} x \ln x = 0.$$

Poser $x = e^{-t}$ ($t \rightarrow +\infty$): $x \ln x = e^{-t}(-t) = -te^{-t} \rightarrow 0$.

$$2) \lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x} = 1.$$

DL: $\ln(1+x) = x - \frac{x^2}{2} + \dots$. Donc $\frac{\ln(1+x)}{x} \rightarrow 1$. (Ou dérivée en 0.)

$$3) \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0.$$

Forme $\frac{\infty}{\infty}$. Par L'Hôpital : $\frac{1/x}{1} = \frac{1}{x} \rightarrow 0$.

$$4) \lim_{x \rightarrow +\infty} \frac{x}{\ln x} = +\infty.$$

C'est l'inverse du cas 3 où $\ln x$ croît beaucoup plus lentement que x .

$$5) \lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x^2} = +\infty.$$

Puisque $\ln(1+x) \sim x$ quand $x \rightarrow 0$, le quotient $\sim \frac{x}{x^2} = \frac{1}{x} \rightarrow +\infty$.

$$6) \lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1.$$

Forme $\frac{0}{0}$. Par L'Hôpital : $\frac{1/x}{1}$ évalué en $x = 1$ donne 1. (Ou dérivée de $\ln$ en 1.)

$$7) \lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{\ln x} = 1.$$

Comme $\sin x \sim x$ quand $x \rightarrow 0$, alors $\ln(\sin x) \sim \ln x \rightarrow$ quotient $\rightarrow 1$.

$$8) \lim_{x \rightarrow +\infty} \frac{\ln(x^2+1)}{\ln(x+1)} = 2.$$

Pour $x \rightarrow \infty$, $\ln(x^2+1) \sim 2\ln x$ et $\ln(x+1) \sim \ln x \Rightarrow$ rapport $\rightarrow 2$.

$$9) \lim_{x \rightarrow 0^+} \frac{\ln(1+x)-x}{x^2} = -\frac{1}{2}.$$

DL: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots$. Donc numérateur $\sim -\frac{x^2}{2}$, divisé par x^2 donne $-\frac{1}{2}$.

$$10) \lim_{x \rightarrow +\infty} \frac{\ln x}{\sqrt{x}} = 0.$$

L'Hôpital : $\frac{1/x}{1/(2\sqrt{x})} = \frac{2\sqrt{x}}{x} = \frac{2}{\sqrt{x}} \rightarrow 0$.

11) $\lim_{x \rightarrow 0^+} x^a \ln x = 0$ pour $a > 0$.

Poser $x = e^{-t}$ ($t \rightarrow +\infty$): $x^a \ln x = e^{-at}(-t) \rightarrow 0$ (exponentielle domine polynôme).

12) $\lim_{x \rightarrow +\infty} (\ln(x+1) - \ln x) = 0$.

$$\ln(x+1) - \ln x = \ln\left(1 + \frac{1}{x}\right) \rightarrow 0.$$

13) $\lim_{x \rightarrow +\infty} \frac{\ln(x^3+2x)}{\ln(x^4+5)} = \frac{3}{4}$.

Pour x grand: $\ln(x^3+2x) \sim 3\ln x$ et $\ln(x^4+5) \sim 4\ln x \Rightarrow$ rapport $\rightarrow 3/4$.

14) $\lim_{x \rightarrow 0^+} \frac{\ln(1+x^2)}{\ln(1+x)} = 0$.

DL: $\ln(1+x^2) \sim x^2$, $\ln(1+x) \sim x$. Donc quotient $\sim x \rightarrow 0$.

15) $\frac{\ln(1+\sin x)}{\ln(1+x)} = 1$.

$\sin x \sim x \Rightarrow \ln(1+\sin x) \sim \sin x \sim x$ et $\ln(1+x) \sim x \rightarrow$ rapport $\rightarrow 1$.

16) $\lim_{x \rightarrow +\infty} \frac{x \ln x}{x^2+1} = 0$.

Diviser numérateur et dénominateur par $x^2 \rightarrow \sim \frac{\ln x}{x}$ qui tend vers 0 (cf. n°3).

17. $\lim_{x \rightarrow 0^+} \frac{\ln(1+e^x)-x}{x^2} = +\infty$.

Remarque: en $x = 0$ le numérateur vaut $\ln(2) \neq 0$. Donc quand $x \rightarrow 0$ le numérateur tend vers $\ln 2$ (valeur non nulle) tandis que $x^2 \rightarrow 0^+$.

Donc le rapport $\rightarrow +\infty$.

(Plus précisément : $\ln(1+e^x) - x = \ln 2 + O(x)$, donc $\sim \frac{\ln 2}{x^2} \rightarrow +\infty$.)

18) $\lim_{x \rightarrow +\infty} \frac{\ln(x!)}{x \ln x} = 1$.

Avec la formule de Stirling : $\ln(x!) \sim x \ln x - x$. Donc $\frac{\ln(x!)}{x \ln x} \sim \frac{x \ln x - x}{x \ln x} =$

$$1 - \frac{1}{\ln x} \rightarrow 1.$$

$$19) \lim_{x \rightarrow +\infty} (\ln(x+1))^{1/\ln x} = 1.$$

Prendre le logarithme:

In de l'expression $= \frac{\ln \ln(x+1)}{\ln x}$. Or $\ln \ln(x+1)/\ln x \rightarrow 0$ quand $x \rightarrow \infty$.

Donc l'expression $\rightarrow e^0 = 1$.

$$21) \lim_{x \rightarrow 1^+} \left(\frac{2x}{x-1} + \ln(x-1) \right)$$

soit $x \in]1; +\infty[$, on a : $\frac{2x}{x-1} + \ln(x-1)$

$$= \frac{1}{x-1} (2x + (x-1)\ln(x-1))$$

posons : $x-1 = t$, on a : $x \rightarrow 1^+ \Leftrightarrow t \rightarrow 0^+$

Donc:

$$\lim_{x \rightarrow 1^+} (x-1)\ln(x-1) = \lim_{t \rightarrow 0^+} t \ln(t) = 0$$

$$\text{Doù : } \lim_{x \rightarrow 1^+} (2x + (x-1)\ln(x-1)) = 2$$

et puisque : $\lim_{x \rightarrow 1^+} \frac{1}{x-1} = +\infty$ alors:

$$\lim_{x \rightarrow 1^+} \frac{1}{x-1} (2x + (x-1)\ln(x-1)) = +\infty$$

$$\text{D'où : } \lim_{x \rightarrow 1^+} \frac{2x}{x-1} + \ln(x-1) = +\infty$$

$$22) \lim_{x \rightarrow +\infty} (2x^2 - x - \ln(x+1))$$

Soit $x \in]0; +\infty[$ on a :

$$2x^2 - x - \ln(x+1) = (x+1) \left(\frac{2x^2-x}{x+1} - \frac{\ln(x+1)}{x+1} \right)$$

$$\text{On a : } \lim_{x \rightarrow +\infty} \frac{2x^2-x}{x+1} = \lim_{x \rightarrow +\infty} \frac{2x^2}{x} = \lim_{x \rightarrow +\infty} 2x = +\infty$$

on pose : $t = x + 1$,

on a : $x \rightarrow +\infty \Leftrightarrow t \rightarrow +\infty$

$$\text{Donc: } \lim_{x \rightarrow +\infty} \frac{\ln(x+1)}{x+1} = \lim_{t \rightarrow +\infty} \frac{\ln t}{t} = 0$$

$$\text{d'où: } \lim_{x \rightarrow +\infty} \left(\frac{2x^2 - x}{x+1} - \frac{\ln(x+1)}{x+1} \right) = +\infty$$

et puisque : $\lim_{x \rightarrow +\infty} (x + 1) = +\infty$ alors :

$$\lim_{x \rightarrow +\infty} (2x^2 - x - \ln(x + 1)) = +\infty$$

$$23) \lim_{x \rightarrow +\infty} (3x - \ln(x^2 + 1))$$

Soit $x \in]0; +\infty[$, on a : $3x - \ln(x^2 + 1) =$

$$\begin{aligned} & 3x - \ln \left(x^2 \left(1 + \frac{1}{x^2} \right) \right) \\ &= 3x - 2\ln(x) - \ln \left(1 + \frac{1}{x^2} \right) \\ &= x \left(3 - 2 \frac{\ln(x)}{x} \right) - \ln \left(1 + \frac{1}{x^2} \right) \end{aligned}$$

On a : $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x^2} \right) = 1$ et la fonction $\ln$ est continue 1 ,

$$\text{donc : } \lim_{x \rightarrow +\infty} \ln \left(1 + \frac{1}{x^2} \right) = \ln(1) = 0$$

Or : $\lim_{x \rightarrow +\infty} \frac{\ln(x)}{x} = 0$ et $\lim_{x \rightarrow +\infty} x = +\infty$ donc

$$\lim_{x \rightarrow +\infty} x \left(3 - \frac{\ln(x)}{x} \right) = +\infty$$

D'où: $\lim 3x - \ln(x^2 + 1) = +\infty$

$$24) \lim_{x \rightarrow 0} \frac{1}{x^2} \cdot \ln(\cos x)$$

$$\text{On a : } \frac{1}{x^2} \cdot \ln(\cos x) = \frac{\ln(\cos x)}{\cos x - 1} \cdot \frac{\cos x - 1}{x^2}$$

Posons :

$t = \cos x$. On a : $x \rightarrow 0 \Leftrightarrow t \rightarrow 1$,
donc:

$$\lim_{t \rightarrow 0} \frac{\ln(\cos x)}{\cos x - 1} = \lim_{t \rightarrow 0} \frac{\ln(t)}{t - 1} = 1$$

$$\text{et on a : } \lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x^2} = -\frac{1}{2}$$

$$\text{donc : } \lim_{x \rightarrow 0} \frac{1}{x^2} \cdot \ln(\cos x) = -\frac{1}{2}$$

$$25) \lim_{x \rightarrow \infty} \frac{\ln(x+1+\sqrt{x^2+1})}{2x}$$

Soit $x \in]0; +\infty[$, on a :

Soit $x \in]0; +\infty[$, on a :

$$\ln(x + 1 + \sqrt{x^2 + 1}) = \ln\left(x \left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right)\right)$$

$$= \ln(x) + \ln\left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right)$$

Donc: $\frac{\ln(x+1+\sqrt{1+x^2})}{2x} = \frac{1}{2} \cdot \frac{\ln(x)}{x} + \frac{1}{2x} \cdot \ln\left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right)$

On a : $\lim_{x \rightarrow -\infty} \frac{\ln(x)}{x} = 0$ et $\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right) = 2$

et la fonction $\ln$ est continue en 2

Donc: $\lim_{x \rightarrow \infty} \ln\left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right) = \ln(2)$ et comme

$\lim_{x \rightarrow -\infty} \frac{1}{2x} = 0$

Alors: $\lim_{x \rightarrow +\infty} \frac{1}{2x} \cdot \ln\left(1 + \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right) = 0$

Par conséquent: $\lim_{x \rightarrow \infty} \frac{\ln(x+1+\sqrt{1+x^2})}{2x} = 0$
--

26) $\lim_{x \rightarrow 2^+} (x-2) \ln(x^2 - 2x)$

Soit $x \in]2; +\infty[$, on a : $(x-2) \ln(x^2 - 2x)$

$= (x-2) \ln(x(x-2))$

$= (x-2)(\ln(x) + \ln(x-2))$

$= (x-2) \ln(x) + (x-2) \ln(x-2)$

posons : $t = x - 2$ on a : $x \rightarrow 2^+ \Leftrightarrow t \rightarrow 0^+$

Donc: $\lim_{x \rightarrow 2^+} (x-2) \ln(x-2) = \lim_{t \rightarrow 0^+} t \ln(t) = 0$

et puisque : $\lim_{x \rightarrow 2^+} \ln(x) = \ln(2)$ et $\lim_{x \rightarrow 2^+} (x-2) = 0$,

alors: $\lim_{x \rightarrow 2^+} (x-2) \ln(x) = 0$

D'où: $\lim_{x \rightarrow 2^+} (x-2) \ln(x^2 - 2x) = 0$

$$27) \lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln^2(x)$$

Soit $x \in]0; +\infty[$, on a : $\ln(\sqrt[6]{x}) = \frac{1}{6}\ln(x)$

c'est-à-dire : $\ln(x) = 6\ln(\sqrt[6]{x})$ et on a : $(\sqrt[6]{x})^2 = \sqrt[3]{x}$

Donc : $\sqrt[3]{x} \cdot \ln^2(x) = (\sqrt[6]{x})^2 \cdot (6\ln(\sqrt[6]{x}))^2 = 36(\sqrt[6]{x} \cdot \ln(\sqrt[6]{x}))^2$

Posons : $t = \sqrt[6]{x}$, on a : $x \rightarrow 0^+ \Leftrightarrow t \rightarrow 0^+$

Donc : $\lim_{x \rightarrow 0^+} \sqrt[6]{x} \cdot \ln(\sqrt[6]{x}) = \lim_{t \rightarrow 0^+} t \ln(t) = 0$, d'où :

$$\lim_{x \rightarrow 0^+} (\sqrt[6]{x} \cdot \ln(\sqrt[6]{x}))^2 = 0$$

Par suite : $\lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln^2(x) = 0$

$$28) \lim_{x \rightarrow -\infty} \frac{\ln(x^2 - x + 1)}{2x + 1}$$

soit $X \in]-\infty; -\frac{1}{2}[$, on a : $\ln(x^2 - x + 1) =$

$$\begin{aligned} &= \ln(x^2) + \ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right) \\ &= 2\ln(-x) + \ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right) \end{aligned}$$

$$\text{Donc : } \frac{\ln(x^2 - x + 1)}{2x + 1} = 2 \times \frac{\ln(-x)}{2x + 1} + \frac{\ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)}{2x + 1}$$

$$= 2 \cdot \frac{\ln(-x)}{(-x)} \cdot \frac{(-x)}{2x + 1} + \frac{\ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)}{2x + 1}$$

Et puisque : $\lim_{x \rightarrow -\infty} \left(1 - \frac{1}{x} + \frac{1}{x^2}\right) = 1$ et la fonction $\ln$ est continue en 1

lors $\lim_{x \rightarrow -\infty} \ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right) = \ln(1) = 0$ et on a : $\lim_{x \rightarrow -\infty} \frac{1}{2x + 1} = 0$

$$\text{denc: } \lim_{1 \rightarrow \infty} \frac{\ln\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)}{2x+1} = \lim_{3 \rightarrow -\infty} \frac{1}{2x+1} \times \left(\ln 1 - \frac{1}{x} + \frac{1}{x^2}\right) = 0$$

rosons : $t = -x$, on a : $x \rightarrow -\infty \Leftrightarrow t \rightarrow +\infty$

$$\text{Donc: } \lim_{x \rightarrow -\infty} \frac{\ln(-x)}{(-x)} = \lim_{t \rightarrow \infty} \frac{\ln(t)}{t} = 0$$

$$\text{ETon } a: \lim_{x \rightarrow -4} \frac{-x}{2x+1} \lim_{x \rightarrow -\infty} = \frac{-x}{2x} = \lim_{x \rightarrow -\infty} \left(-\frac{1}{2}\right) = -\frac{1}{2}$$

$$\text{Doù: } \lim_2 \frac{\ln(-x)}{(-x)} \frac{(-x)}{2x+1} = 0, \text{ par suite : } \lim \frac{\ln(x^2-x+1)}{2x+1} = 0$$

$$29) \lim_{x \rightarrow +\infty} \frac{x^2}{\ln x} :$$

On a: $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2} = 0$ et $(\forall x \in]1; +\infty[); \frac{\ln x}{x^2} > 0$, donc: $\lim_{x \rightarrow +\infty} \frac{x^2}{\ln x} = +\infty$

Calculons $\lim_{x \rightarrow +\infty} \ln(x^2 + 1) - x$:
soit x un élément de $]0; +\infty[$, on a :

$$\begin{aligned} \ln(x^2 + 1) - x &= \ln\left(x^2 \left(1 + \frac{1}{x^2}\right)\right) - x = \ln(x^2) + \ln\left(1 + \frac{1}{x^2}\right) - x \\ &= 2\ln(x) - x + \ln\left(1 + \frac{1}{x^2}\right) = x\left(2 \cdot \frac{\ln x}{x} - 1\right) + \ln\left(1 + \frac{1}{x^2}\right) \end{aligned}$$

et comme: $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x^2}\right) = 1$ et la fonction $\ln$ est continue en 1 ;

$$\text{alors: } \lim_{x \rightarrow +\infty} \ln\left(1 + \frac{1}{x^2}\right) = \ln(1) = 0$$

$$\text{et on a : } \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0 \text{ donc: } \lim_{x \rightarrow +\infty} x\left(2 \cdot \frac{\ln x}{x} - 1\right) = -\infty$$

$$\text{Ainsi : } \lim_{x \rightarrow +\infty} \ln(x^2 + 1) - x = -\infty$$

$$30) \lim_{x \rightarrow +\infty} \frac{(x+1)\ln x}{x^2} :$$

Soit x un élément de $]0; +\infty[$, on a :

$$\frac{(x+1)\ln x}{x^2} = \frac{x+1}{x} \cdot \frac{\ln x}{x} = \left(\frac{x}{x} + \frac{1}{x}\right) \cdot \frac{\ln x}{x} = \left(1 + \frac{1}{x}\right) \cdot \frac{\ln x}{x}$$

$$\text{et comme : } \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0 \text{ et } \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right) = 1$$

$$\text{alors : } \lim_{x \rightarrow +\infty} \frac{(x+1)\ln x}{x^2} = 0$$

$$31) \lim_{x \rightarrow +\infty} x \ln \left(\frac{x+1}{x-1} \right):$$

$$\text{En posant : } X = \frac{x+1}{x-1}$$

$$\text{On a d'une part: } \lim_{x \rightarrow +\infty} \frac{x+1}{x-1} = \lim_{x \rightarrow +\infty} \frac{x}{x} = \lim_{x \rightarrow +\infty} 1 = 1$$

$$\text{donc: } x \rightarrow +\infty \Rightarrow X \rightarrow 1$$

$$\text{et d'autre part: } X = \frac{x+1}{x-1} \Leftrightarrow x+1 = X(x-1)$$

$$\begin{aligned} \Leftrightarrow x+1 &= Xx - X \\ \Leftrightarrow x - Xx &= -1 - X \\ \Leftrightarrow x(1-X) &= -(1+X) \\ \Leftrightarrow x &= \frac{X+1}{X-1} \end{aligned}$$

Donc:

$$\lim_{x \rightarrow +\infty} x \ln \left(\frac{x+1}{x-1} \right) = \lim_{X \rightarrow 1} \frac{X+1}{X-1} \ln(X) = \lim_{X \rightarrow 1} (X+1) \times \frac{\ln(X)}{X-1} = 2$$

$$\text{car : } \lim_{X \rightarrow 1} \frac{\ln(X)}{X-1} = 1 \text{ et } \lim_{X \rightarrow 1} (X+1) = 2$$

$$\text{Donc: } \lim_{x \rightarrow +\infty} x \ln \left(\frac{x+1}{x-1} \right) = 2$$

$$32) \lim_{x \rightarrow e} \frac{(\ln x)^2 + 2\ln x - 3}{x - e} :$$

En posant : $X = \frac{x}{e}$

On a : $x \rightarrow e \Rightarrow X \rightarrow 1$ et $x = eX$

donc:

$$\begin{aligned} \frac{(\ln x)^2 + 2\ln x - 3}{x - e} &= \frac{(\ln(eX))^2 + 2\ln(eX) - 3}{eX - e} = \frac{(\ln X + 1)^2 + 2(1 + \ln X) - 3}{e(X - 1)} \\ &= \frac{(\ln X)^2 + 2\ln X + 1 + 2 + 2\ln X - 3}{e(X - 1)} = \frac{(\ln X)^2 + 4\ln X}{e(X - 1)} \\ &= \frac{\ln X}{X - 1} \times \frac{\ln X + 4}{e} \end{aligned}$$

et comme : $\lim_{X \rightarrow 1} \frac{\ln X}{X - 1} = 1$ et $\lim_{X \rightarrow 1} \frac{\ln X + 4}{e} = \frac{4}{e}$

alors: $\lim_{x \rightarrow e} \frac{(\ln x)^2 + 2\ln x - 3}{x - e} = \frac{4}{e}$

$$33) \lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{1+x}{1-x} \right)$$

Soit $x \in]-1, 0[\cup]0, 1[$, on a : $\frac{1}{x} \ln \left(\frac{1+x}{1-x} \right) = \frac{1}{x} (\ln(1+x) - \ln(1-x))$

$$\begin{aligned} &= \frac{\ln(1+x)}{x} - \frac{\ln(1-x)}{x} \\ &= \frac{\ln(1+x)}{x} + \frac{\ln(1+(-x))}{(-x)} \end{aligned}$$

et comme : $\lim_{x \rightarrow 0} \frac{\ln(1+(-x))}{(-x)} = \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$ (on pose : $t = -x$)

Alors: $\lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{1+x}{1-x} \right) = 2$

$$34) \lim_{x \rightarrow -\infty} \ln\left(\frac{2x-1}{x+2}\right) :$$

On a: $\lim_{x \rightarrow -\infty} \frac{2x-1}{x-2} = \lim_{x \rightarrow -\infty} \frac{2x}{x} = \lim_{x \rightarrow -\infty} 2 = 2$

et la fonction $\ln$ est continue en 2 ;

$$\text{Alors: } \lim_{x \rightarrow -\infty} \ln\left(\frac{2x-1}{x-2}\right) = \ln 2$$

$$35) \lim_{x \rightarrow 2^+} \left(\frac{3x}{x-2} + \ln(x-2) \right)$$

$$\lim_{x \rightarrow 2^+} \left(\frac{3x}{x-2} + \ln(x-2) \right)$$

$$= \lim_{\substack{t \rightarrow 0^+ \\ t=x-2}} \left(\frac{3(t+2)}{t} + \ln t \right)$$

$$= 3 + \lim_{t \rightarrow 0^+} \left(\frac{6}{t} + \ln t \right)$$

$$= 3 + \lim_{t \rightarrow 0^+} \frac{1}{t} (6 + t \ln t)$$

$$= 3 + (+\infty)(6 + 0^-) = +\infty$$

$$36) \lim_{x \rightarrow +\infty} \ln(\sqrt{x^2+3} - x)$$

$$\lim_{x \rightarrow +\infty} \ln(\sqrt{x^2+3} - x)$$

$$= \lim_{x \rightarrow +\infty} \ln\left(\frac{(\sqrt{x^2+3} - x)(\sqrt{x^2+3} + x)}{\sqrt{x^2+3} + x}\right)$$

$$= \lim_{x \rightarrow +\infty} \ln\left(\frac{x^2+3-x^2}{\sqrt{x^2+3} + x}\right)$$

$$= \lim_{x \rightarrow +\infty} \ln\left(\frac{3}{\sqrt{x^2+3} + x}\right)$$

$$= \lim_{t \rightarrow 0^+} (\ln t) = -\infty$$

$$t = \frac{3}{\sqrt{x^2+3} + x}$$

$$37) \lim_{x \rightarrow (-1)^+} (x+1) \ln(x^2+x)$$

$$\begin{aligned} & \lim_{x \rightarrow (-1)^+} (x+1) \ln(x^2+x) \\ &= \lim_{\substack{t \rightarrow 0^+ \\ t=x+1}} t \ln(t^2-t) \\ &= \lim_{t \rightarrow 0^-} (t^2-t) \ln(t^2-t) \cdot \left(\frac{t}{t^2-t} \right) \\ &= \left(\lim_{\substack{u \rightarrow 0^+ \\ u=t^2-t}} (u \ln u) \right) \cdot \left(\lim_{t \rightarrow 0^-} \left(\frac{t}{t^2-t} \right) \right) \\ &= (0^-) \cdot \lim_{t \rightarrow 0^-} \left(\frac{1}{t-1} \right) \\ &= (0^-) \times \left(\frac{1}{0-1} \right) = 0^+ \end{aligned}$$

$$38) \lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln x$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \sqrt[3]{x} \cdot \ln x &= \lim_{x \rightarrow 0^+} x^{\frac{1}{3}} \cdot \ln \left(\left(x^{\frac{1}{3}} \right)^3 \right) \\ &= \lim_{x \rightarrow 0^+} x^{\frac{1}{3}} \cdot 3 \cdot \ln \left(x^{\frac{1}{3}} \right) \\ &= 3 \lim_{\substack{t \rightarrow 0^+ \\ t = x^{\frac{1}{3}}}} t \ln t = 3 \times 0^- = 0^- = 0 \end{aligned}$$

$$39) \lim_{x \rightarrow +\infty} x \cdot \ln \left(\frac{x-1}{x+1} \right)$$

$$\begin{aligned}
 & \lim_{x \rightarrow +\infty} x \cdot \ln \left(\frac{x-1}{x+1} \right) \\
 &= \lim_{x \rightarrow 0^+} x \ln \left(1 - \frac{2}{x+1} \right) \\
 &= \lim_{t \rightarrow 1^-} \left(\frac{2}{1-t} - 1 \right) \ln t \\
 &= -2 \lim_{t \rightarrow 1^-} \left(\frac{\ln t}{t-1} \right) - \lim_{t \rightarrow 1^-} \ln t \\
 &= -2 \lim_{t \rightarrow 1^-} \left(\frac{\ln t - \ln 1}{t-1} \right) - \ln(1^-) \\
 &= -2((\ln t)'_{t=1}) - 0 \\
 &= -2 \left(\frac{1}{t} \right)_{t=1} - 0 = -2 \cdot \frac{1}{1} - 0 = -2
 \end{aligned}$$

Solution : limites fonction exponentielle

1 Calculons : $\lim_{x \rightarrow +\infty} 3xe^{-x}$

Soit x de $\mathbb{R}$, on a : $3xe^{-x} = 3 \times \frac{x}{e^x}$

Puisque : $\lim_{x \rightarrow +\infty} \frac{e^x}{x} = +\infty$, alors : $\lim_{x \rightarrow +\infty} \frac{x}{e^x} = 0$, donc : $\lim_{x \rightarrow +\infty} 3 \times \frac{x}{e^x} = 0$

D'où : $\lim_{x \rightarrow +\infty} 3xe^{-x} = 0$

2 Calculons : $\lim_{x \rightarrow -\infty} (x+1)e^x$

Soit x de $\mathbb{R}$, on a : $(x+1)e^x = xe^x + e^x$

Puisque : $\lim_{x \rightarrow -\infty} e^x = 0$ et $\lim_{x \rightarrow -\infty} xe^x = 0$, alors : $\lim_{x \rightarrow -\infty} (xe^x + e^x) = 0$

Donc : $\lim_{x \rightarrow -\infty} (x+1)e^x = 0$

3 Calculons : $\lim_{x \rightarrow -\infty} x + 2 + xe^x$

On a : $\lim_{x \rightarrow -\infty} x + 2 = -\infty$ et $\lim_{x \rightarrow -\infty} xe^x = 0$, donc : $\lim_{x \rightarrow -\infty} x + 2 + xe^x = -\infty$

4 Calculons : $\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3}$

Soit x et $\mathbb{R}'$, on a : $\frac{e^x - 1}{x^3} = \frac{e^x - 1}{x} \times \frac{1}{x^2}$

Puisque : $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$ et $\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$, alors : $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \frac{1}{x^2} = +\infty$

Donc : $\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} = +\infty$

5 Calculons : $\lim_{x \rightarrow -\infty} \frac{e^x + e^{-x}}{3 + e^x}$

On a : $\lim_{x \rightarrow -\infty} e^x = 0$ et $e^x > 0$ pour tout x de $\mathbb{R}$, alors $\lim_{x \rightarrow -\infty} e^{-x} = \lim_{x \rightarrow -\infty} \frac{1}{e^x} = +\infty$

D'où : $\lim_{x \rightarrow -\infty} e^x + e^{-x} = +\infty$, et on a : $\lim_{x \rightarrow -\infty} 3 + e^x = 3$

Par suite : $\lim_{x \rightarrow -\infty} \frac{e^x + e^{-x}}{3 + e^x} = +\infty$

6 Calculons : $\lim_{x \rightarrow -\infty} \frac{e^x - 1}{2e^x + 1}$

On a : $\lim_{x \rightarrow -\infty} e^x = 0$, donc : $\lim_{x \rightarrow -\infty} e^x - 1 = -1$ et $\lim_{x \rightarrow -\infty} 2e^x + 1 = 1$

D'où : $\lim_{x \rightarrow -\infty} \frac{e^x - 1}{2e^x + 1} = -1$

7 Calculons: $\lim_{x \rightarrow +\infty} \frac{2e^x + 1}{e^x - 1}$

Soit x de $]0; +\infty[$, on a : $\frac{2e^x + 1}{e^x - 1} = \frac{e^x(2 + e^{-x})}{e^x(1 - e^{-x})} = \frac{2 + e^{-x}}{1 - e^{-x}}$

Puisque : $\lim_{x \rightarrow +\infty} e^x = +\infty$, alors: $\lim_{x \rightarrow +\infty} \frac{1}{e^x} = 0$, c'est-à-dire : $\lim_{x \rightarrow +\infty} e^{-x} = 0$

Donc: $\lim_{x \rightarrow +\infty} 2 + e^{-x} = 2$ et $\lim_{x \rightarrow +\infty} 1 - e^{-x} = 1$

D'où: $\lim_{x \rightarrow +\infty} \frac{2 + e^{-x}}{1 - e^{-x}} = 2$, c'est-à-dire : $\lim_{x \rightarrow +\infty} \frac{2e^x + 1}{e^x - 1} = 2$

8 Calculons : $\lim_{x \rightarrow +\infty} 2x - e^x$

Soit x de $\mathbb{R}^*$, on a : $2x - e^x = x \left(2 - \frac{e^x}{x} \right)$

Puisque : $\lim_{x \rightarrow +\infty} \frac{e^x}{x} = +\infty$, alors : $\lim_{x \rightarrow +\infty} 2 - \frac{e^x}{x} = -\infty$

Et on a : $\lim_{x \rightarrow +\infty} x = +\infty$, donc : $\lim_{x \rightarrow +\infty} x \left(2 - \frac{e^x}{x} \right) = -\infty$

D'où: $\lim_{x \rightarrow +\infty} 2x - e^x = -\infty$

9 Calculons: $\lim_{x \rightarrow +\infty} x^3 - e^x$

Soit x de $\mathbb{R}^*$, on a : $x^3 - e^x = x^3 \left(1 - \frac{e^x}{x^3} \right)$

Puisque : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3} = +\infty$, alors: $\lim_{x \rightarrow +\infty} 1 - \frac{e^x}{x^3} = -\infty$

Et on a : $\lim_{x \rightarrow +\infty} x^3 = +\infty$, donc: $\lim_{x \rightarrow +\infty} x^3 \left(1 - \frac{e^x}{x^3} \right) = -\infty$

D'où: $\lim_{x \rightarrow +\infty} x^3 - e^x = -\infty$

10 Calculons: $\lim_{x \rightarrow +\infty} e^{2x} - x^2$

Soit x de $\mathbb{R}^*$, on a : $e^{2x} - x^2 = x^2 \left(\frac{e^{2x}}{x^2} - 1 \right) = x^2 \left(\left(\frac{e^x}{x} \right)^2 - 1 \right)$

Puisque : $\lim_{x \rightarrow +\infty} \frac{e^x}{x} = +\infty$, alors: $\lim_{x \rightarrow +\infty} \left(\frac{e^x}{x} \right)^2 = +\infty$, c'est-à-dire : $\lim_{x \rightarrow +\infty} \frac{e^{2x}}{x^2} = +\infty$

Ainsi : $\lim_{x \rightarrow +\infty} \frac{e^{2x}}{x^2} - 1 = +\infty$

Et on a: $\lim_{x \rightarrow +\infty} x^2 = +\infty$; donc : $\lim_{x \rightarrow +\infty} x^2 \left(\frac{e^{2x}}{x^2} - 1 \right) = +\infty$

D'où: $\lim_{x \rightarrow +\infty} e^{2x} - x^2 = +\infty$

(11) Calculons : $\lim_{x \rightarrow 0} e^{1 - \frac{1}{x^2}}$

On a : $\lim_{x \rightarrow 0} 1 - \frac{1}{x^2} = -\infty$ et $\lim_{x \rightarrow -\infty} e^x = 0$, donc, par composée. $\lim_{x \rightarrow 0} e^{1 - \frac{1}{x^2}} = 0$

12 Calculons : $\lim_{x \rightarrow +\infty} xe^{2x} - e^x$

Soit x de $\mathbb{R}$, on a : $xe^{2x} - e^x = (xe^x - 1)e^x$

Puisque : $\lim_{x \rightarrow +\infty} e^x = +\infty$ et $\lim_{x \rightarrow +\infty} x = +\infty$, alors : $\lim_{x \rightarrow +\infty} xe^x - 1 = +\infty$

Donc, $\lim_{x \rightarrow +\infty} (xe^x - 1)e^x = +\infty$ c'est-à-dire : $\lim_{x \rightarrow +\infty} xe^{2x} - e^x = +\infty$

13 Calculons : $\lim_{x \rightarrow -\infty} (x^4 - x)e^x$

Soit x de $\mathbb{R}$, on a : $(x^4 - x)e^x = x^4e^x - xe^x$

Puisque $\lim_{x \rightarrow -\infty} xe^x = 0$ et $\lim_{x \rightarrow -\infty} x^4e^x = 0$, alors : $\lim_{x \rightarrow -\infty} x^4e^x - xe^x = 0$

Donc : $\lim_{x \rightarrow -\infty} (x^4 - x)e^x = 0$

14 Calculons $\lim_{x \rightarrow -\infty} \frac{e^x}{1+x^2}$:

On a : $\lim_{x \rightarrow -\infty} e^x = 0$ et $\lim_{x \rightarrow -\infty} 1 + x^2 = +\infty$, donc : $\lim_{x \rightarrow -\infty} \frac{e^x}{1+x^2} = 0$

15 Calculons : $\lim_{x \rightarrow -\infty} (2 - x)e^x - x$

Soit x de $\mathbb{R}$, on a : $(2 - x)e^x - x = 2e^x - xe^x - x$

Puisque $\lim_{x \rightarrow -\infty} 2e^x = 0$ et $\lim_{x \rightarrow -\infty} xe^x = 0$, alors : $\lim_{x \rightarrow -\infty} 2e^x - xe^x = 0$

Et on a : $\lim_{x \rightarrow -\infty} (-x) = +\infty$; donc : $\lim_{x \rightarrow -\infty} 2e^x - xe^x - x = +\infty$

D'où : $\lim_{x \rightarrow -\infty} (2 - x)e^x - x = +\infty$

16 Calculons : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - x + 1}$

Soit x de $\mathbb{R}$, on a : $\frac{e^x}{x^2 - x + 1} = \frac{e^x}{x^2} \times \frac{x^2}{x^2 - x + 1}$

Puisque : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2} = +\infty$ et $\lim_{x \rightarrow +\infty} \frac{x^2}{x^2 - x + 1} = \lim_{x \rightarrow +\infty} \frac{x^2}{x^2} = \lim_{x \rightarrow +\infty} 1 = 1$

alors : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2} \times \frac{x^2}{x^2 - x + 1} = +\infty$, donc : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - x + 1} = +\infty$

17 Calculons : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3 + 1}$

Soit $x > 0$, on a : $\frac{e^x}{x^3 + 1} = \frac{e^x}{x^3} \times \frac{x^3}{x^3 + 1}$

Puisque $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3} = +\infty$ et $\lim_{x \rightarrow +\infty} \frac{x^3}{x^3 + 1} = \lim_{x \rightarrow +\infty} \frac{x^3}{x^3} = \lim_{x \rightarrow +\infty} 1 = 1$

alors : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3} \times \frac{x^3}{x^3 + 1} = +\infty$, donc : $\lim_{x \rightarrow +\infty} \frac{e^x}{x^3 + 1} = +\infty$

18 Calculons : $\lim_{x \rightarrow -\infty} e^{x^2-3x-5}$

On a : $\lim_{x \rightarrow -\infty} x^2 - 3x - 5 = \lim_{x \rightarrow -\infty} x^2 = +\infty$ et $\lim_{x \rightarrow +\infty} e^x = +\infty$, donc par composée $\lim_{x \rightarrow -\infty} e^{x^2-3x-5} = +\infty$

19 Calculons : $\lim_{x \rightarrow -\infty} x(e^{2x} - e^x)$

Soit x de $\mathbb{R}$, on a : $x(e^{2x} - e^x) = xe^x(e^x - 1)$

Puisque : $\lim_{x \rightarrow -\infty} xe^x = 0$ et $\lim_{x \rightarrow -\infty} e^x - 1 = -1$, alors : $\lim_{x \rightarrow -\infty} xe^x(e^x - 1) = 0$

Donc : $\lim_{x \rightarrow -\infty} x(e^{2x} - e^x) = 0$

20 Calculons : $\lim_{x \rightarrow -\infty} x^2(e^x - e^{2x})$

Soit x de $\mathbb{R}$, on a : $x^2(e^x - e^{2x}) = x^2e^x(1 - e^x)$

Et on a : $\lim_{x \rightarrow -\infty} x^2e^x = 0$ et $\lim_{x \rightarrow -\infty} 1 - e^x = 1$, car : $\lim_{x \rightarrow -\infty} e^x = 0$

Donc : $\lim_{x \rightarrow -\infty} x^2e^x(1 - e^x) = 0$, d'où : $\lim_{x \rightarrow -\infty} x^2(e^x - e^{2x}) = 0$

$$21) \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x\sin 2x}} - e^{\cos 2x}}{x^2}$$

$$\begin{aligned}
 21. \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x\sin 2x}} - e^{\cos 2x}}{x^2}, & \frac{0}{0} \\
 = \lim_{x \rightarrow 0} \frac{e^{\cos 2x} (e^{\sqrt{1+x\sin 2x} - \cos 2x} - 1)}{x^2} \\
 = e \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x\sin 2x} - \cos 2x} - 1}{\sqrt{1+x\sin 2x} - \cos 2x} \left(\frac{\sqrt{1+x\sin 2x} - \cos 2x}{x^2} \right) \\
 = e \times 1 \times \lim_{x \rightarrow 0} \frac{\sqrt{1+x\sin 2x} - \cos 2x}{x^2} \\
 = e \lim_{x \rightarrow 0} \frac{1 + x\sin 2x - \cos^2 2x}{x^2(\sqrt{1+x\sin 2x} + \cos 2x)} \\
 = \frac{e}{2} \lim_{x \rightarrow 0} \left(\frac{x\sin 2x}{x^2} + \frac{1 - \cos^2 2x}{x^2} \right) \\
 = \frac{e}{2} \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} (2) + \frac{1 - \cos 2x}{x^2} (1 + \cos 2x) \right) \\
 = \frac{e}{2} \left(1 \times 2 + \frac{2^2}{2} (1 + 1) \right) = \frac{e}{2} (2 + 2 \times 2) = 3e \\
 \therefore \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x\sin 2x}} - e^{\cos 2x}}{x^2} = 3e
 \end{aligned}$$

$$22) \lim_{x \rightarrow 0} \frac{e^{\tan^2 x + \cos 2x} - e^{1 + \sin^2 3x}}{x^2}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^{\tan^2 x + \cos 2x} - e^{1 + \sin^2 3x}}{x^2}, \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{e^{1 + \sin^2 3x} (e^{\tan^2 x + \cos 2x - 1 - \sin^2 3x} - 1)}{x^2} \\ &= e^{1+0} \lim_{x \rightarrow 0} \frac{e^{\tan^2 x + \cos 2x - 1 - \sin^2 3x} - 1}{\tan^2 x + \cos 2x - 1 - \sin^2 2x} \\ & \quad \times \frac{\tan^2 x - \sin^2 3x (1 - \cos 2x)}{x^2} \\ &= e \times 1 \times \lim_{x \rightarrow 0} \left[\left(\frac{\tan x}{x} \right)^2 - \left(\frac{\sin 3x}{3x} \right)^2 \times 9 - \frac{1 - \cos 2x}{x^2} \right] \\ &= e \left[1^2 - 1^2 \times 9 - \frac{2^2}{2} \right] = e(1 - 9 - 2) = -10e \\ &\therefore \lim_{x \rightarrow 0} \frac{e^{\tan^2 x + \cos 2x} - e^{1 + \sin^2 3x}}{x^2} = -10e \end{aligned}$$

$$23) \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^{2x} - 1) \dots (e^{nx} - 1)}{x^n}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^{2x} - 1) \dots (e^{nx} - 1)}{x^n}, \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \frac{e^{2x} - 1}{x} \times \dots \times \frac{e^{nx} - 1}{x} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \lim_{2x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 \times \dots \times \lim_{nx \rightarrow 0} \frac{e^{nx} - 1}{nx} \times n \\ &= 1 \times 2 \times 3 \times \dots \times n = n! \\ &\therefore \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^{2x} - 1) \dots (e^{nx} - 1)}{x^n} = n! \end{aligned}$$

$$24) \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\tan 2x}}{x^3}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\tan 2x}}{x^3}, \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{e^{\tan 2x} (e^{\sin 2x - \tan 2x} - 1)}{x^3} \\ &= e^0 \lim_{x \rightarrow 0} \frac{e^{\sin 2x - \tan 2x} - 1}{\sin 2x - \tan 2x} \times \frac{\sin 2x - \tan 2x}{x^3} \\ &= 1 \times \lim_{x \rightarrow 0} \frac{\tan 2x (1 - \cos 2x)}{x^3} \\ &= -2 \lim_{x \rightarrow 0} \left(\frac{\tan 2x}{2x} \right) \times \frac{1 - \cos 2x}{x^2} \\ &= -2 \lim_{2x \rightarrow 0} \frac{\tan 2x}{2x} \times \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \\ &= -2(1) \times \left(\frac{2^2}{2} \right) = -4 \\ &\therefore \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\tan 2x}}{x^3} = -4 \end{aligned}$$

$$25) \lim_{x \rightarrow 0} \frac{1 - e^x \cos 2x}{\tan^2 x}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{1 - e^x \cos 2x}{\tan^2 x}, \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{1 - e^x(1 - 1 + \cos 2x)}{\tan^2 x} \\ &= \lim_{x \rightarrow 0} \frac{1 - e^x + e^x(1 - \cos 2x)}{\tan^2 x} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{\tan^2 x} + \lim_{x \rightarrow 0} \frac{e^x(1 + \cos 2x)}{\tan^2 x} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{x^2} \times \left(\frac{x}{\tan x}\right)^2 + e^0 \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \times \left(\frac{x}{\tan x}\right)^2 \\ &= \frac{1}{0} \times (1) \times (1)^2 + 1 \times \frac{2^2}{2} \times 1^2 = \infty + 2 = \infty \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} \frac{1 - e^x - \cos 2x}{\tan^2 x} = \infty$$

$$26. \lim_{x \rightarrow 0} \frac{e^{x^2+2x} - e^{x^2-2x}}{x}, \frac{0}{0}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{e^{x^2-2x}(e^{x^2+2x-x^2+2x} - 1)}{x} = \lim_{x \rightarrow 0} \frac{e^{x^2-2x}(e^{4x} - 1)}{x} \\ &= e^0 \lim_{x \rightarrow 0} \frac{(e^{4x} - 1)}{4x} \times 4 = \lim_{4x \rightarrow 0} \frac{(e^{4x} - 1)}{4x} \times 4 = 4 \end{aligned}$$

$$- - - - \frac{4x}{x} = 4$$

$$\therefore \lim_{x \rightarrow 0} \frac{e^{x^2+2x} - e^{x^2-2x}}{x} = 4$$

x

$$\begin{aligned}
& 27. \lim_{x \rightarrow 0} \frac{1 - e^{\sin^2 x} \sqrt{\cos x}}{x^2}, \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \frac{(1 - e^{2\sin^2 x} \sqrt{\cos x})(1 + e^{\sin^2 x} \sqrt{\cos x})}{x^2(1 + e^{\sin^2 x} \sqrt{\cos x})} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{(1 - e^{2\sin^2 x} \sqrt{\cos x})}{x^2} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{1 - e^{2\sin^2 x} (1 - 1 + \cos x)}{x^2} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left[\frac{-(e^{2\sin^2 x} - 1)}{x^2} + \frac{e^{2\sin^2 x} (1 - \cos x)}{x^2} \right]}{-\frac{1}{2} \lim_{x \rightarrow 0} \frac{e^{2\sin^2 x} - 1}{2\sin^2 x} \times 2 \left(\frac{\sin^2 x}{x} \right)^2 + \frac{1}{2} e^0 \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}} \\
&= -\frac{1}{2} (1)(2)(1)^2 + \frac{1}{2} \times \left(\frac{1^2}{2} \right) = -1 + \frac{1}{4} = -\frac{3}{4} \\
&\quad \frac{1}{\sin^2 x} - \frac{1}{\cos x} \\
&\therefore \lim_{x \rightarrow 0} \frac{1 - e^{\sin^2 x}}{x^2} = -\frac{3}{4}
\end{aligned}$$

$$\begin{aligned}
28). \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x^2}} - e^{\cos 2x}}{x^2}, \frac{0}{0} \\
= \lim_{x \rightarrow 0} \frac{e^{\cos 2x} (e^{\sqrt{1+x^2} - \cos 2x} - 1)}{x^2} \\
= \lim_{x \rightarrow 0} e^{\cos 0} \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x^2} - \cos 2x} - 1}{x^2} \\
= e \times \lim_{x \rightarrow 0} \frac{e^{\sqrt{1+x^2} - \cos 2x} - 1}{\sqrt{1+x^2} - \cos 2x} \times \frac{\sqrt{1+x^2} - \cos 2x}{x^2} \\
= \frac{e}{2} \times \lim_{x \rightarrow 0} \left[\frac{1+x^2 - \cos^2 2x}{x^2} \right] \\
= \frac{e}{2} \times \lim_{x \rightarrow 0} \left[1 + \frac{1 - \cos^2 x}{x^2} \right] \\
= \frac{e}{2} \times \left[1 + \lim_{x \rightarrow 0} \frac{1 - \cos^2 2x}{x^2} \times (1 + \cos 2x) \right] \\
= \frac{e}{2} \times \left[1 + \frac{2^2}{2} \times (1 + \cos 0) \right] \\
= \frac{e}{2} [1 + 2(2)] = \frac{5e}{2} \\
\therefore \lim_{x \rightarrow 0} \frac{5}{x^2}
\end{aligned}$$

$$\begin{aligned}
29). \lim_{x \rightarrow \infty} \left(2^{\frac{1}{x}} + 3^{\frac{1}{x}} - 2 \right) x, 0 \times \infty \\
= \lim_{x \rightarrow \infty} \left[\left(2^{\frac{1}{x}} - 1 \right) + \left(3^{\frac{1}{x}} - 1 \right) \right] x \\
= \lim_{x \rightarrow \infty} \left[\frac{2^{\frac{1}{x}} - 1}{\frac{1}{x}} + \frac{3^{\frac{1}{x}} - 1}{\frac{1}{x}} \right], \text{ use } \lim_{x \rightarrow \infty} \frac{a^x - 1}{x} = \ln a \\
= \ln^2 2 + \ln 3 = \ln 6 \\
\therefore \lim_{x \rightarrow \infty} \left(2^{\frac{1}{x}} + 3^{\frac{1}{x}} - 2 \right) x = \ln 6
\end{aligned}$$

$$\begin{aligned}
30. \lim_{x \rightarrow 0} \frac{e^{1-\sqrt{\cos 2x}} - e^{\tan^2 x}}{x^2}, \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \left[\frac{e^{1+\sqrt{\cos 2x}} - 1}{x^2} - \frac{e^{\tan^2 x} - 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{e^{1-\sqrt{\cos 2x}} - 1}{1 - \sqrt{\cos 2x}} \times \frac{1 - \sqrt{\cos 2x}}{x^2} - \lim_{x \rightarrow 0} \frac{e^{\tan^2 x} - 1}{\tan^2 x} \times \frac{\tan^2 x}{x^2} \right] \\
&= 1 \times \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos 2x}}{x^2} - 1 \times \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^2 \\
&= \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \times \frac{1}{1 + \sqrt{\cos 2x}} - 1^2 \\
&= \frac{2^2}{2} \times \frac{1}{1 + \sqrt{\cos 0}} - 1 = 2 \times \frac{1}{2} = \frac{2}{2} - 1 = 0 \\
&\therefore \lim_{x \rightarrow 0} \frac{e^{1-\sqrt{\cos 2x}} - e^{\tan^2 x}}{x^2} = 0
\end{aligned}$$